

Computer Networks

Module 1

Introduction to Networking

Outline

- Communication System
- Basic Concepts
 - Line Configuration
 - Transmission mode
- Types of Networks
- Network Topology

What is Networking?

In simple term “ *A network is simply a collection of computers or other hardware devices that are connected together, either physically or logically using special hardware and software in order to exchange information.*”

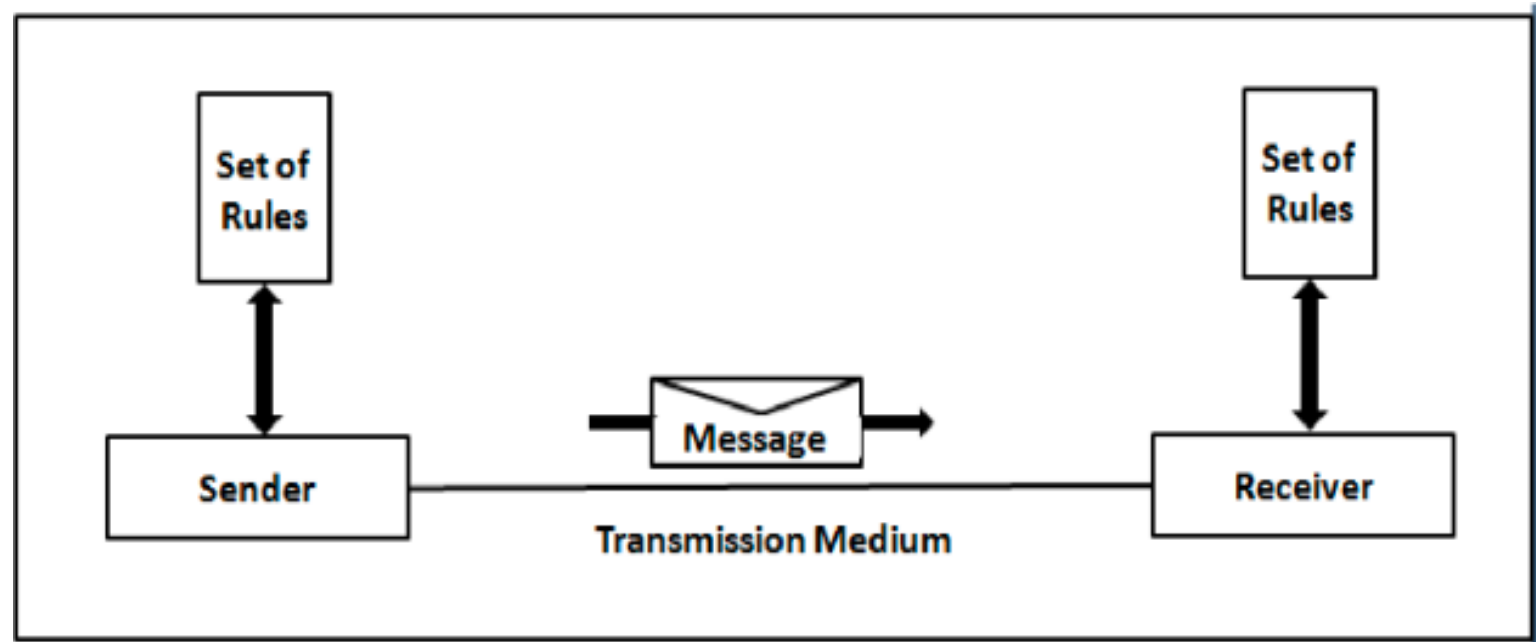
And **Networking** is term that describes the *process involved in design, implementation, management of network* using network technology.

Introduction to Communication System

- Data Communication is a process of exchanging data or information and in case of computer networks it done in between two or more devices over a transmission medium.
- Communication system consists of Hardware and Software.
- **Hardware:** Sender, Receiver and Intermediate devices
- **Software:** Set of rules and protocols that need to satisfied.

Components of Data Communication

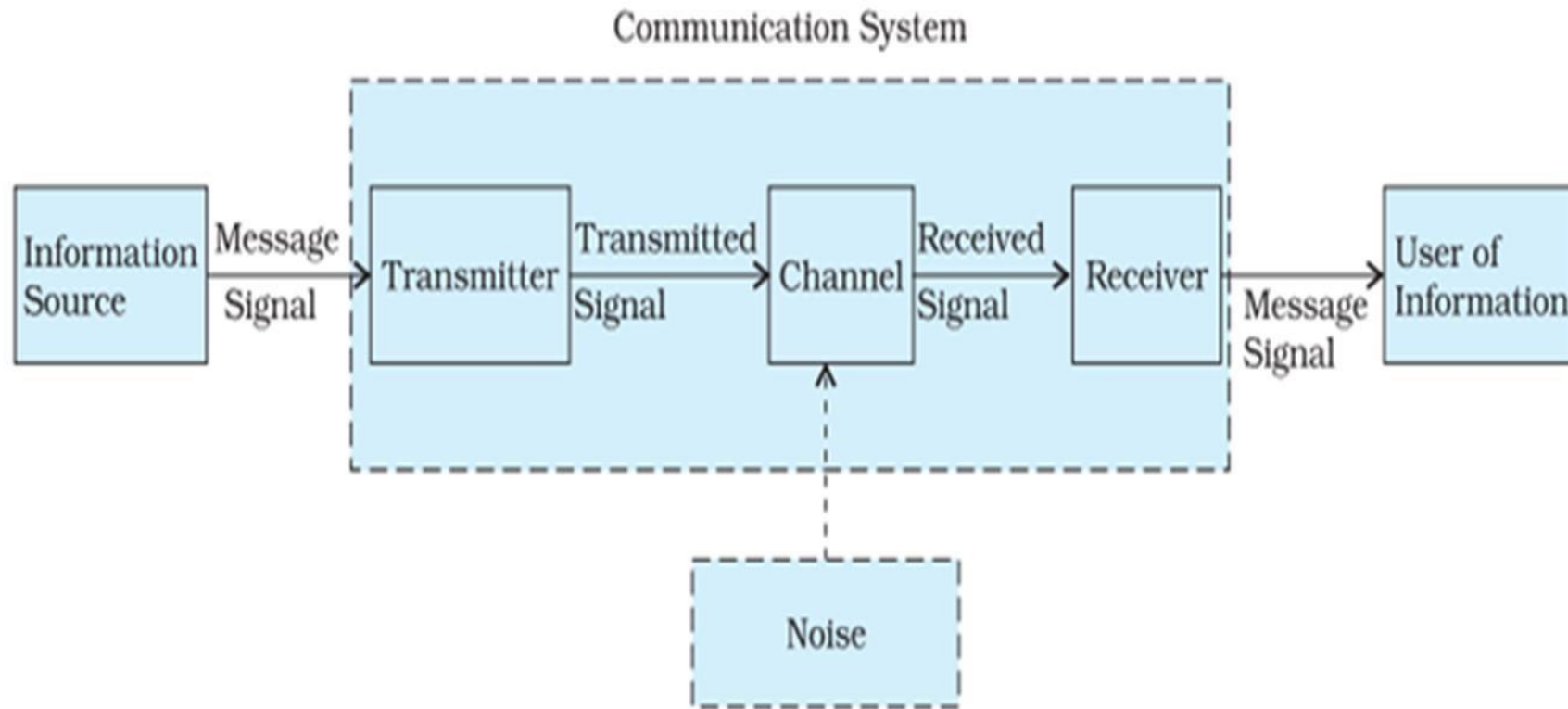
- Components:
 - Message
 - Sender
 - Receiver
 - Transmission Medium
 - Protocols



Effectiveness of data communication system depends on four fundamental charact :

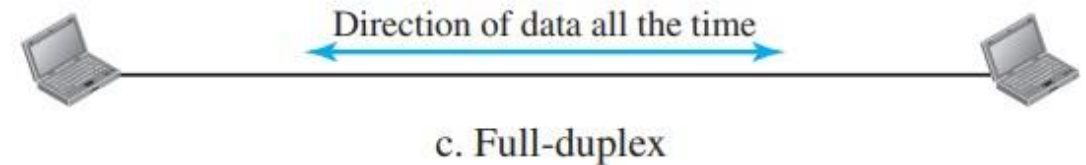
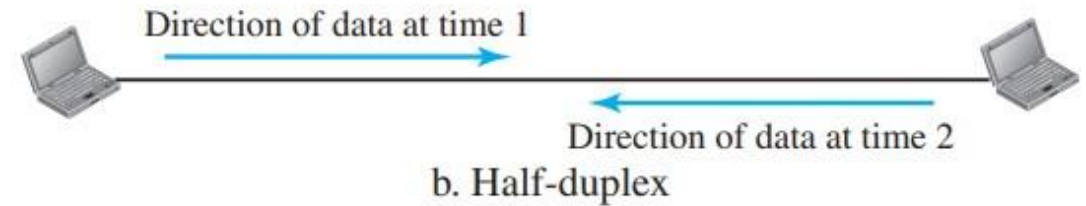
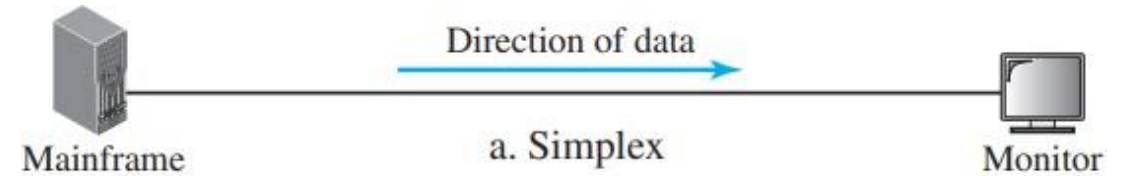
1. Delivery
2. Accuracy
3. Timeliness
4. Jitter

Elements Of Communication System



Data Flow

- Simplex
- Half-Duplex
- Full Duplex



Network Criteria

- **Performance:**

- Can be measured in terms of

- Transit time
 - Response time

- Evaluated by two networking metrics: Throughput & Delay

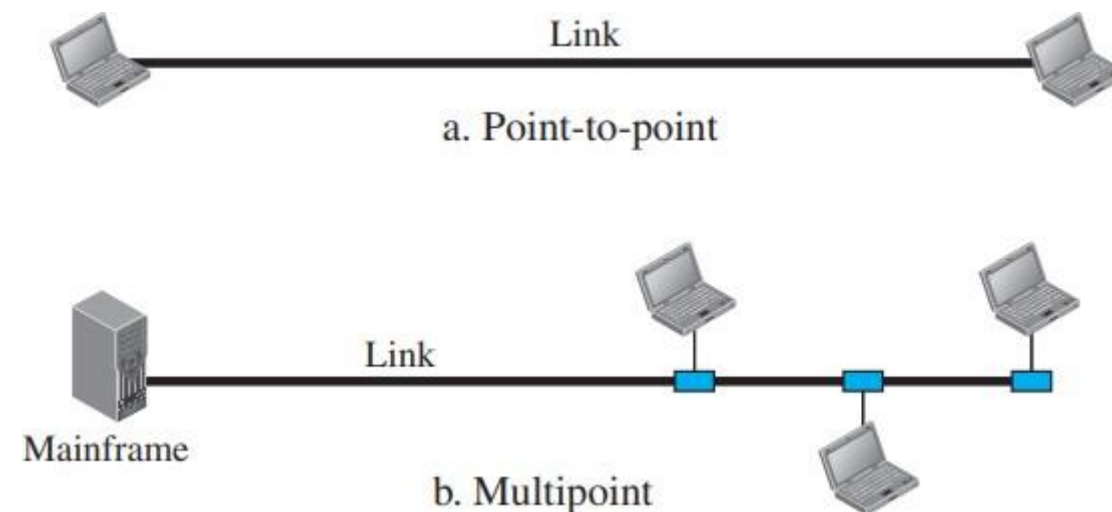
- **Reliability:** measured by frequency of failure, time taken by link to recover from failure and network robustness ion catastrophe

- **Security:** Securing data from unauthorized access, protecting data from damage, implementing policies, recovery from breaches and data loss.

Physical Structures

Type of Connections:

- Point-to-Point connection:
 - Provides dedicated link between two devices
 - Entire link capacity is reserved between two devices
- Multipoint (Multidrop):
 - More than two devices share a single link.
 - Capacity of channel is shared
 - Channel is shared either *Spatially* or *Timeshared*.



Physical Topology

Physical Topology refers to a way in which network is laid out physically.

Topology is geometric representation of relationship of all the links and linking devices (also called nodes) to one another.

There are Four basic Topologies:

- Mesh
- Star
- Bus
- Ring

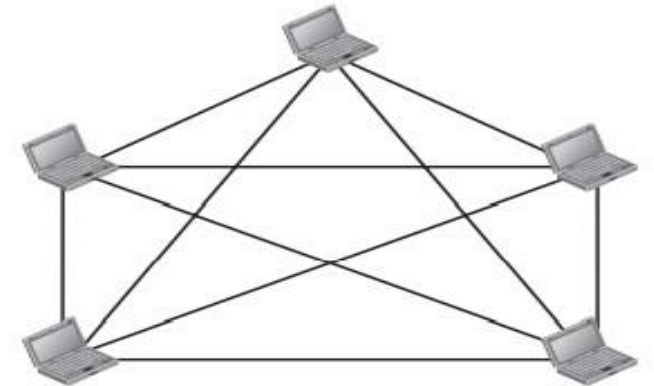
Mesh Topology

- Every device has a dedicated point-to-point link to every other device.
- Link carries traffic only between the two devices it connects.
- Total number of physical links in a fully connected mesh network with n nodes is equal to **$n(n-1)$** .
- In case of communication in both direction : **$n(n-1)/2$ duplex node links**.
- Every device on network must have **$(n-1)$** input/output (I/O) ports.

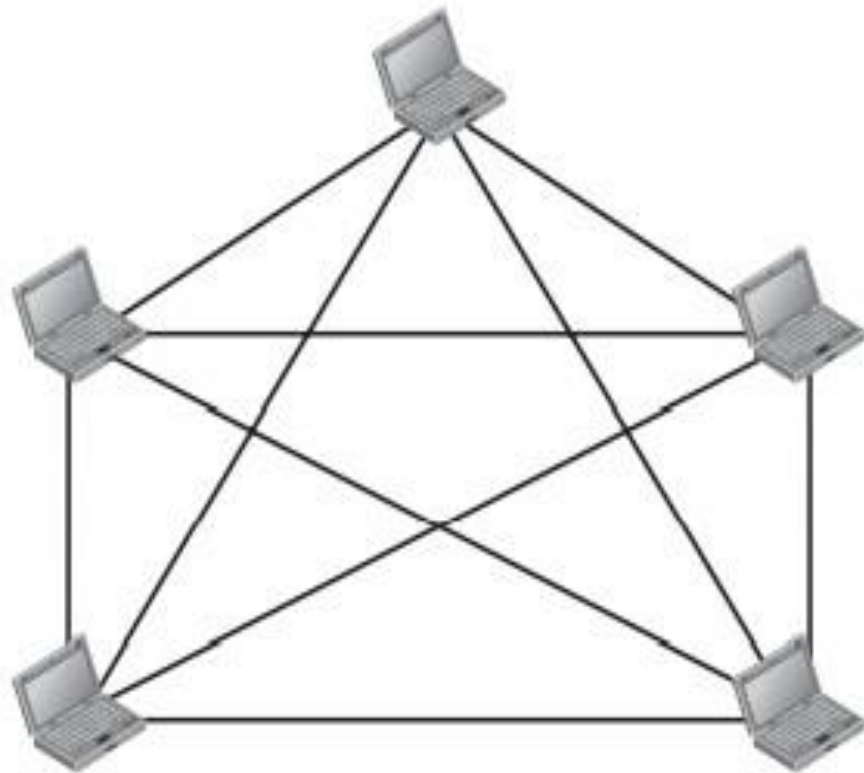
Mesh Topology

- **Advantages**
 - No traffic problems
 - Robust
 - Privacy and security
 - Point to point link: fault identification and isolation is easy
- **Disadvantages**
 - Installation and reconnection difficult as devices interconnected.
 - Bulk wiring
 - Expensive: hardware cost

$n = 5$
10 links.



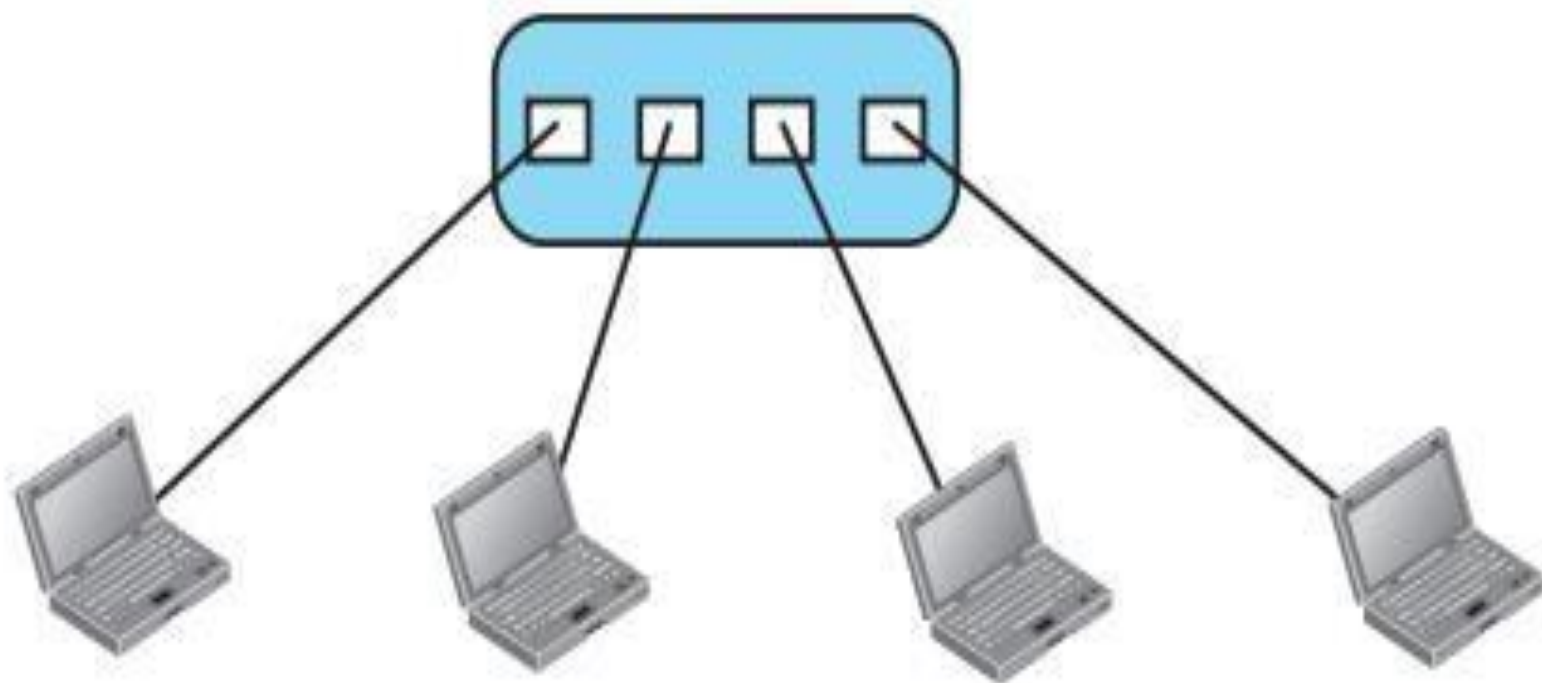
$n = 5$
10 links.



Star Topology

- each device has a dedicated point-to-point link only to a central controller, usually called a **hub**.
- devices are not directly linked to one another.
- star topology does not allow direct traffic between devices.
- The controller acts as an exchange, it relays the data to the other connected device.
- The star topology is used in local-area networks (LANs)

Hub



Star Topology

Advantages

- Less expensive than mesh
- Reconfiguration and installation is easy.
- Robustness
- Easy fault identification as long as hub is working

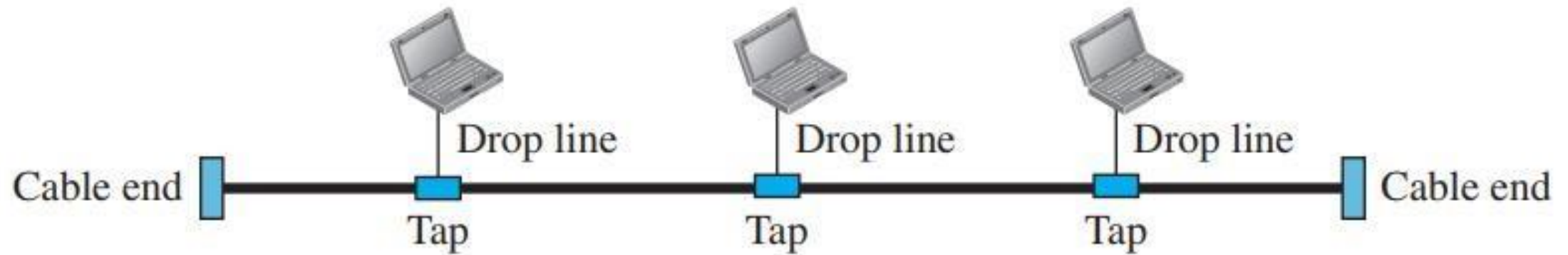
Disadvantages

- Dependency on single point controller.
- Each node must be linked to hub: more cabling required compared to other topologies except mesh.

Bus Topology

- A bus topology is multipoint connection.
- One long cable acts as a backbone to link all the devices in a network.
- Nodes are connected to the bus cable by drop lines and taps.
- A drop line is a connection running between the device and the main cable.
- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.

Bus Topology



Bus Topology

Advantages

- Ease of installation: less cabling as compared to mesh and star.
- A bus uses less cabling than mesh or star topologies

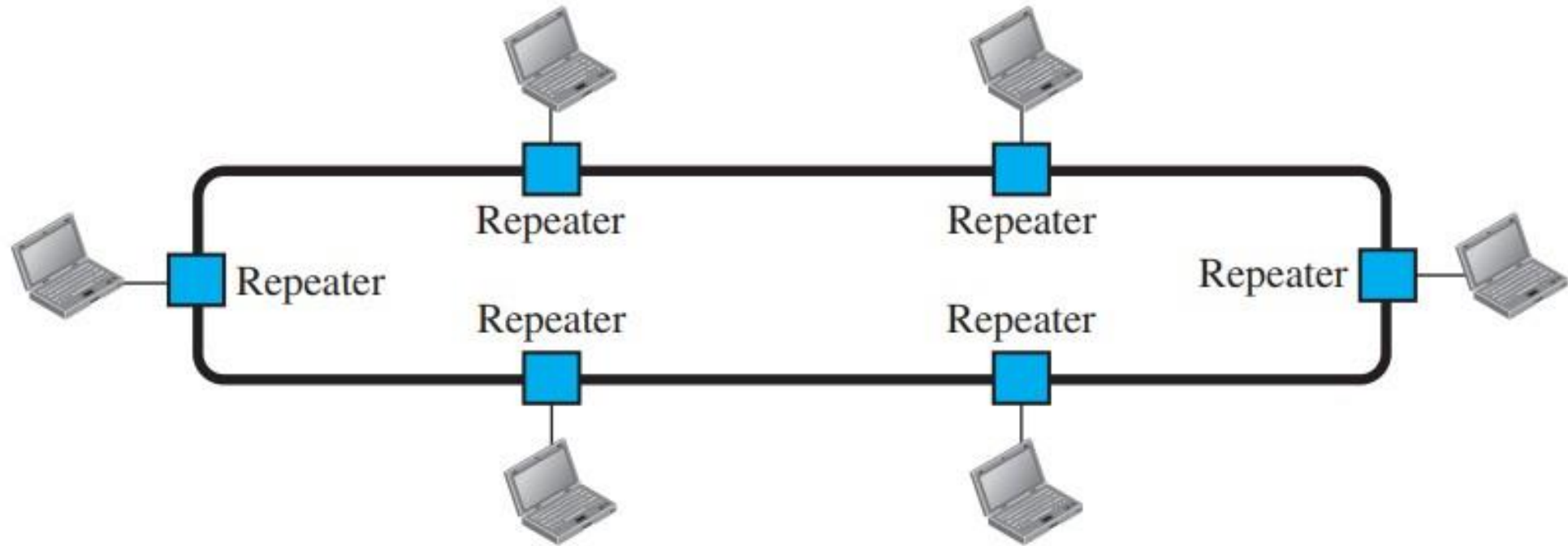
Disadvantages

- It is Difficult to add new devices.
- Difficult reconfiguration and fault isolation.
- A fault in Backbone stops all transmission
- Limited cable length and number of nodes that can be connected.

Ring Topology

- Each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater.

Ring Topology



Ring Topology

Advantages

- Easy to install and reconfigure.
- Easy adding and deleting of connections.
- Fault isolation simplified: devices can raise the alarm.

Disadvantages

- Unidirectional traffic can be a disadvantage.
- Breakdown in ring disables the entire network.

Types of Networks

- **Local Area Network (LAN)**
- **Metropolitan Area Network(MAN)**
- **Wide Area Network(WAN)**

Local Area Networks

- A *local-area network* (LAN) is a computer network that spans a relatively small area.
- A local area network (LAN) is usually privately owned and links the devices in a single office, building, or campus.
- LAN size is limited to a few kilometers.

- LANs are designed to allow resources to be shared between personal computers or workstations.

Example: hardware or software

- One of the computers may be given a large capacity disk drive and may become a server to clients.
- LAN extends up to 10m to 1km

Local Area Networks

- LANs are distinguished from other types of networks by their transmission media and topology.
- Speeds: 100 , 1000 Mbps.

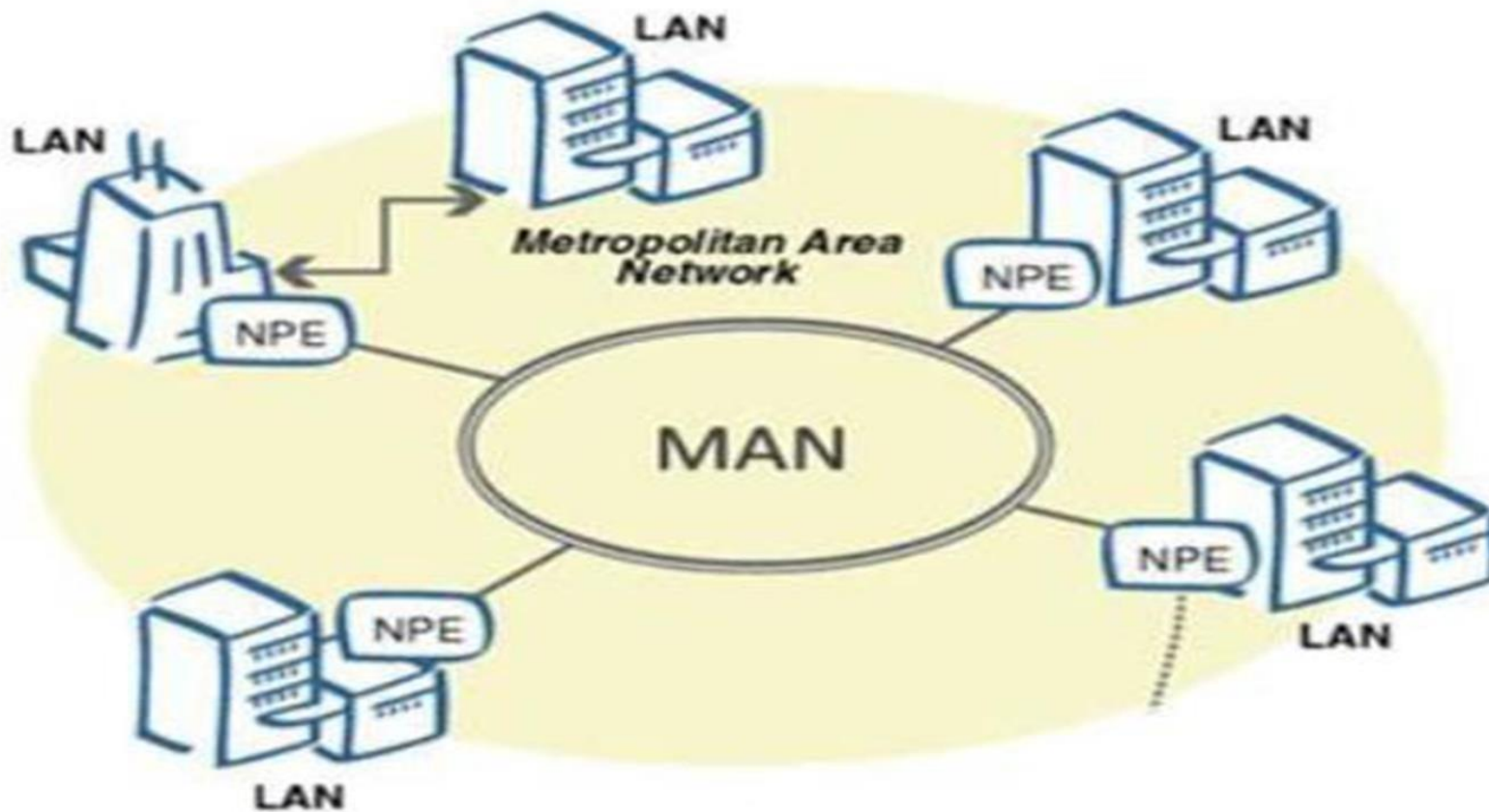
Metropolitan Area Networks

- Metropolitan Area Network usually covers area inside a town or a city.
- It is designed for customers who need a high-speed connectivity, normally to the Internet, and have endpoints spread over a city or part of city.

Example: cable TV network

- MAN extends up to 30-40 km.
- Speed of 34–155 Mbit/s.
- MAN uses Guided Media or Unguided media.

Metropolitan Area Networks

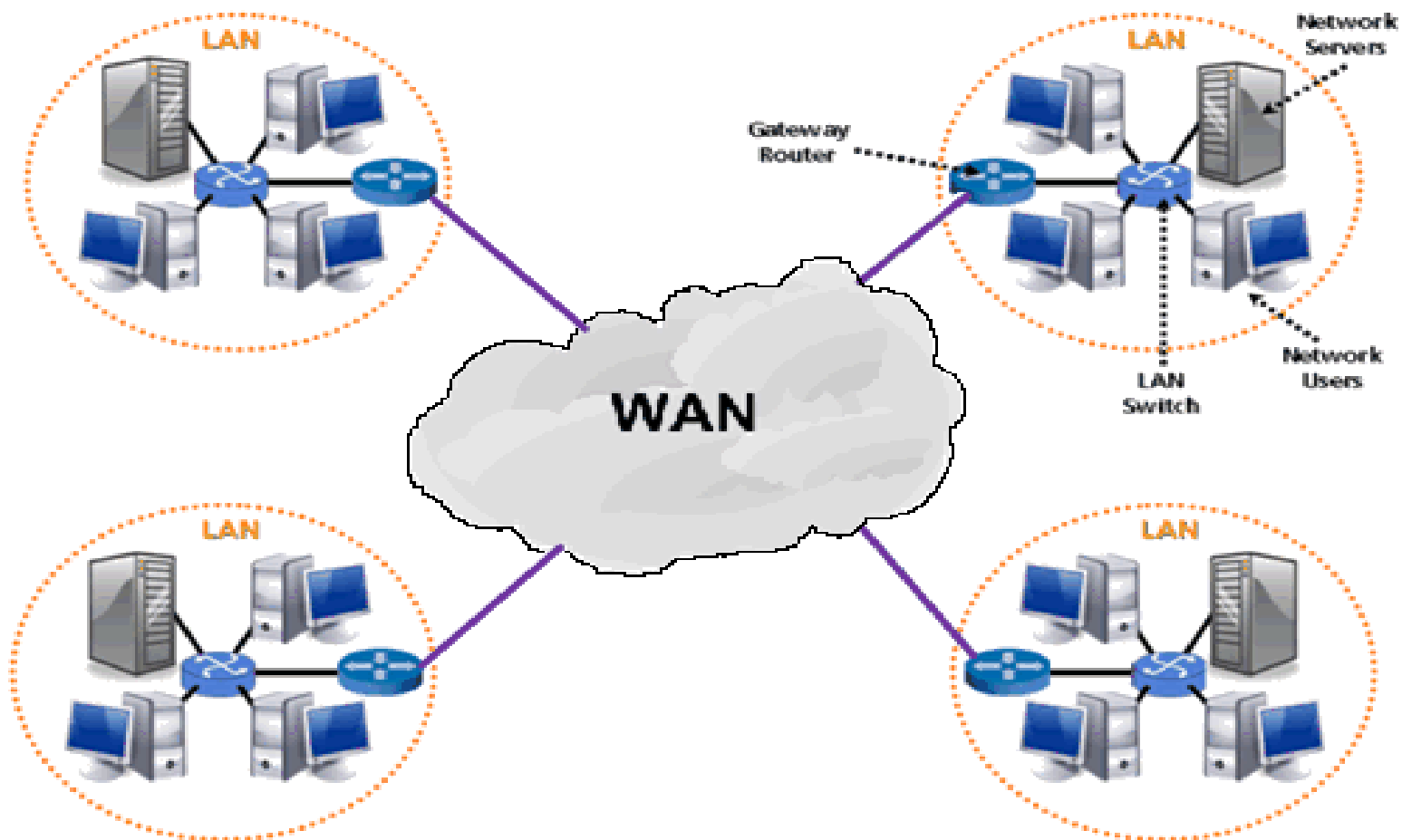


Wide Area Networks

- A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent or even the whole world.
- WAN (Wide Area Network) is a group of computers and other network devices which are connected and not restricted to a geographical location.
- Internet is WAN.

- WAN speed varies based on geographical location of the servers. WAN connects several LANs.
- WAN connection speeds can be 10Mbps or 100Mbps.
- WAN mainly uses Guided Media or Unguided media.

Wide Area Networks



Switching

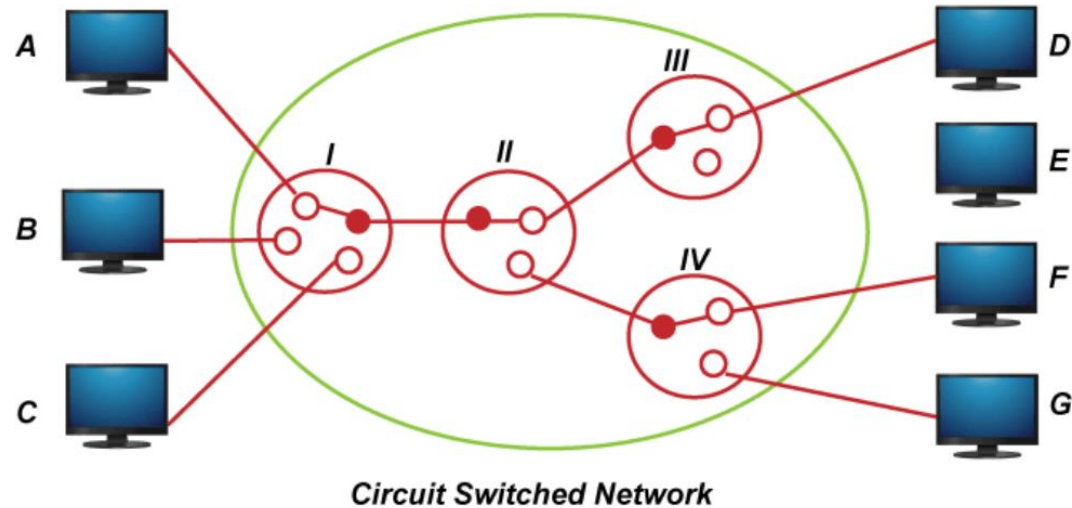
An internet is a switched network in which a switch connects at least two links together.

The two most common types of switched networks are circuit- switched and packet-switched networks.

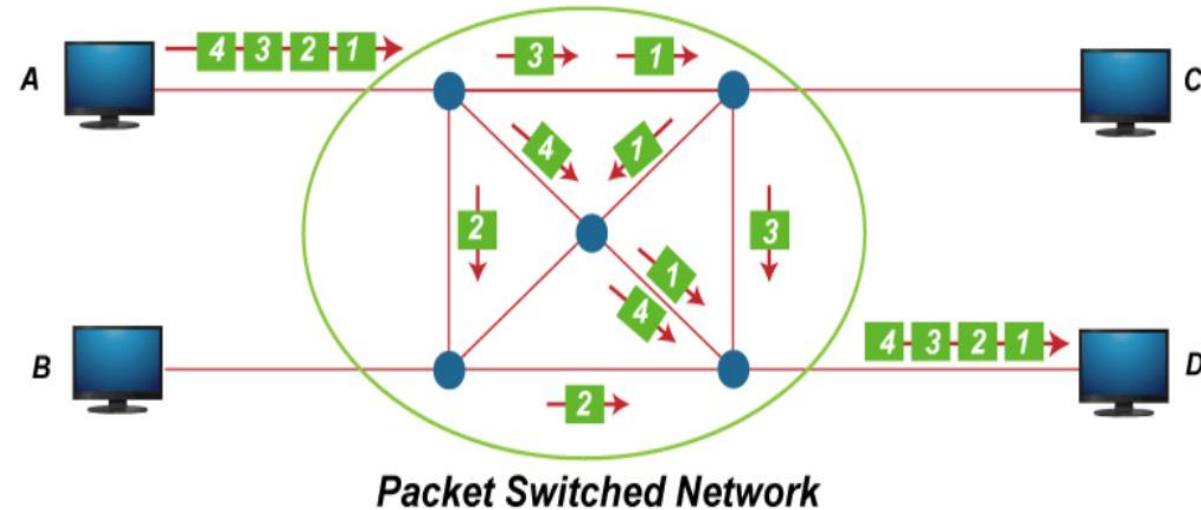
- Circuit-Switched Network
- Packet Switched network
- A circuit-switched network is one of the simplest data communication methods in which a dedicated path is established between the sending and receiving device. In this physical links connect via a set of switches.
- In the Packet switching Network, the message is divide into packets. Each packet contains a header which includes the source address, destination address, and control information.

Switching

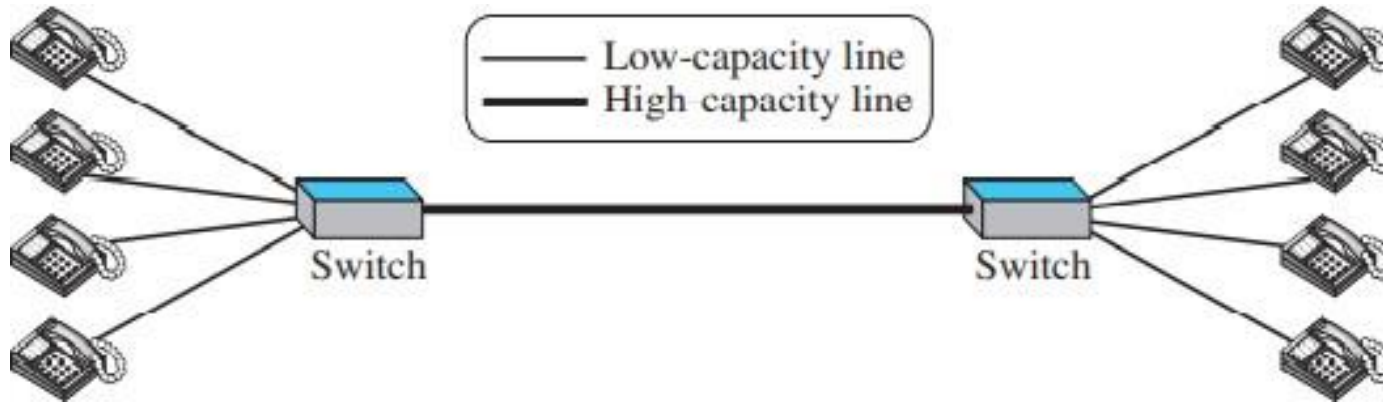
- Circuit-Switched Network**



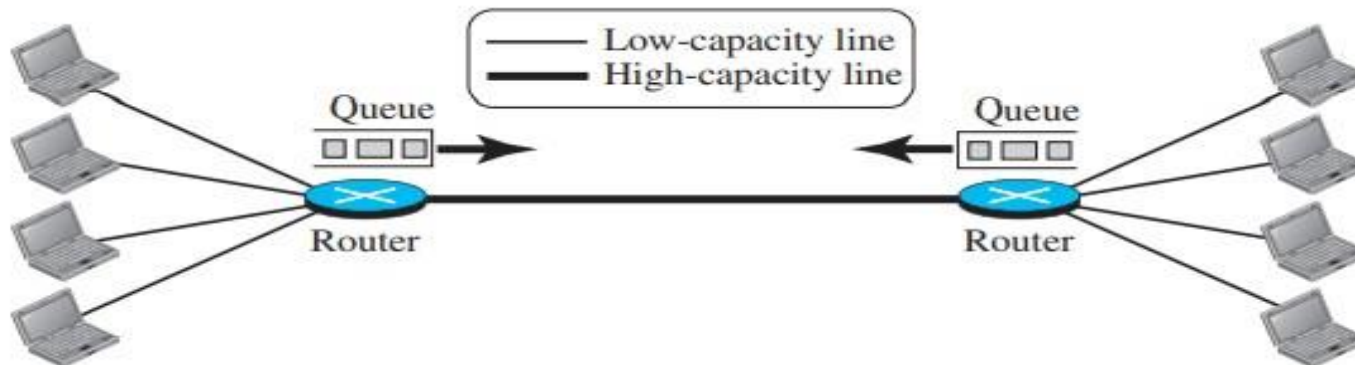
- Packet Switched network**



Switching

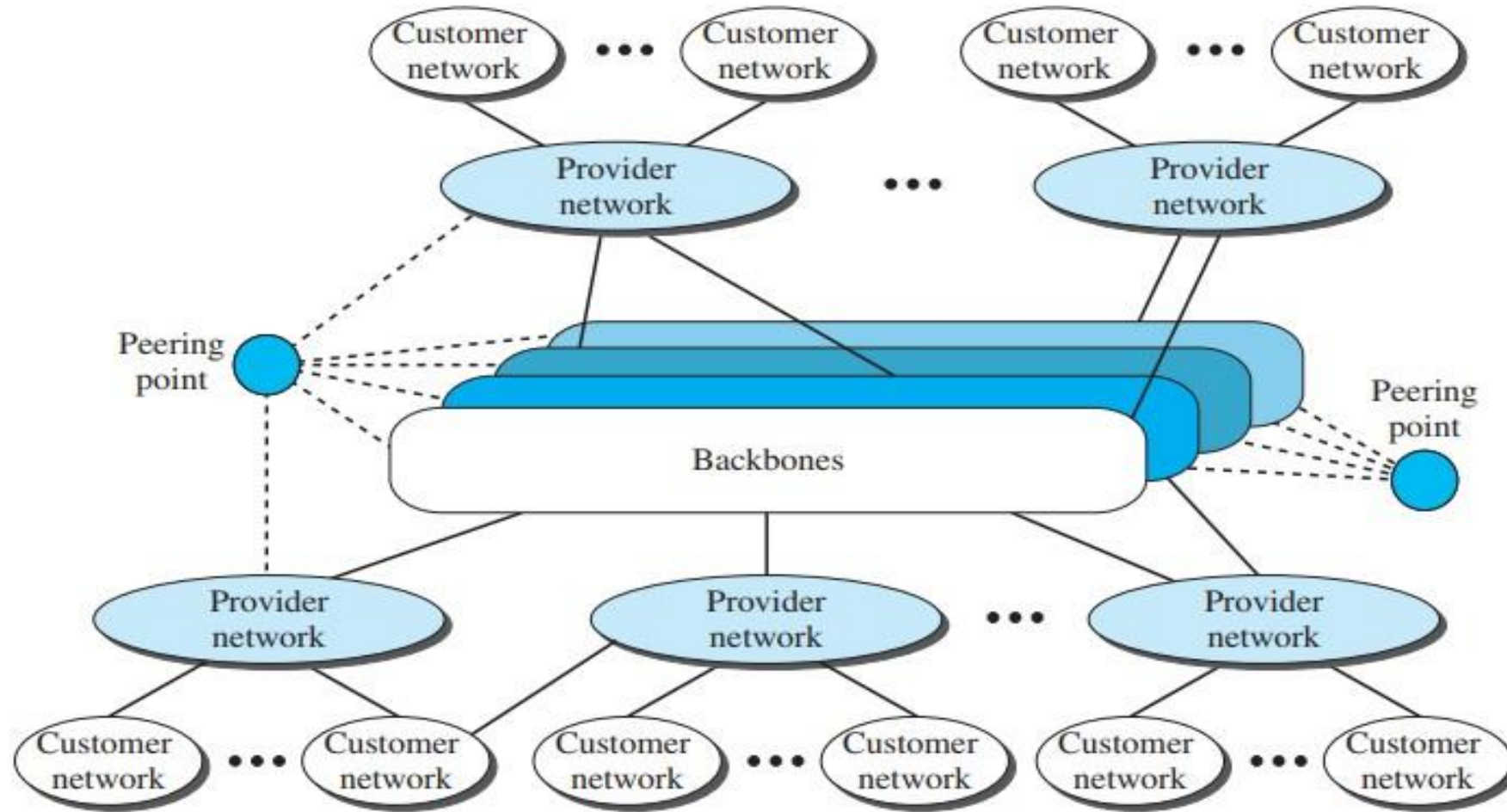


A Circuit Switched network



A Packet Switched network

The Internet Today



Connection Less Services

- No session connection between sender and receiver.
- No reliability
- Short messages
- Does not maintain state information.
- Less overhead
- Example: Walkie-Talkie

Connection Oriented Services

- Requires session connection (analogous to a phone call).
- Reliable network service.
- Set up virtual links between the end systems through a network.
- Long messages.
- High overhead and places greater demands on B/W.
- Example: Email

Protocol Layering:

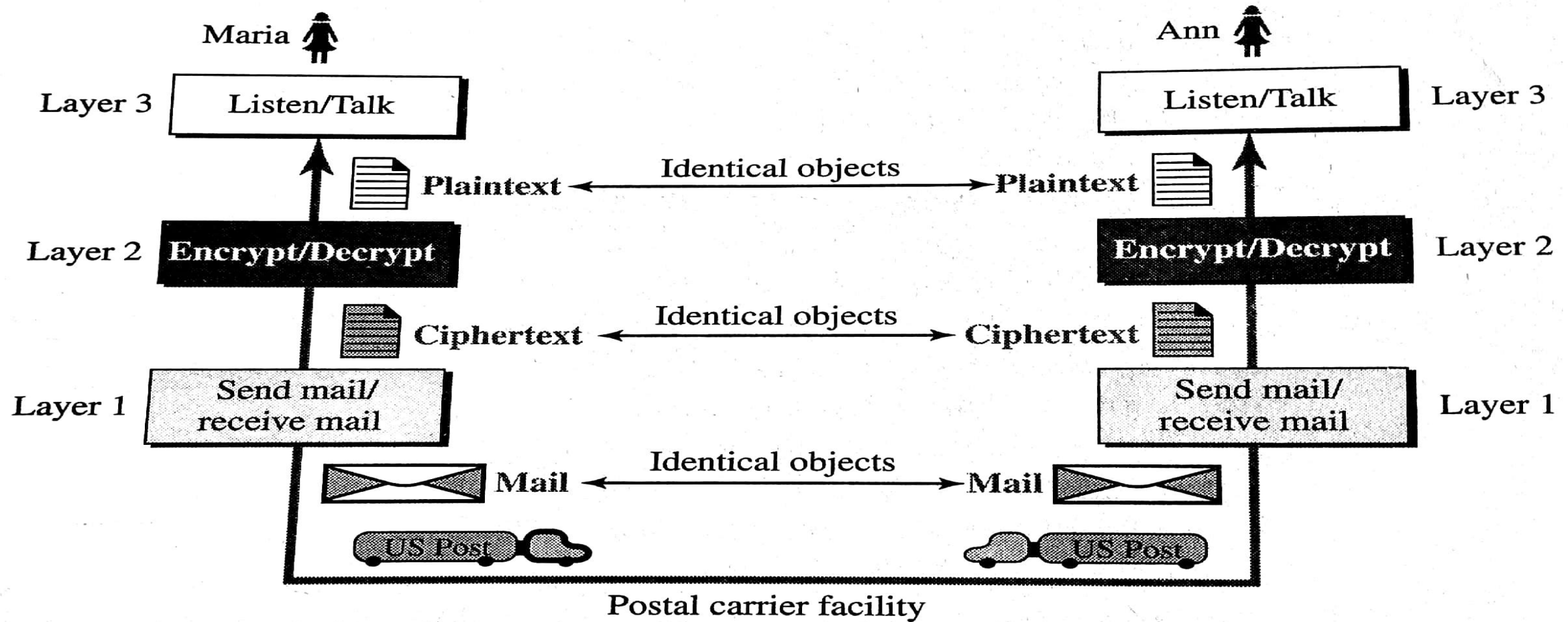
- ISO OSI Model
- TCP/IP model

Why Layered Architecture ?

- Assume Maria and Ann are neighbors with a lot of common ideas.
- Communication between Maria and Ann takes place in one layer, face-to-face, in the same language
- Rules
 - Greetings
 - Vocabulary level between them
 - One should refrain when other is speaking
 - Conversation should be dialog
 - Greetings at the end



- Now assume that Ann is offered a higher-level position her job, but needs to move to another branch, far from Maria
- They are planning to start a new business so they need to communicate
- So they want to continue the conversation using regular mail through post office
- They don't want their ideas to be revealed by other people.
- They agree on encryption/ decryption technique



At Senders Side

Lets assume Maria sends the first letter to Ann

Maria talks to machine at layer 3 as if its Ann

Machine listens to Maria and creates a plain text which is passed to second layer machine

The second layer machine takes the plain text, encrypts it, and creates the cipher text which is passed to first layer machine

The first layer machine, a robot, takes the cipher text, puts it in an envelope , adds sender and receiver address and mails it

At receivers Side

**The first layer machine picks up the letter from Ann's mail box
It takes out the cipher text from the letter and delivers it to the
second layer machine**

**The second layer machine decrypts the message, creates the plain
text, and passes it to the third layer machine**

**The third layer machine takes the plain text and reads it as though
Maria is speaking**

Modularity

We could have used one machine to do the job of all the three machines

**However if both Maria and Ann decide that the encryption/decryption done by the machine is not enough to protect the secrecy, they need to change the whole machine
In present situation they need to change only the second layer machine**

This is called modularity

Why layering?

Communication does not always use two end systems, there are intermediate systems that need only some layers, but not all layers. If we do not use protocol layering, we would have to make each intermediate system as complex as the end systems, which makes the whole system more expensive.

Principles of Protocol Layering

First Principle:

Each layer should be able to perform two opposite tasks for bidirectional communication

Example : Third layer Listen (one direction) Talk (another direction)

Second Layer encrypt and decrypt

First layer send and receive mail

Second Principle:

Two objects under each layer at both sides should be identical

Example: Third layer both sides plain text

Second layer both sides cipher text

First layer both sides a piece of mail

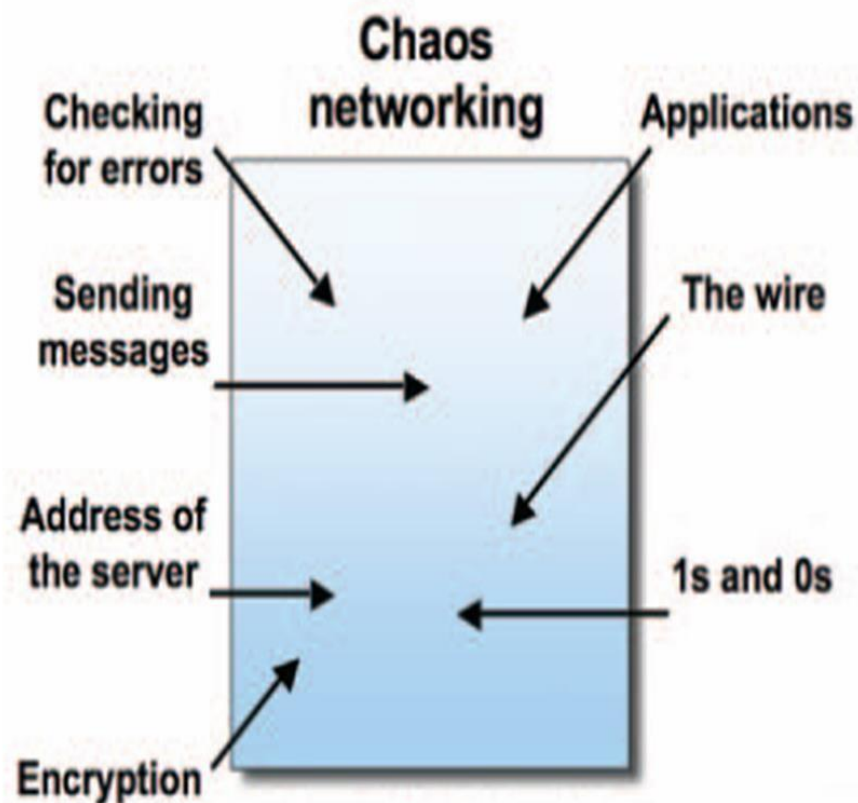
The OSI model

- Over the past couple of decades many of the networks that were built used different hardware and software implementations, as a result they were incompatible, and it became difficult for networks using different specifications to communicate with each other.
- The **Open System Interconnection (OSI)** model includes a set of protocols that allows any two systems to communicate regardless of their architecture.

The OSI model

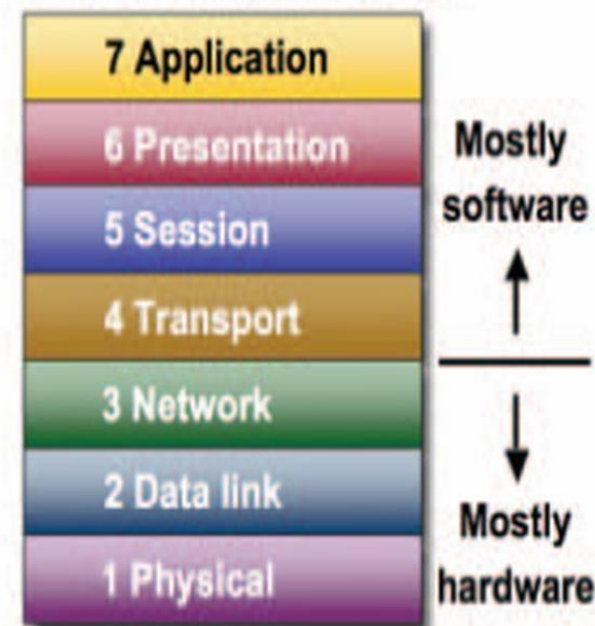
- The OSI model is a concept that describes, how data communications should take place.
- OSI is also called as the framework for design of network systems.
- It divides the process into seven steps called layers.

The OSI model



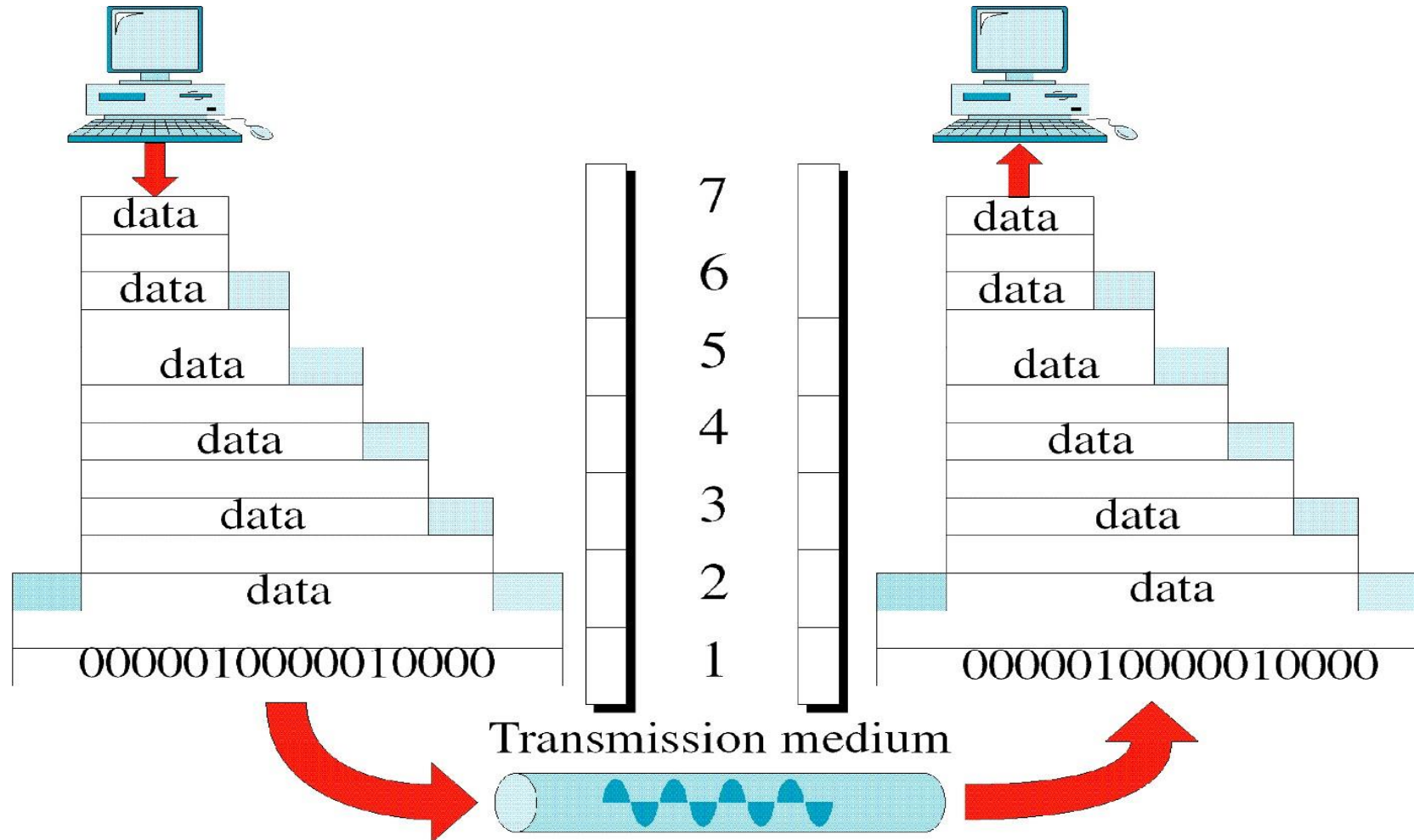
Without the OSI model, networks would be very difficult to understand and implement.

Networking OSI seven-layer model



With the OSI model, networks can be broken up into manageable pieces. The OSI model provides a common language to explain components and their functionality.

The OSI model



Physical Layer

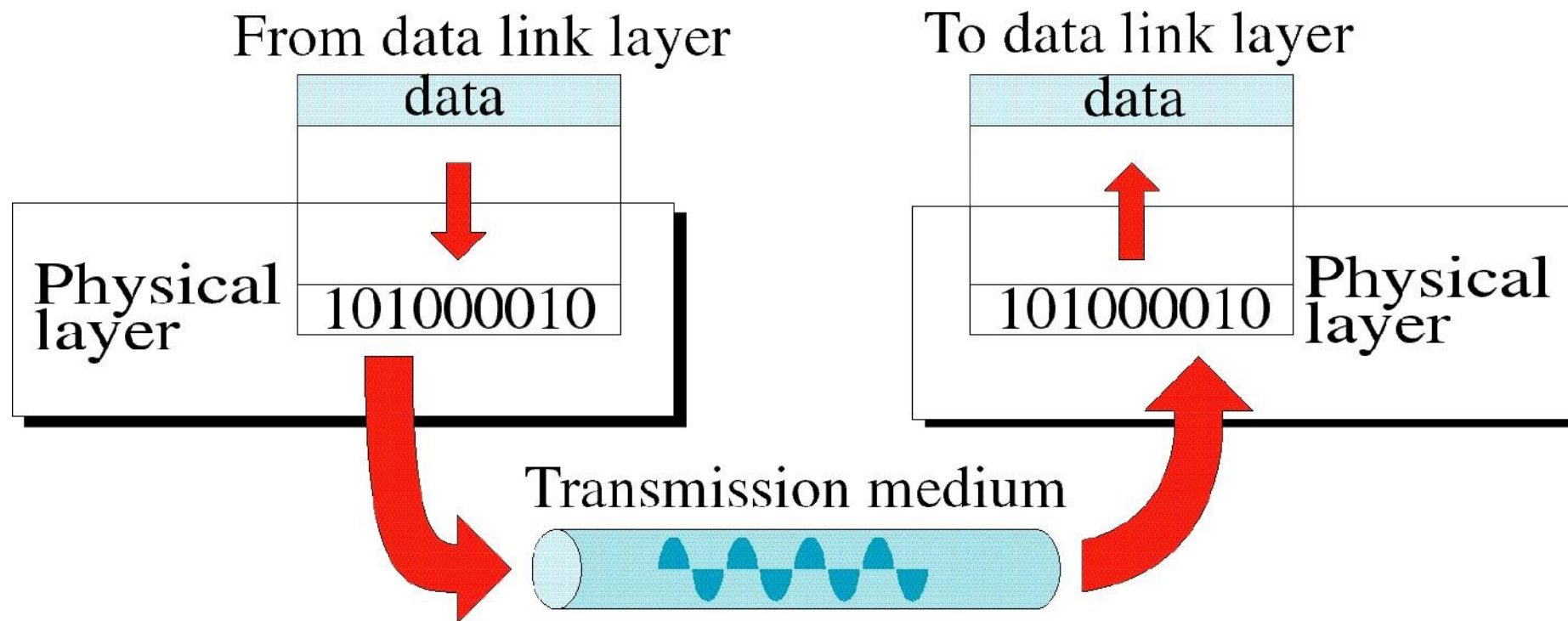
- The physical layer deals with the physical characteristics of the transmission medium.
- This layer consists of simply the wire or media by which the network signals are conducted.
- The physical layer of the OSI model defines connector and interface specifications, as well as the medium (cable) requirements for transmission to occur.

Physical Layer

Functions of physical layer:

- Physical characteristics of interfaces and medium.
- Representation of bits: Conversion from binary to electrical or optical.
- Data rate.
- Synchronization of bits.
- Physical topology
- Line configuration: point to point or multipoint.

Physical Layer



Data Link Layer

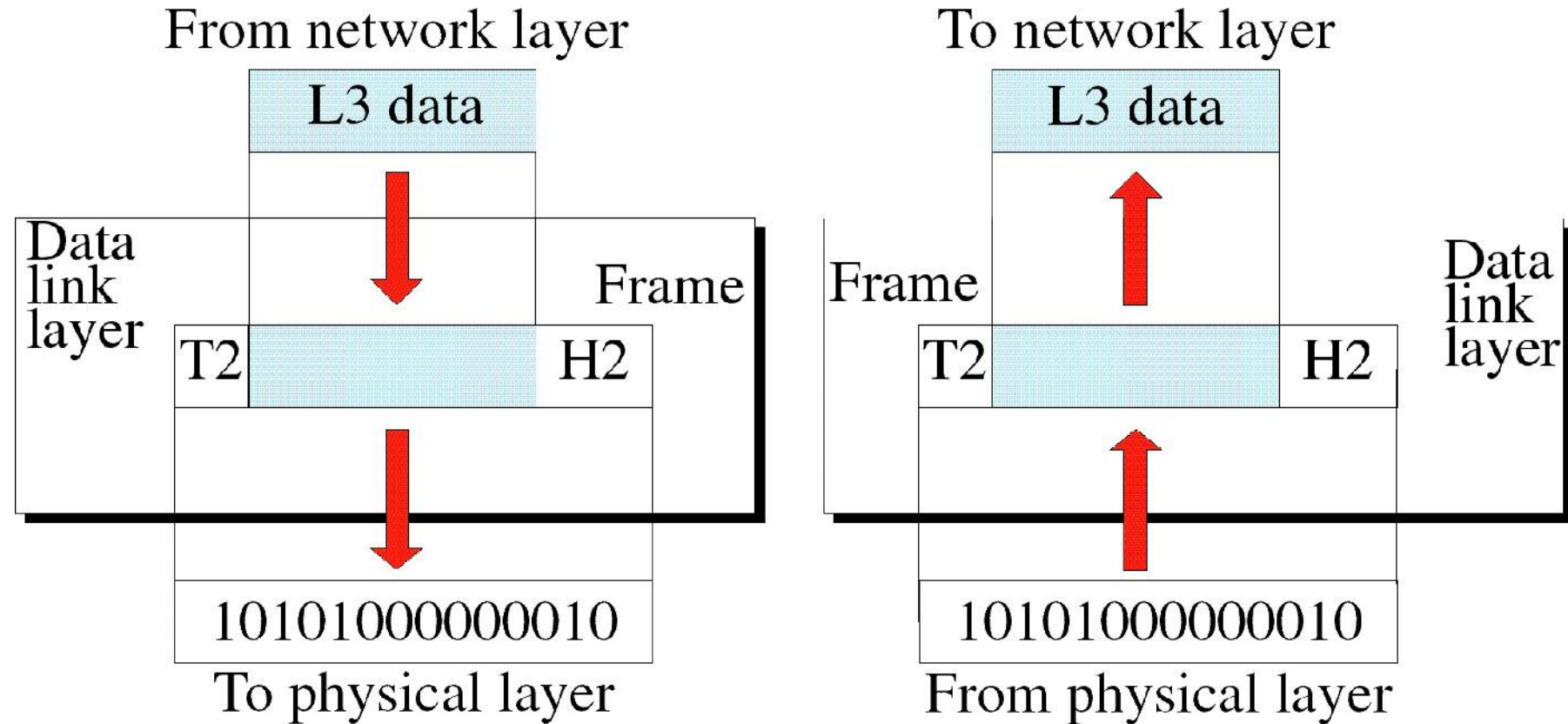
- The data link layer transforms the physical layer, a raw transmission facility, to a reliable link.
- *Node to node delivery (Hop to Hop Delivery)*
- The data link layer uses the **MAC address** to define a hardware or data link address for multiple stations to share the same medium and still uniquely identify each other.
- Concerned with network access, error notification, ordered delivery of frames, and flow control.

Data Link Layer

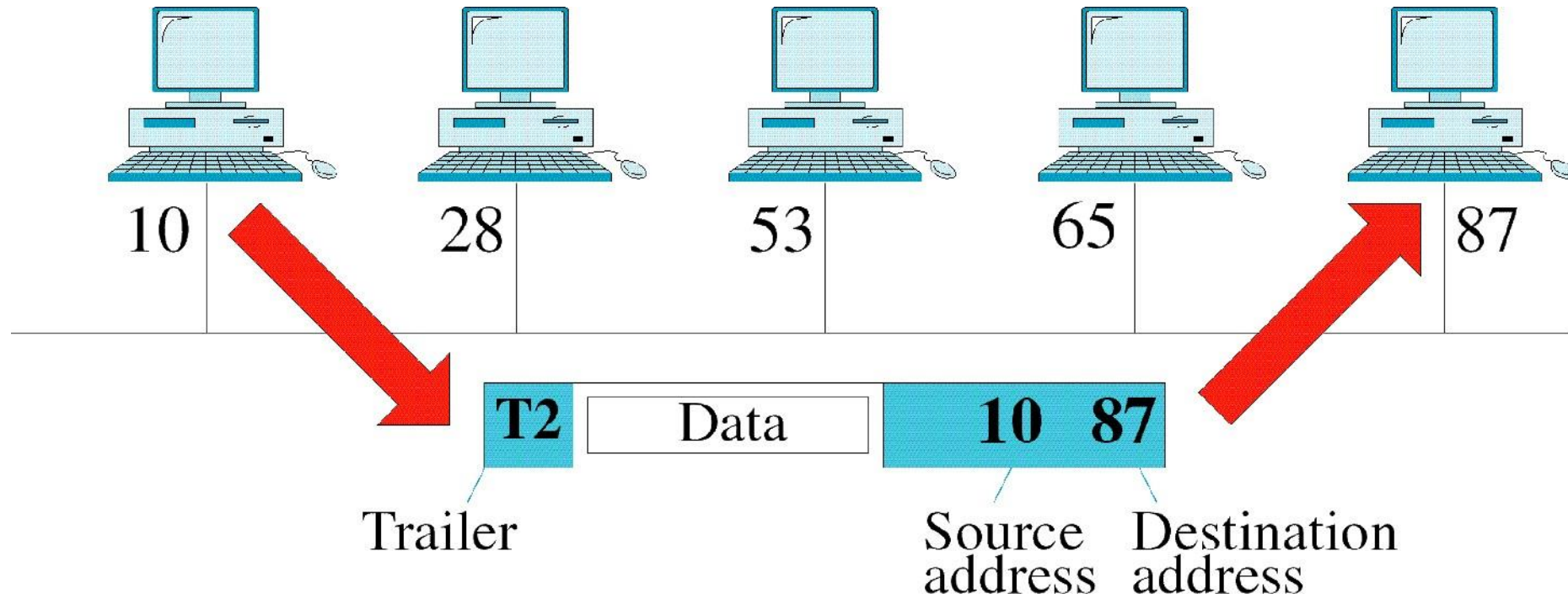
Functions of Data Link layer:

- Framing
- Physical addressing: adding header to frame to identify sender and receiver address.
- Flow control
- Error control
- Access control

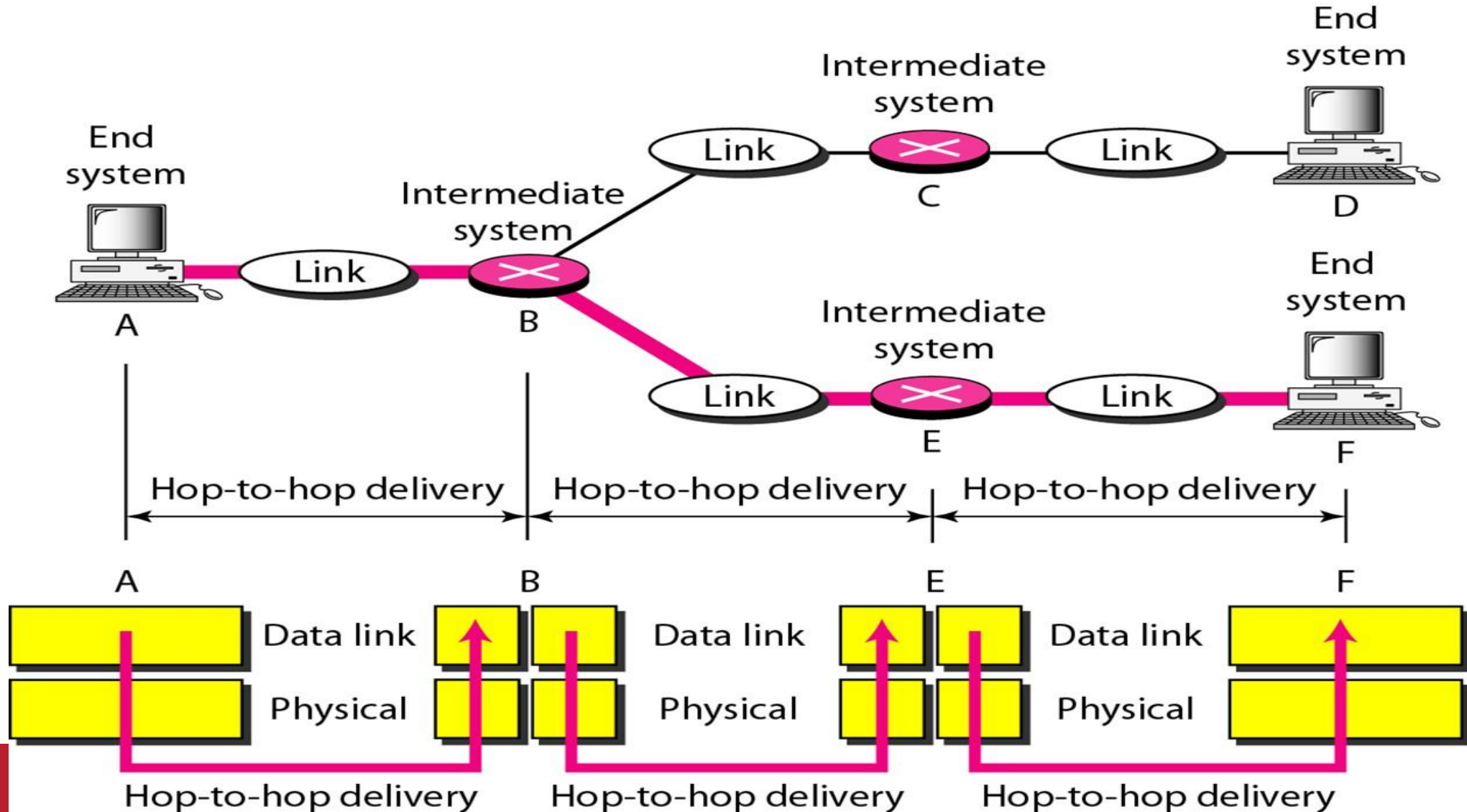
Data Link Layer



Data Link Layer



Hop-to-Hop Delivery



Network Layer

- This layer establishes the route between the sending and receiving stations.
- Responsible for delivery of individual packets from source to destination.
- Two systems connected on same link-no network layer.
- Two systems on different link-need network layer

Network Layer

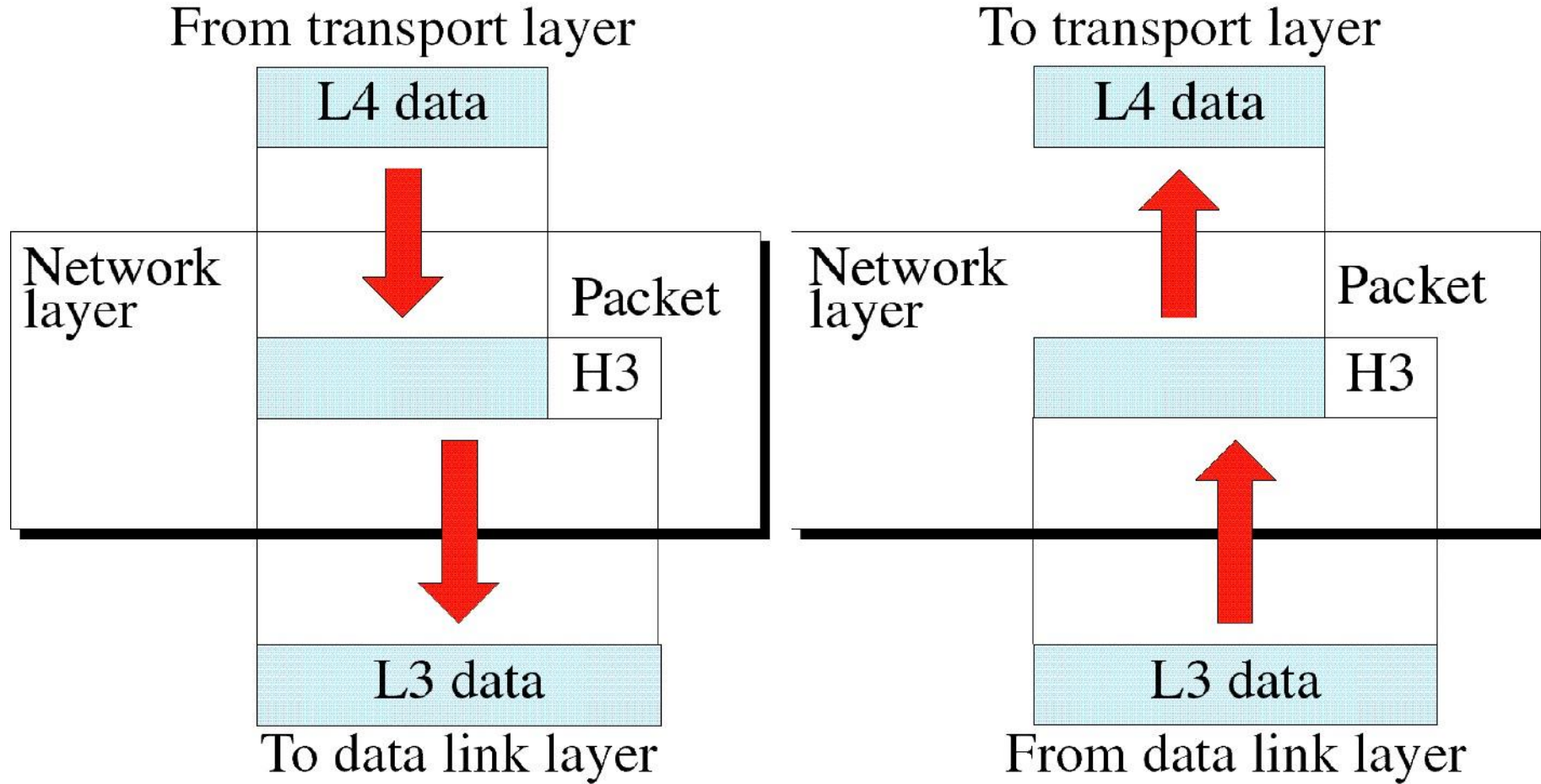
- It handles the routing of data (sending in the right direction to the right destination on outgoing transmissions and receiving incoming transmission at the packet). The layer does routing & forwarding of data.
- Network layer addresses can also be referred to as logical addresses.

Network Layer

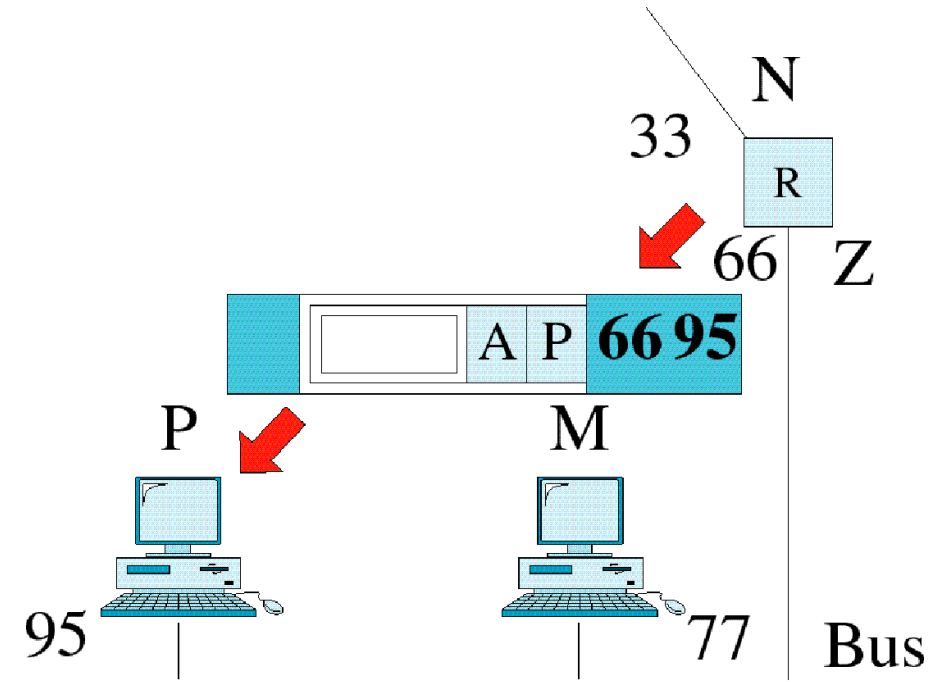
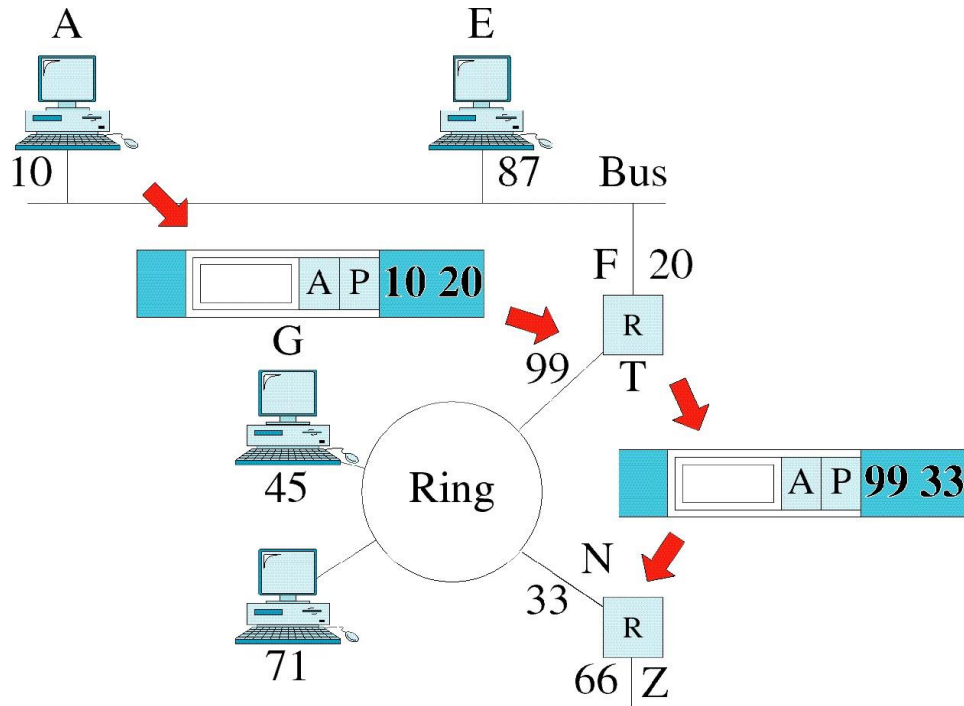
Functions of network layer:

- **Logical addressing:**
 - Data link layer physical addressing-handles addresses locally.
 - Packet crosses network boundary-IP addressing.
- **Routing** : Routing packets to destination via connecting devices over large networks.

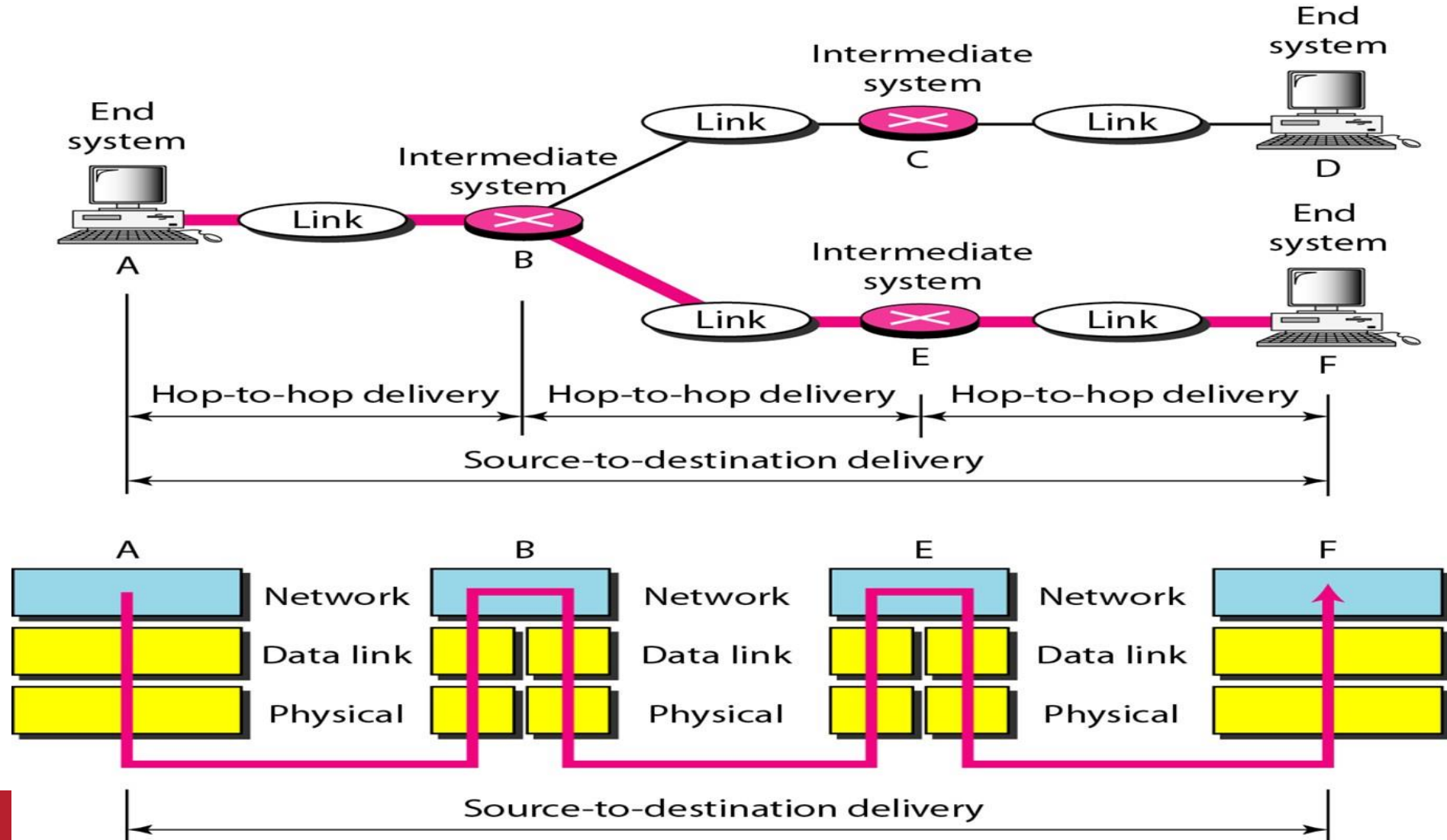
Network Layer



Network Layer Example



Source-to-destination delivery



Transport Layer

- The transport layer is responsible for the delivery of a message from one process to another.
- Transport layer recognizes relationship between the packets and makes sure that the whole message arrives intact and in order.
- It is responsible for constructing stream of data segments, sending and checking for correct delivery.

Transport Layer

- If data is sent incorrectly, this layer has the responsibility to ask for retransmission of the data.
- This layer acts as an interface between the bottom and top three layers.

Transport Layer

Responsibilities of transport layer:

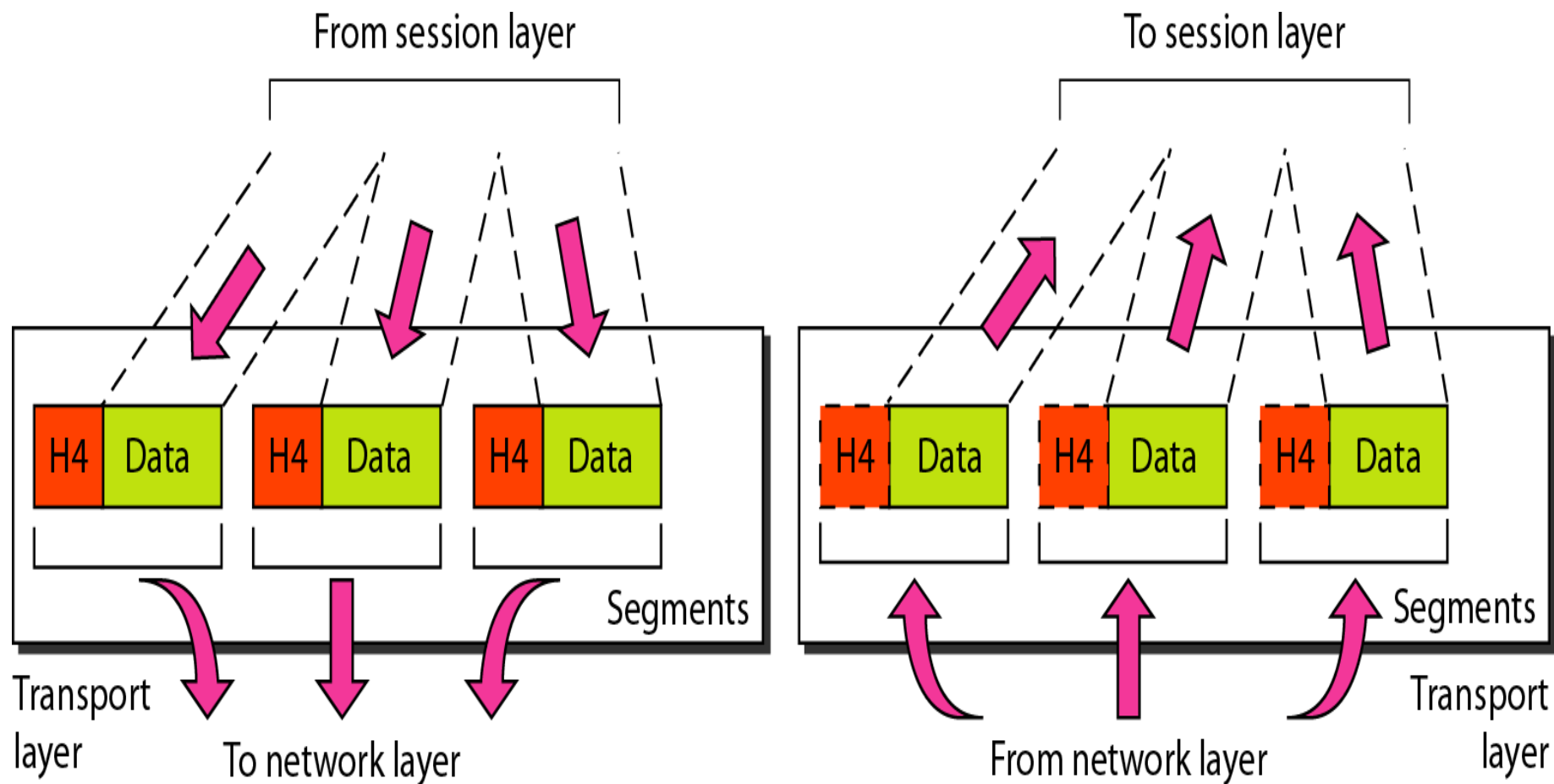
- Port addressing: Port Address(process to process).
 - Network layer gets each packet to correct computer.
 - Transport layer gets each packet to correct process on that computer.
- Segmentation and reassembly: Message divided into segments-sequence number
- Error control and Flow Control for process to process.

Transport Layer

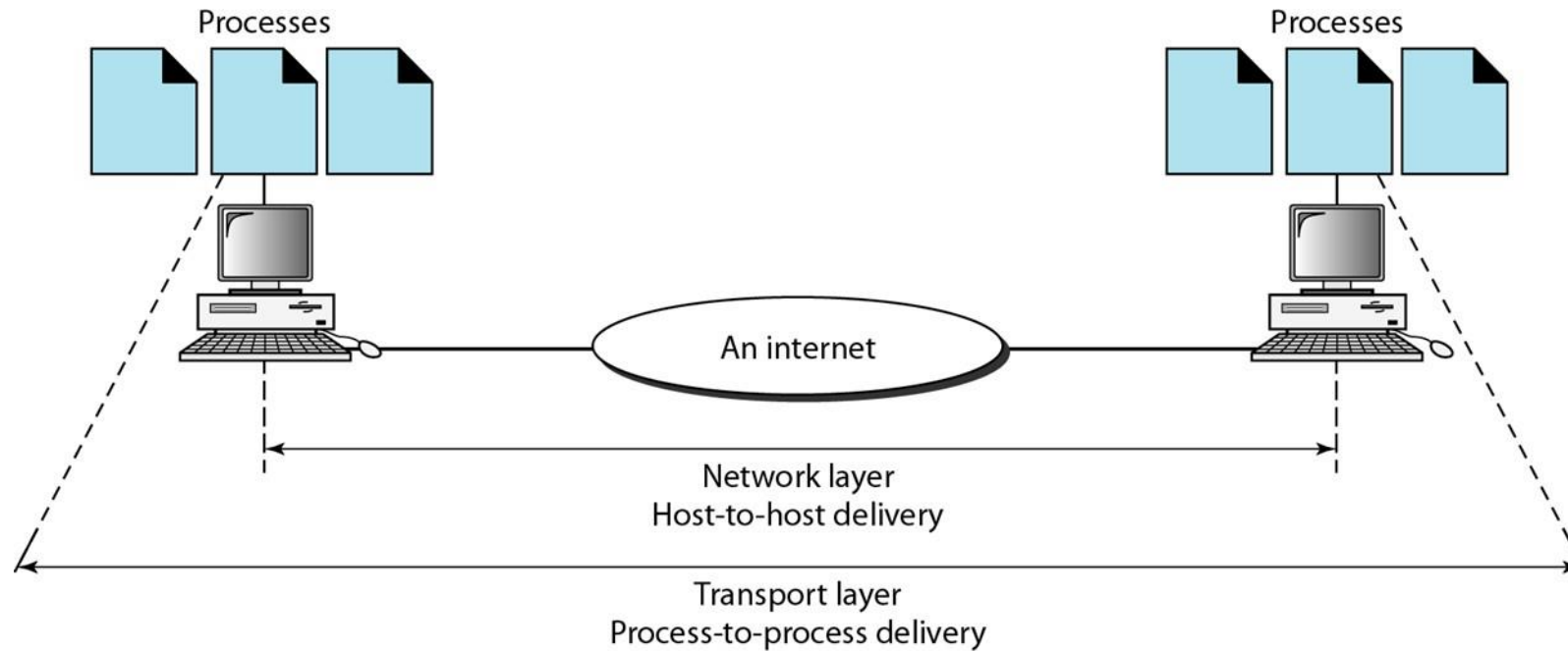
Responsibilities of transport layer:

- Connection control: *Connection-less or connection-oriented.*
- Flow control: *flow control at this layer is performed end to end rather than across a single link.*
- Error control: *error control at this layer is performed process-to-process rather than across a single link.*

Transport Layer



Reliable process-to-process delivery of a message



Session Layer

- The services provided by the first three layers (physical, data link, and network) are not sufficient for some processes.
- The session layer defines how to start, control and end conversations (called sessions) between applications.
- It provides coordination of the communication in an ordering manner.

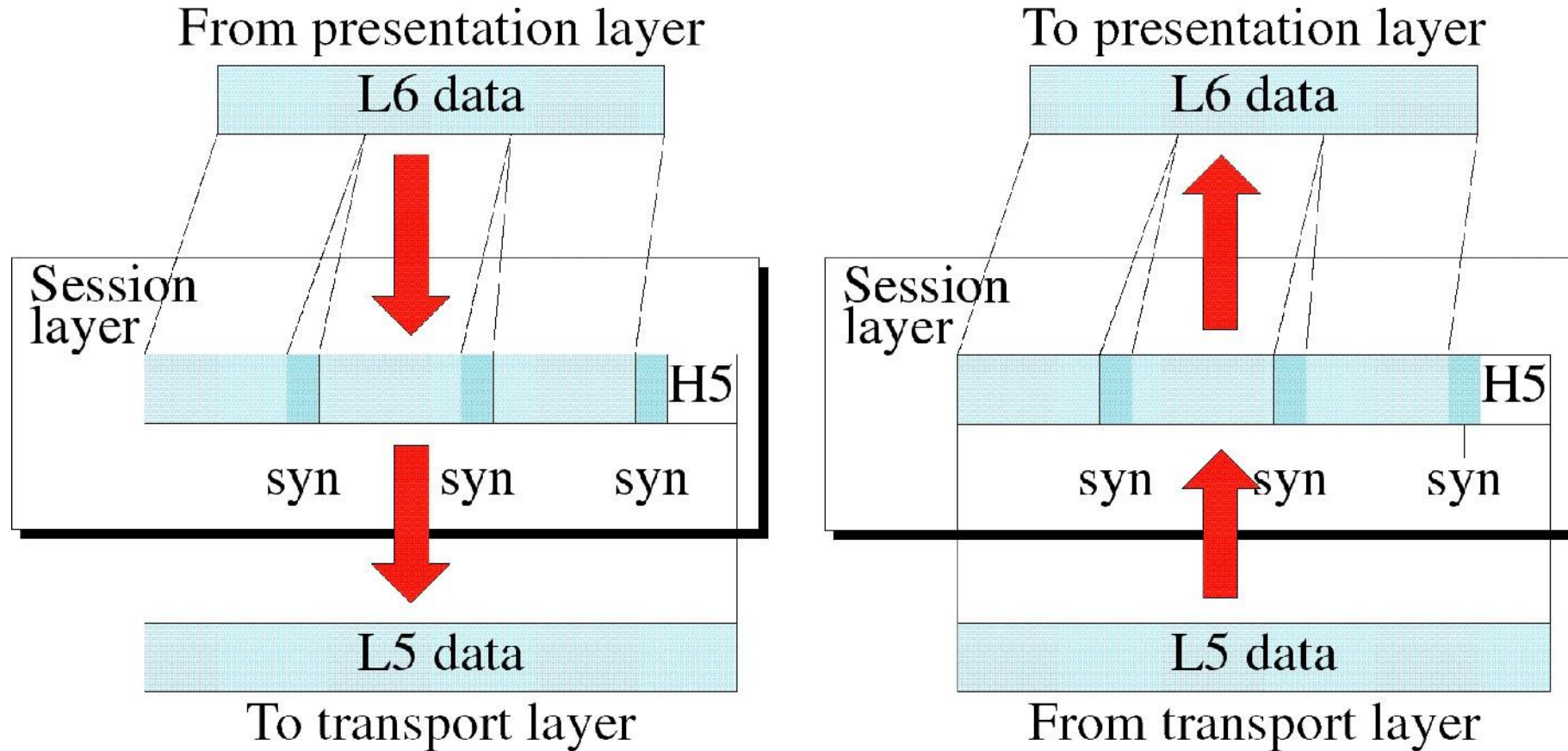
Session Layer

- The session layer offers provisions for efficient data transfer.

Responsibilities of the session layer:

- Dialog control: The session layer allows two systems to enter into a dialog.
- Synchronization: allows to add checkpoints.

Session Layer



Presentation Layer

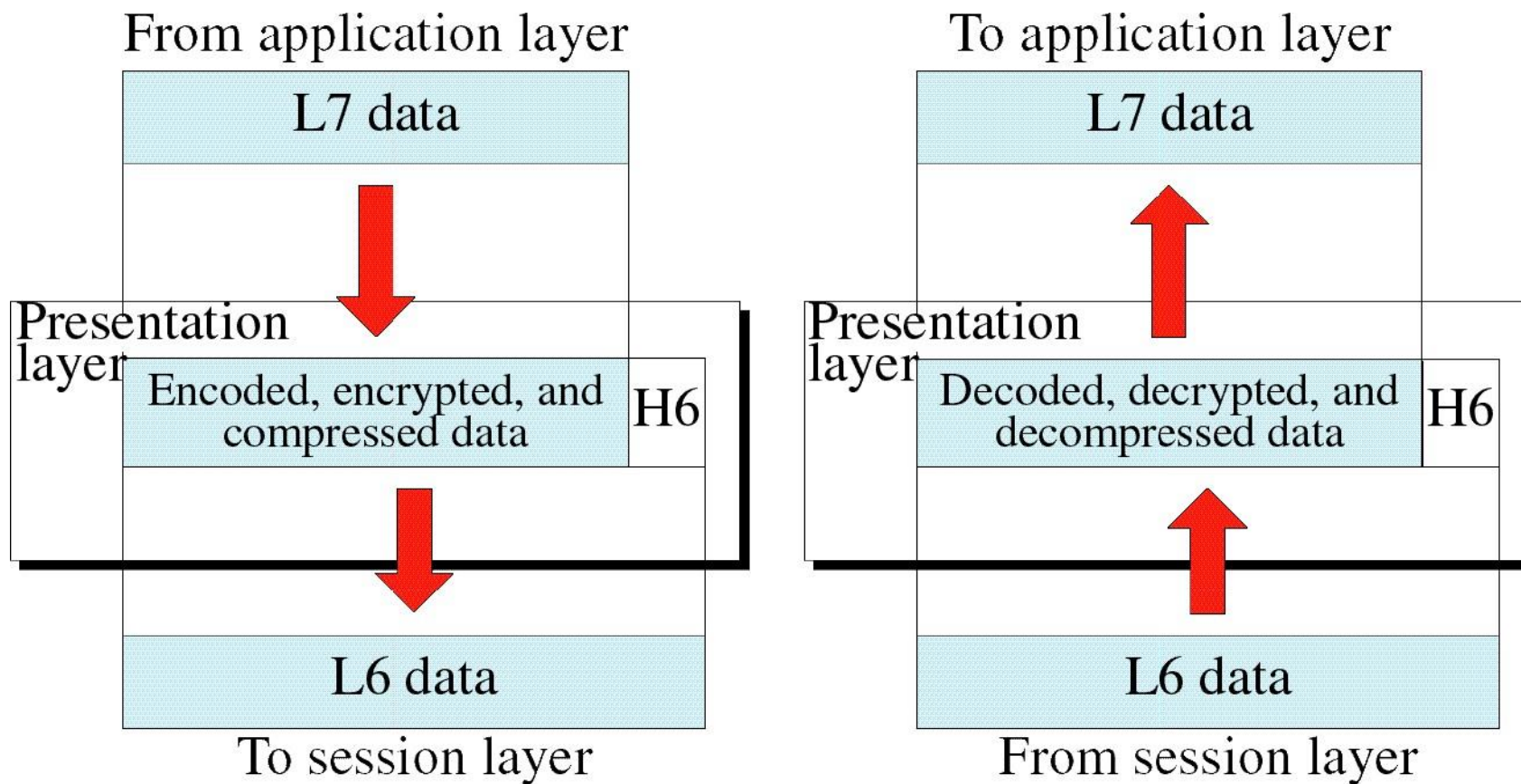
- The presentation layer ensures that the information that the application layer of one system sends out is readable by the application layer of another system.
- If necessary, the presentation layer translates between multiple data formats by using a common format.
- The presentation layer basically allows an application to read (or understand) the message.

Presentation Layer

Responsibilities of presentation layer:

- Translation: Changes data so that another computer can read it.
- Compression: Makes data smaller to move data in same amount of time.
- Encryption: Encodes data to protect from interception.

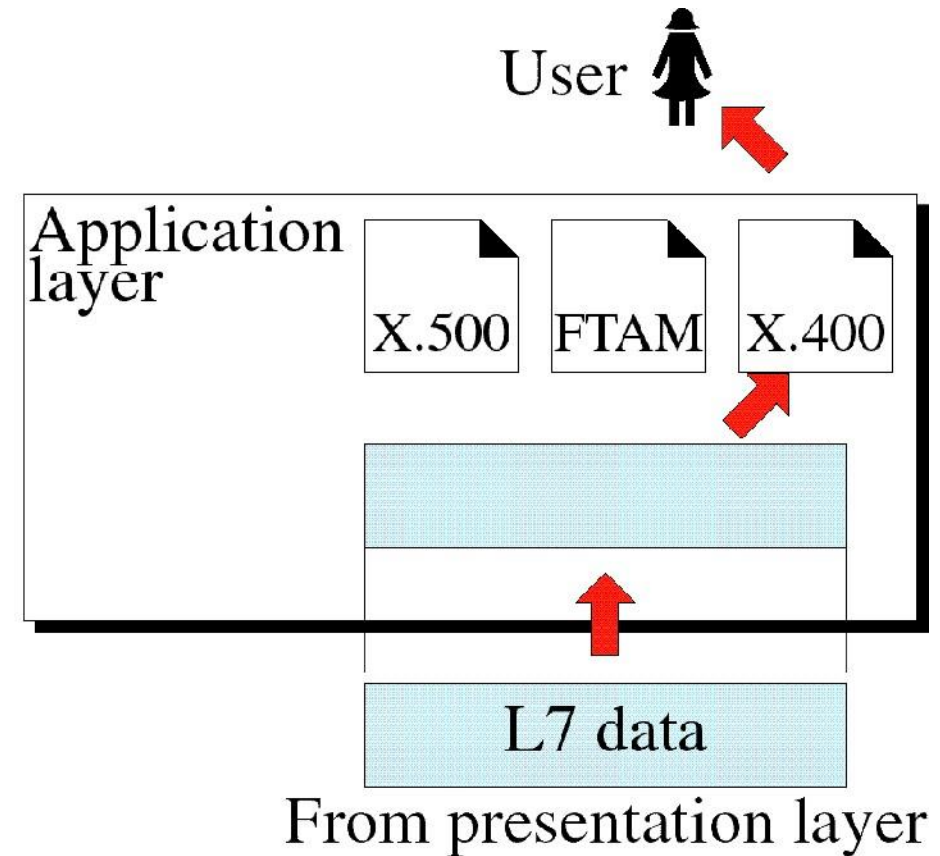
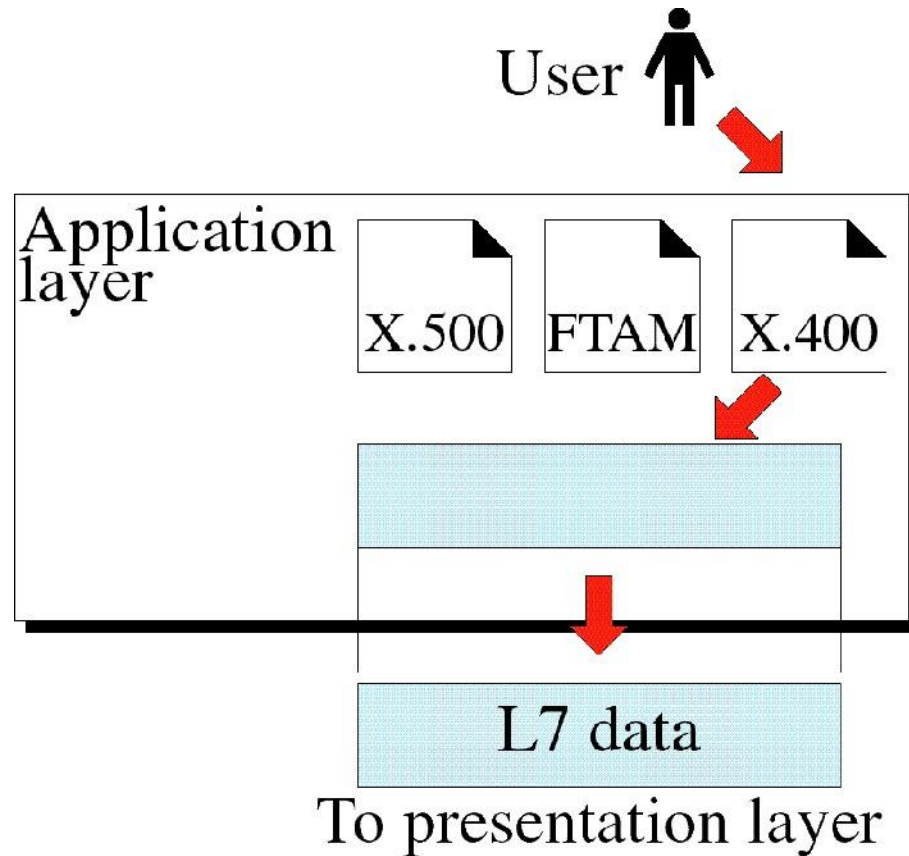
Presentation Layer



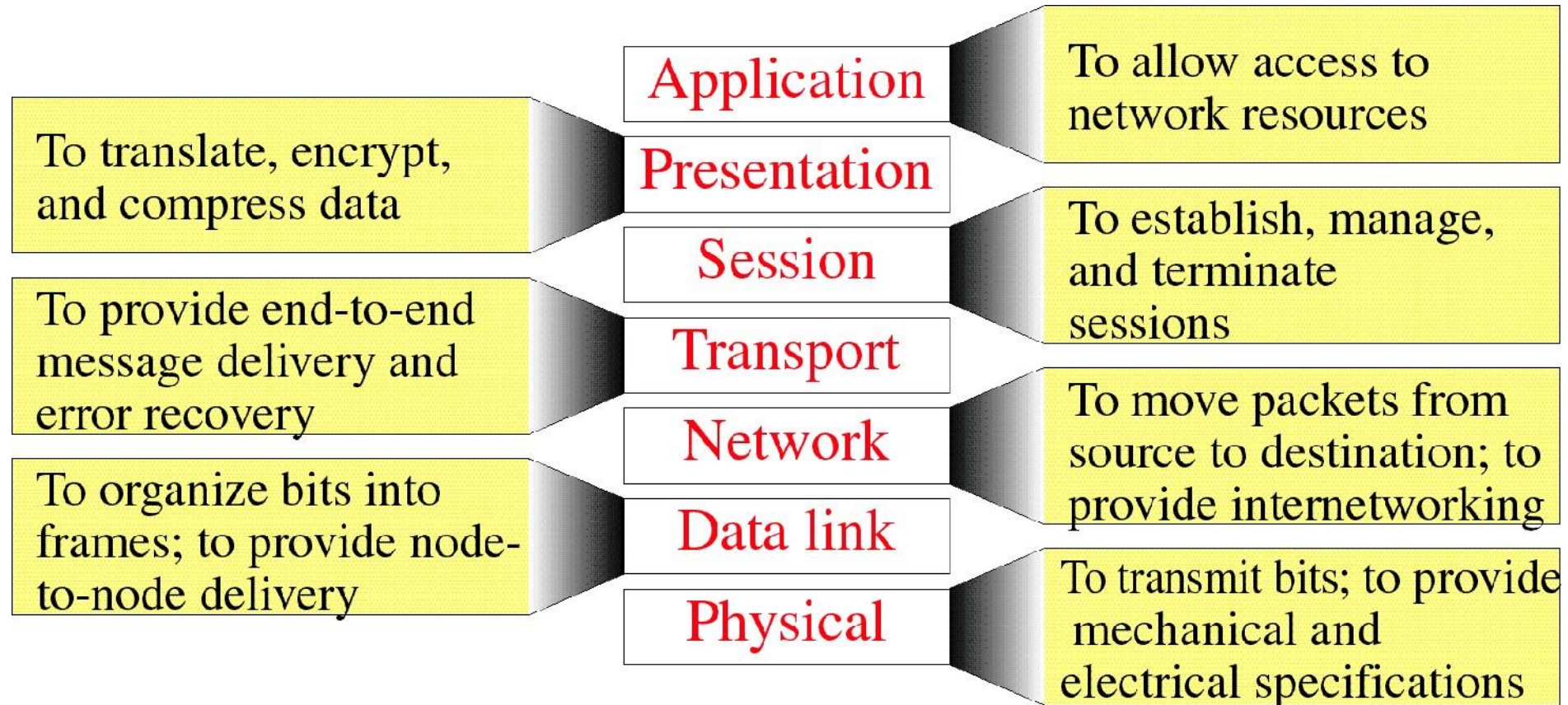
Application Layer

- The application layer enables the user, whether human or software, to access the network.
- It provides user interfaces and support – email, shared database , access to remote file etc.
- It differs from the other layers in that it does not provide services to any other OSI layer, but rather, only to applications outside the OSI model.

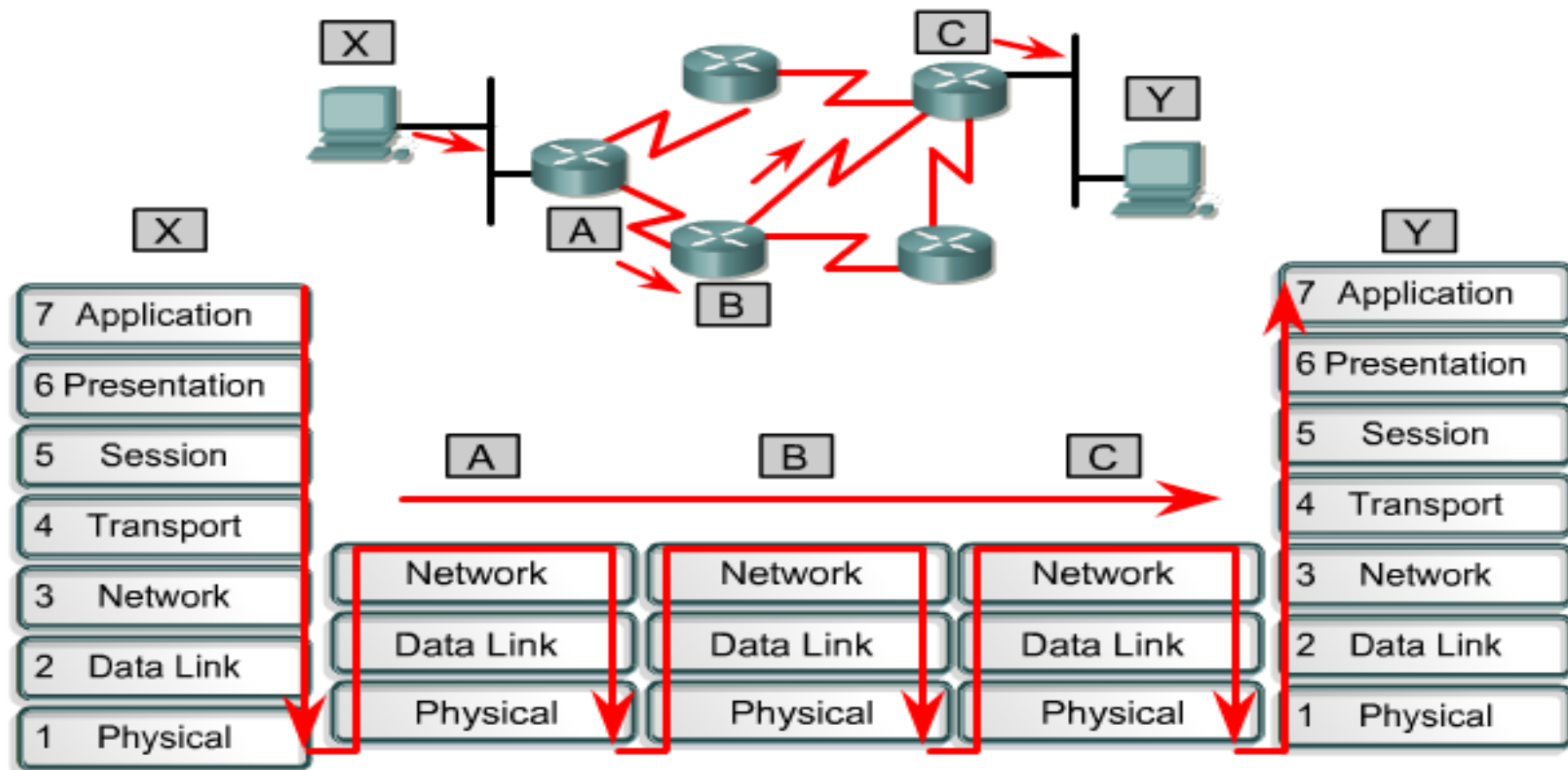
Application Layer



Summary of Layer Functions



Data Flow Through a Network



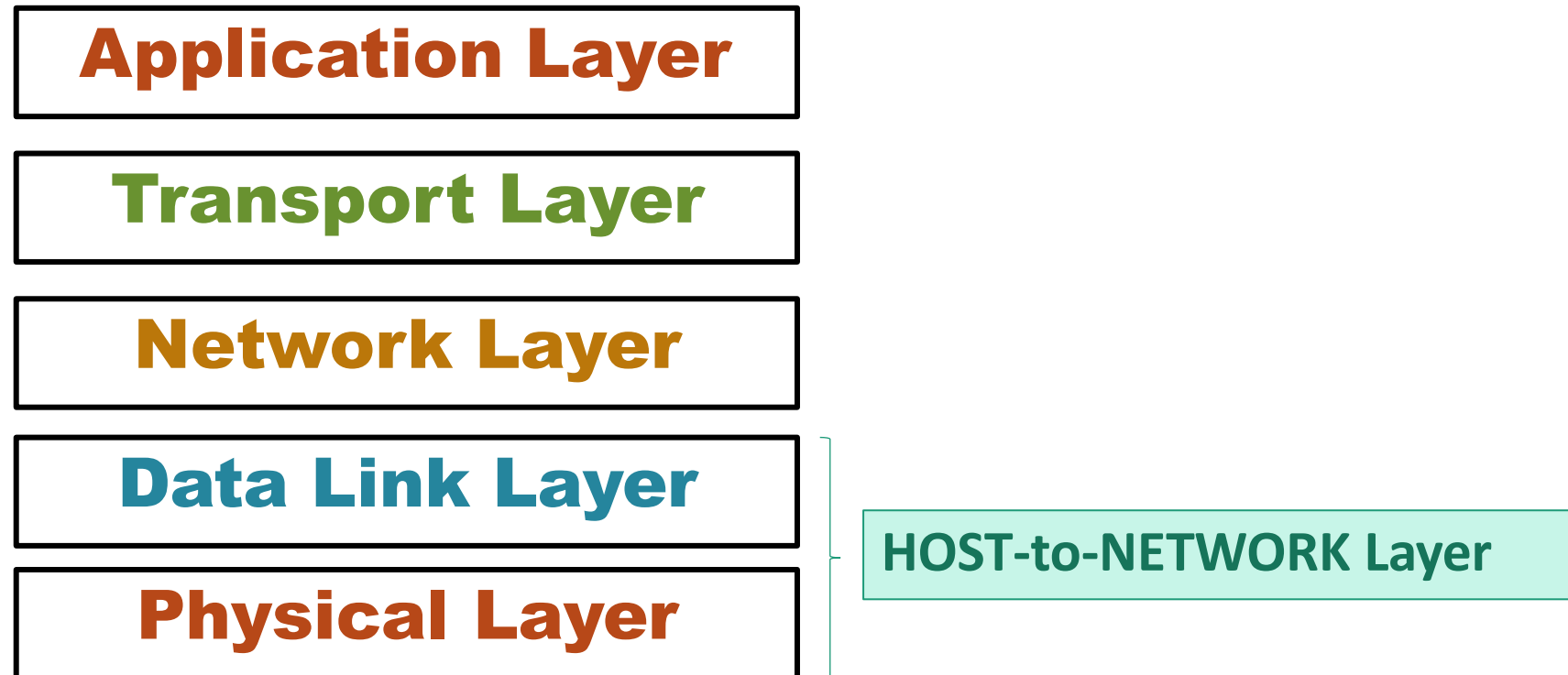
Data flow in a network focuses on layers one, two and three of the OSI model. This is after being transmitted by the sending host and before arriving at the receiving host.

TCP/IP Model

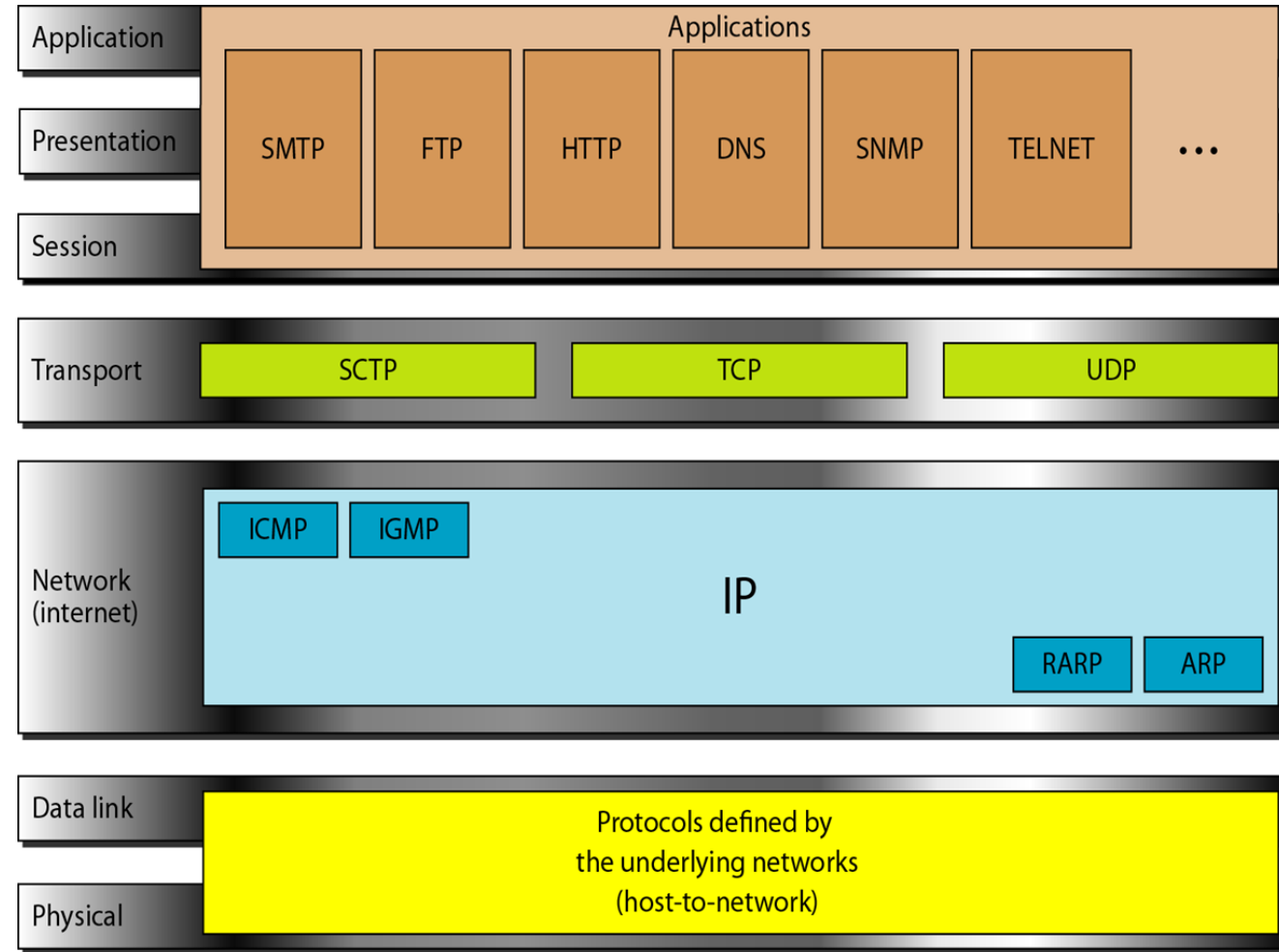
- Developed prior to OSI model
- TCP/IP is a hierarchical protocol made up of interactive modules, each providing specific functionality, but they are not interdependent

TCP/IP PROTOCOL SUITE

- TCP/IP protocol suite is made of five layers:



TCP/IP PROTOCOL SUITE



Network Layer

- TCP/IP does not support any specific protocol
- All standard and proprietary protocols supported at this level.
- At this level TCP/IP supports Internetworking protocol(IP).

IP in turn uses 4 supporting protocols.

- ARP
- RARP
- ICMP
- IGMP

Network Layer

- **Internetworking Protocol (IP)**
 - ✓ IP is transmission mechanism used by TCP/IP
 - ✓ Unreliable and connection-less protocol- best effort delivery
 - ✓ No error checking or tracking.
 - ✓ Transports data in packets called datagrams.
 - ✓ Does not keep track of routes and no facility reordering.

Network Layer

Address Resolution Protocol (ARP)

- Associates logical address with physical address.
- ARP is used to find physical address of node when logical address is known.

Reverse Address Resolution Protocol (RARP)

- RARP is used to find logical address of node when physical address is known.
- Usually used when computer is connected to network for the first time.

Network Layer

Internet Control Message Protocol (ICMP)

- Used by hosts and gateways to send notification of datagram problem to sender.

Internet Group Message Protocol (IGMP)

- Used to facilitate simultaneous transmission of message to group of recipients.

Transport Layer

- IP is host to host protocol

Transport layer has three protocols:

- UDP (User Datagram Protocol)
- TCP (Transmission Control Protocol)
- SCTP (Stream Control Transmission Protocol)

Transport Layer

User Datagram Protocol (UDP):

- Process to Process
- Adds only port address , error control , checksum etc. to data from upper layers.

Transport Layer

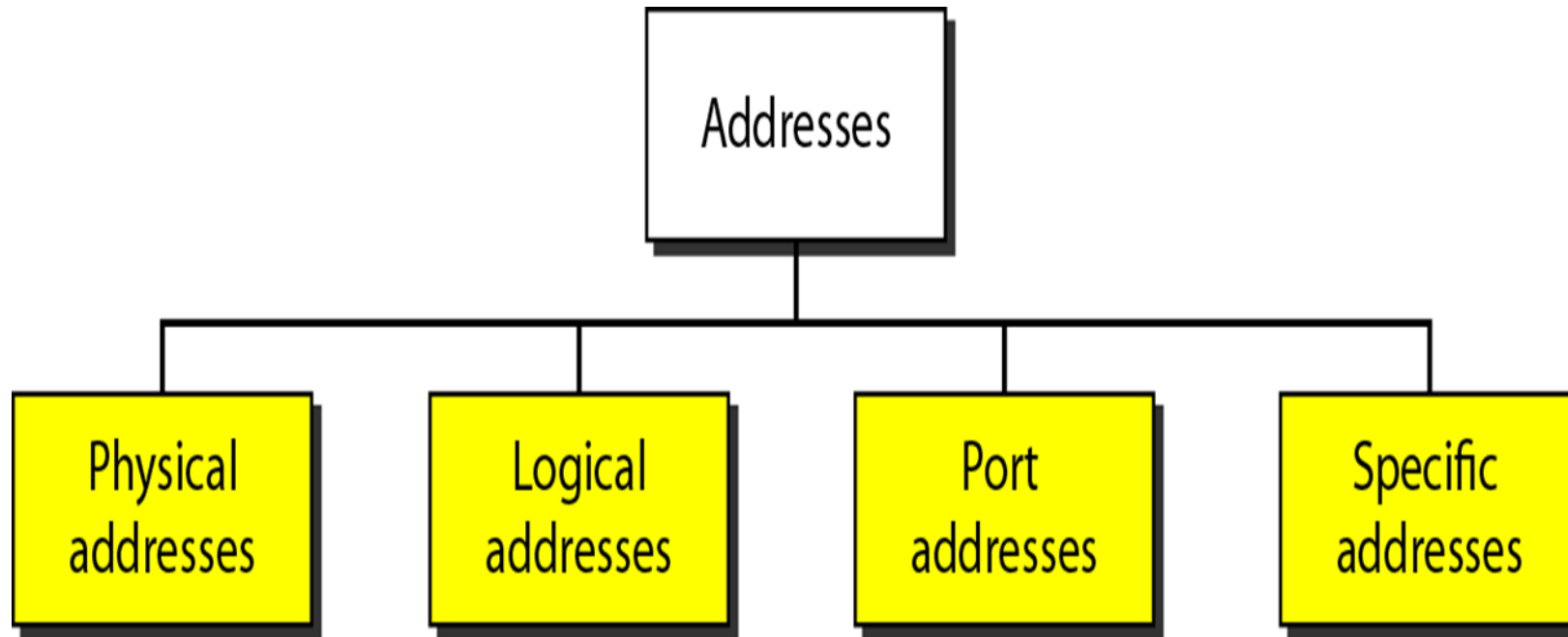
Transmission Control Protocol (TCP)

- Reliable stream protocol.
- Data is divided into segments.
- Segment contain sequence number for reordering.
- At receiving end , TCP collects each datagram and reorders it based on sequence numbers

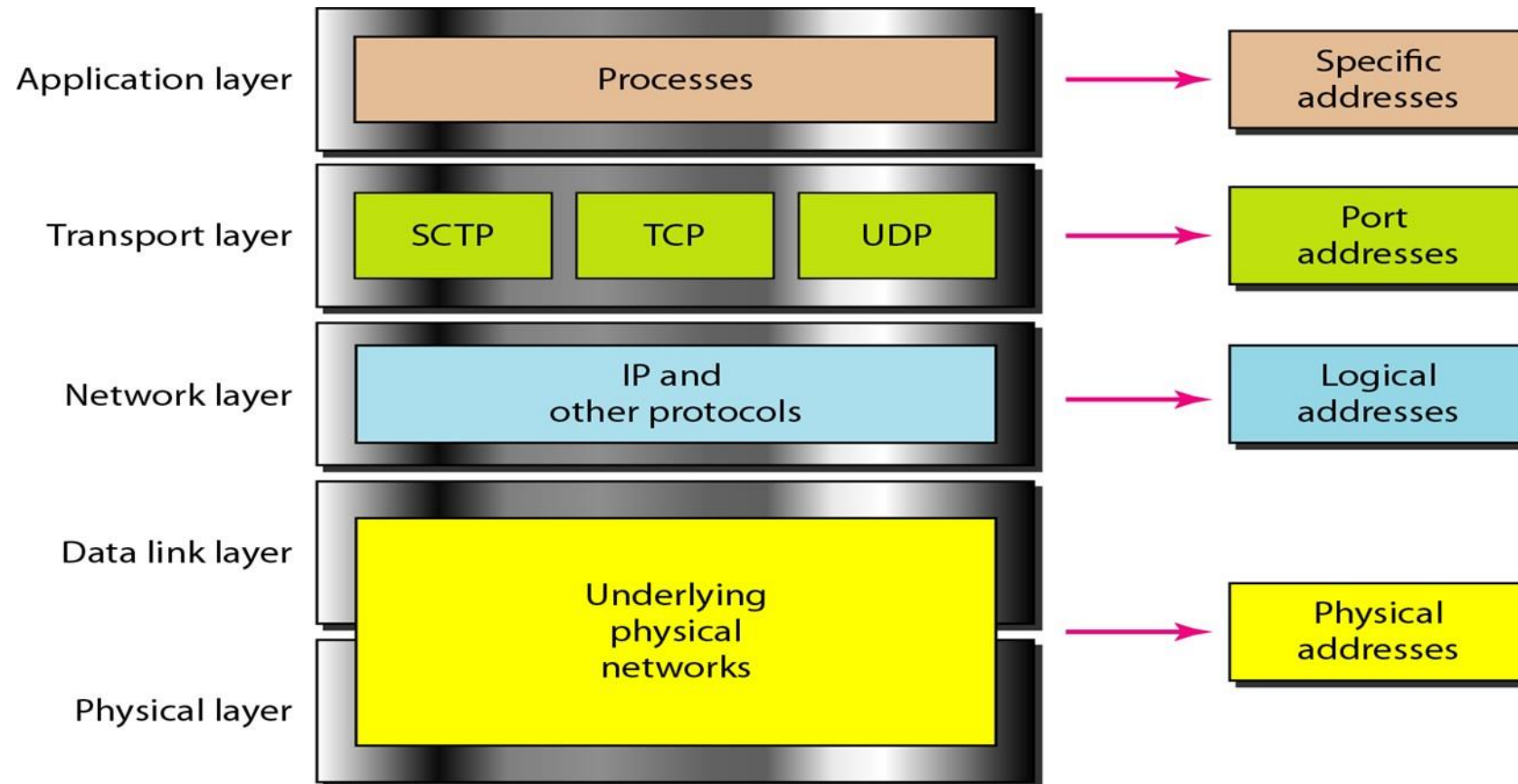
Stream Control Transmission Protocol (SCTP)

- Combines features of TCP and UDP

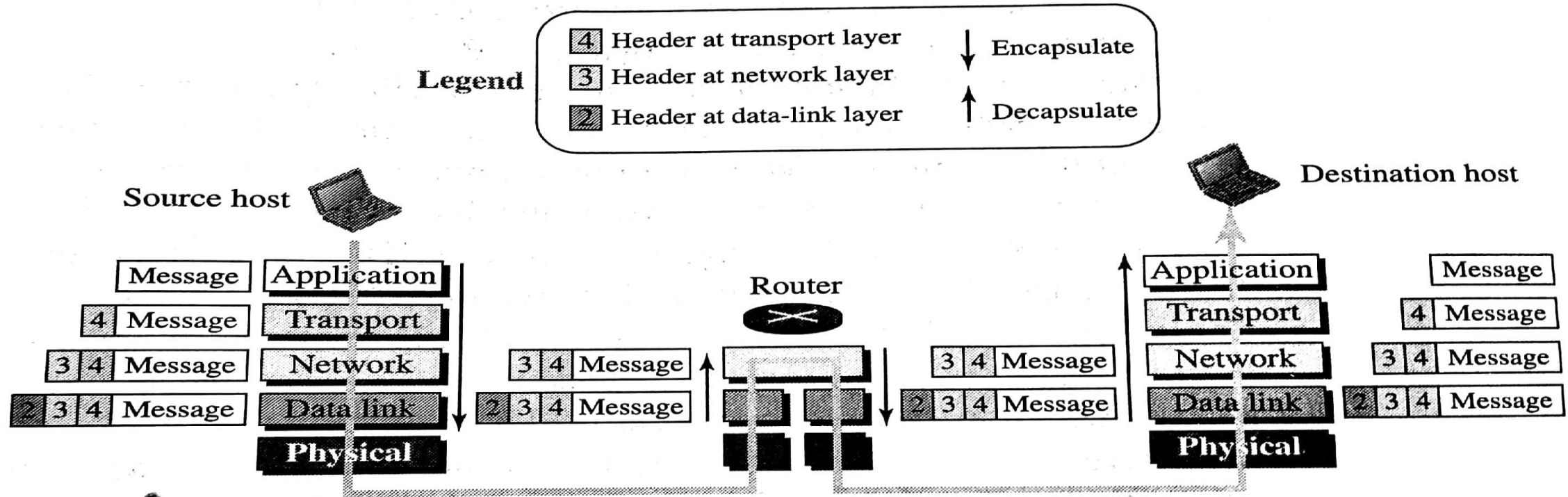
Addressing



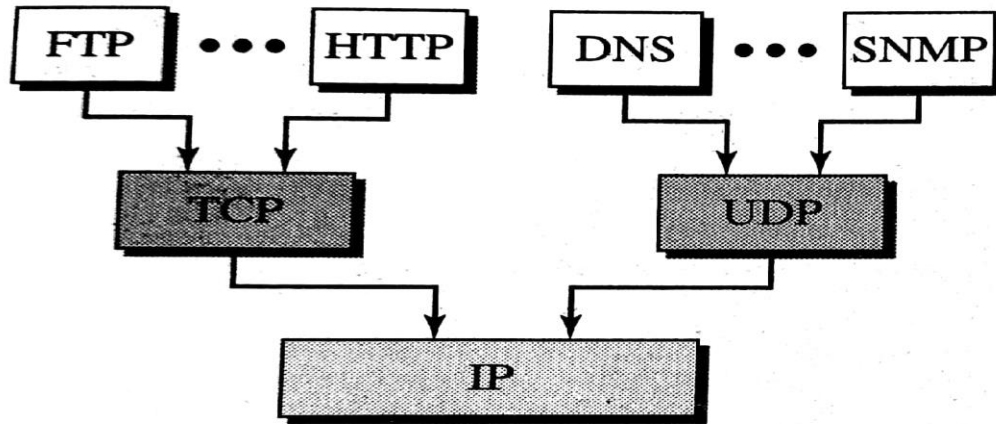
Relationship of layers and addresses in TCP/IP



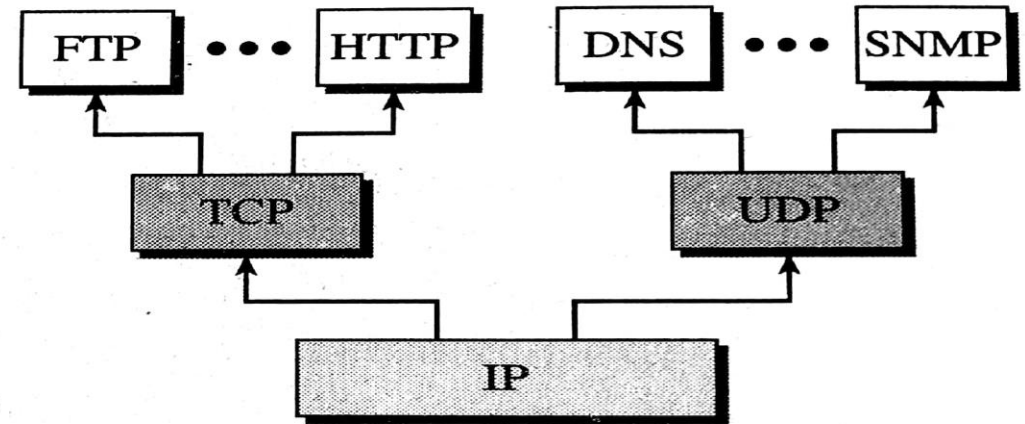
Data Encapsulation and Decapsulation



Multiplexing and Demultiplexing



a. Multiplexing at source



b. Demultiplexing at destination

MAC Address

Example MAC Address

3A-34-52-C4-69-B8

Organizationally
Unique Identifier
(OUI)

Network Interface
Controller
(NIC)

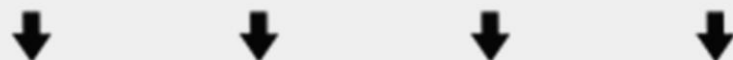
```
wikihow@wikihow: ~  
wikihow@wikihow:~$ ifconfig -a  
eth0      Link encap:Ethernet  HWaddr 08:00:27:15:84:10  
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0  
          inet6 addr: fe80::a00:27ff:fe15:8410/64 Scope:Link  
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1  
          RX packets:90 errors:0 dropped:0 overruns:0 frame:0  
          TX packets:133 errors:0 dropped:0 overruns:0 carrier:0  
          collisions:0 txqueuelen:1000  
          RX bytes:30496 (30.4 KB)  TX bytes:10000 (10.0 KB)  
  
lo        Link encap:Local Loopback  
          inet addr:127.0.0.1  Mask:255.0.0.0  
          inet6 addr: ::1/128 Scope:Host  
          UP LOOPBACK RUNNING  MTU:65536  Metric:1  
          RX packets:88 errors:0 dropped:0 overruns:0 frame:0  
          TX packets:88 errors:0 dropped:0 overruns:0 carrier:0  
          collisions:0 txqueuelen:0  
          RX bytes:8214 (8.2 KB)  TX bytes:8214 (8.2 KB)  
  
wikihow@wikihow:~$
```

Network Connection Details	
Network Connection Details:	
Property	Value
Physical Address	08-00-27-CE-76-FE
IP Address	10.0.2.15
Subnet Mask	255.255.255.0
Default Gateway	10.0.2.2
DHCP Server	10.0.2.2
Lease Obtained	9/18/2014 6:34:11
Lease Expires	9/19/2014 6:34:11
DNS Servers	208.67.222.222 69.179.243.70
WINS Server	

IP Address

An IPv4 address (dotted-decimal notation)

172 . 16 . 254 . 1



10101100 . 00010000 . 11111110 . 00000001



One byte = Eight bits

Thirty-two bits (4 × 8), or 4 bytes

```
muser@TD8610: ~
File Edit View Terminal Help
muser@TD8610:~$ ifconfig
eth0    Link encap:Ethernet  HWaddr 00:0c:29:eb:05:55
        inet addr:192.168.17.129 Bcast:192.168.17.255 Mask:255.255.255.0
        inet6 addr: fe00::20c:29ff:feeb:555/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:107 errors:0 dropped:0 overruns:0 frame:0
        TX packets:44 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:13438 (13.4 KB)  TX bytes:7409 (7.4 KB)
        Interrupt:19 Base address:0x2000

lo      Link encap:Local Loopback
        inet addr:127.0.0.1 Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING  MTU:16436  Metric:1
        RX packets:4 errors:0 dropped:0 overruns:0 frame:0
        TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:240 (240.0 B)  TX bytes:240 (240.0 B)
```

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 8 . 8 . 8 . 8

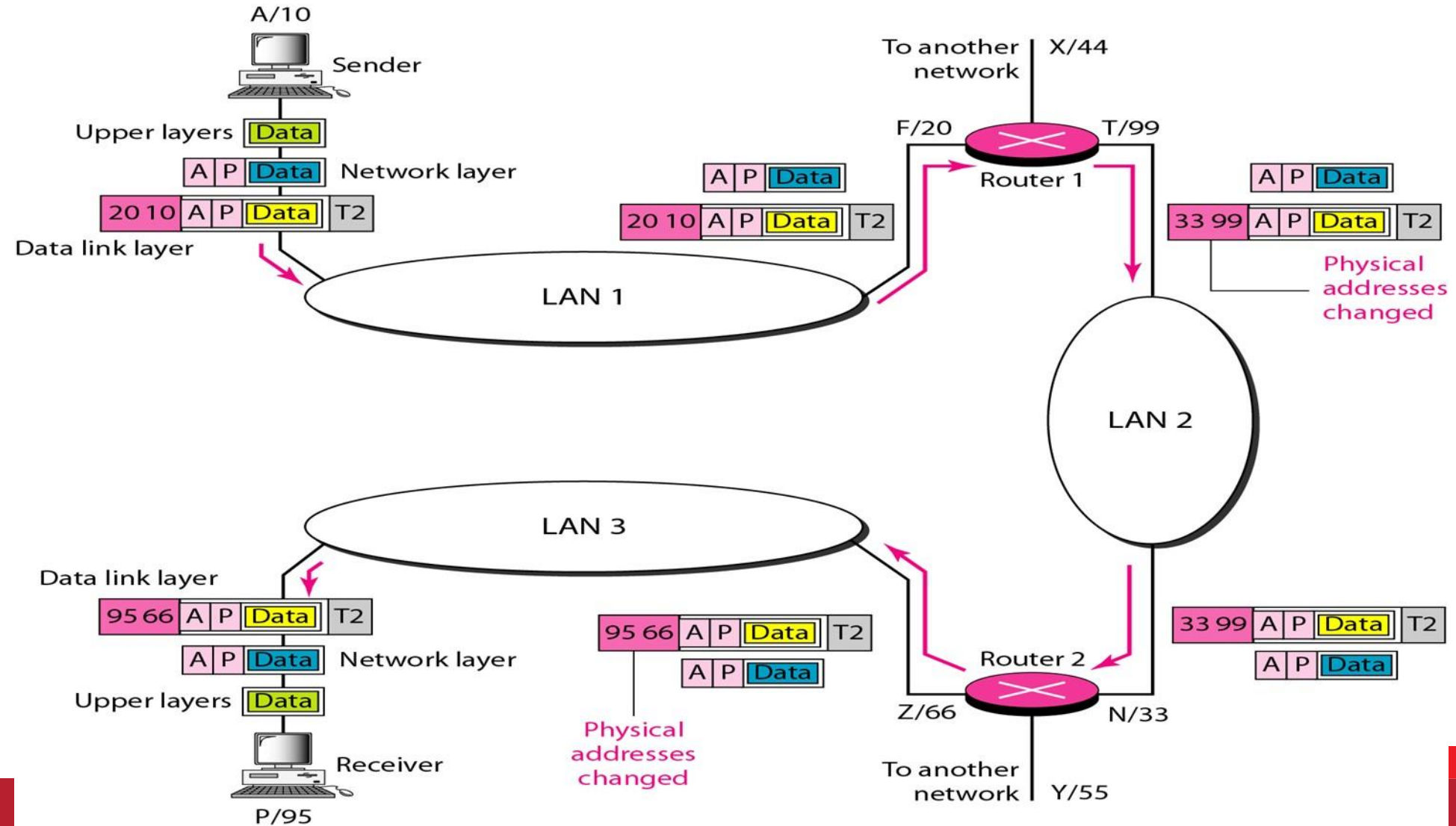
Alternate DNS server: 4 . 2 . 2 . 1

☐ Validate settings upon exit

Advanced...

OK Cancel

Addresses (Example)



Connecting Devices

*LANs or WANs do not normally operate in isolation. They are connected to one another or to the Internet. To connect LANs or WANs, we use **connecting devices**. Connecting devices can operate in different layers of the Internet model. Repeaters and hubs operate in the first layer of the Internet model. Bridges and two-layer switches operate in the first two layers. Routers and three-layer switches operate in the first three layers*

The topics discussed in this section include:

Repeaters

Hubs

Bridges

Router

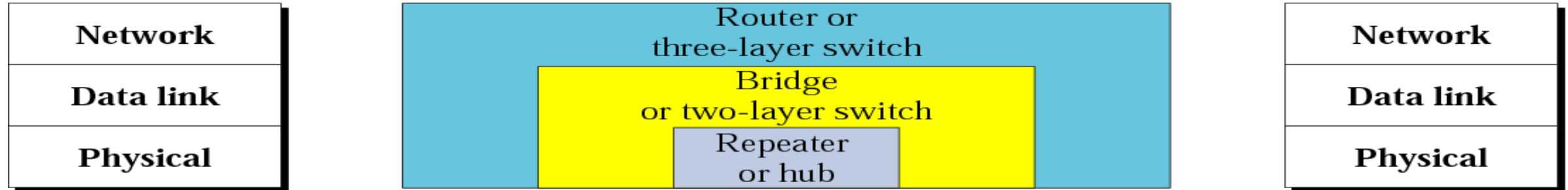
Hub,Switch ,Router:

https://www.youtube.com/watch?v=1z0ULvg_pW8

Hub,Bridge,Switch,Router:

<https://www.youtube.com/watch?v=eMamgWlIRFY>

Figure 3.28 *Connecting devices*



Repeater

Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network.

An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength. It is a 2 port device.

Figure 3.29 Repeater

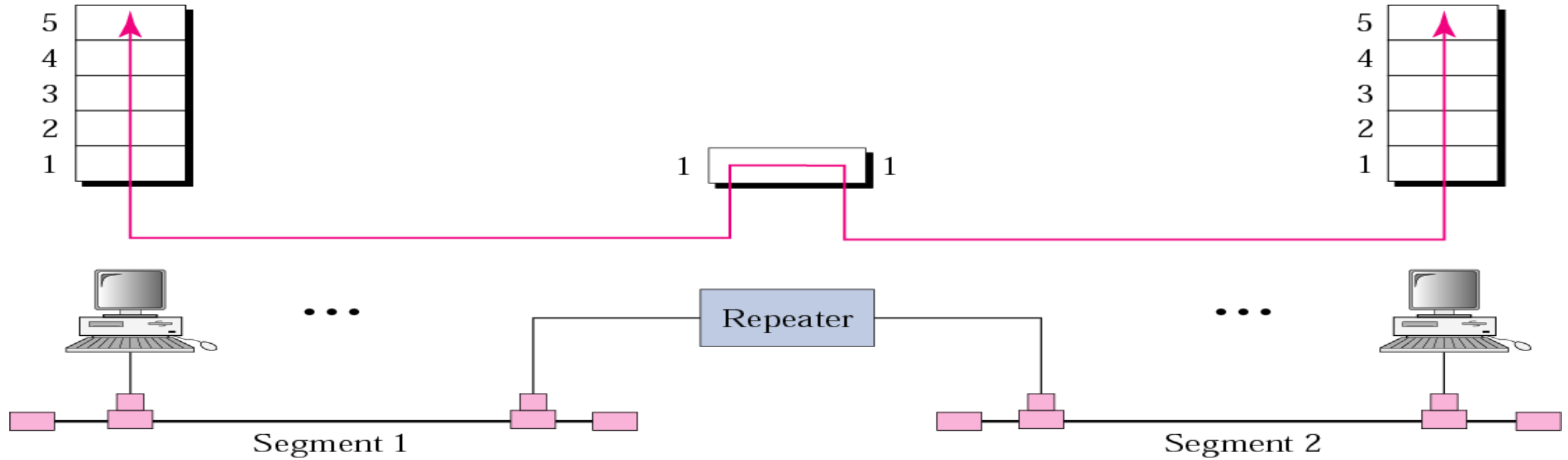
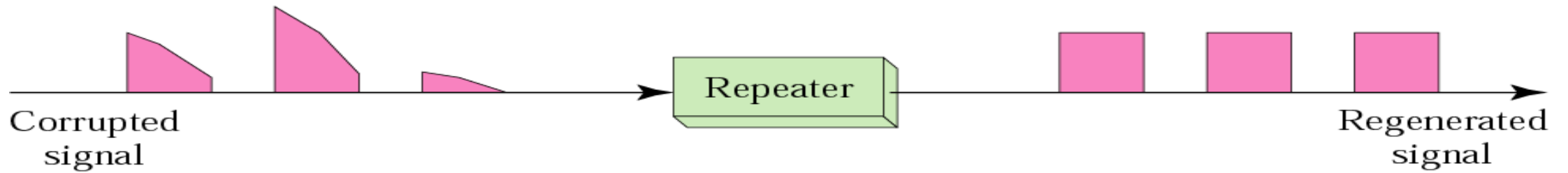


Figure 3.30 *Function of a repeater*



a. Right-to-left transmission.



b. Left-to-right transmission.

HUB

Hub – A hub is basically a multiport repeater.

A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations.

Hubs cannot filter data, so data packets are sent to all connected devices. They do not have the intelligence to find out best path for data packets which leads to inefficiencies and wastage.

Types of Hub

There are mainly two types of Hub, they are:

Active Hub: An Active hub is also known as Concentrator. It requires a power supply and can work as a repeater. Thus, it can analyze the data packets and can amplify the transmission signals, if needed.

Passive Hub: A passive hub does not need any power supply to operate. It only provides communication between the networking devices and does not amplify the transmission signals. In other words, it just forwards the data as it is.

HUB

Following are the advantages of using a Hub:

It is simple to implement.

The implementation cost is low.

It does not require any special system administration configuration. We can just plug and play it.

Following are the disadvantages of using a Hub:

It can connect devices of the same network only.

It uses a half-duplex mode of communication.

It is less secure, as it broadcasts the data packets.

It can be used in a limited network size only.

Broadcasting induces unnecessary traffic on the channel.



Repeater



Hub

Bridge

A bridge is a layer-2 network connecting device, i.e., it works on the physical and data-link layer of the OSI model. It interprets data in the form of data frames. In the physical layer, the bridge acts as a Repeater which regenerates the weak signals, while in the data-link layer, it checks the MAC(Media Access Control) address of the data frames for its transmission.

A bridge connects the devices which are present in the same network. It is mainly used to segment a network to allow a large network size. It has two types of port - incoming and outgoing. It uses the incoming port to receive the data frames and outgoing port to send the data frames to other devices.

Bridge

A Bridge has filtering capacity. It means that it can discard the faulty data frames and will allow only the errorless data frames in the network. Also, it can check the destination MAC address of a frame and decides the port from which the frame should be sent out. For this, it maintains a table containing the physical(MAC) addresses of all the devices in the network. Whenever a data frame arrives at the incoming port of the bridge, it first checks the data frame for any kind of errors. If the frame is errorless, it directs the data frame to the specified MAC address(taking instance from the address table) using its outgoing port. It does not change the physical(MAC Address) of the frames during transmission. In other words, a Bridge is a Repeater with filtering capability.

Bridge

There are mainly two types of Bridge, they are:

Transparent Bridge: Transparent bridge simply works as a transmission medium between two devices. They are actually transparent(they are present but are not functionally visible to the devices) to the networking devices.

Routing Bridge: Routing bridges have their unique identity, they can be easily identified by the network devices. The source station or the sender can send the data packets through specific bridges(using the unique identity of bridges).

Bridge

Following are the advantages of using a Bridge:

It is not so complex to implement.

The implementation cost is medium.

It does not require any special system administration configuration. We can just plug and play it.

Improves security by limiting the scope of data frames.

It has the filtering capability.

It can be used in a large network.

Following are the disadvantages of using a Bridge:

It can connect devices of the same network only.

There is a delay in forwarding the frames due to error checking.

There is a need to maintain an Address table.

Switch

A switch is a layer-2 network connecting device, i.e., it works on the physical and data-link layer of the OSI model. It interprets data in the form of data frames. A switch acts as a multiport bridge in the network. It provides the bridging functionality with greater efficiency.

A switch maintains a Switch table which has the MAC addresses of all the devices connected to it. It is preferred more over the hub, as it reduces any kind of unnecessary traffic in the transmission channel. A switch can connect the devices only in the same network. It uses the full-duplex mode of communication and saves bandwidth. The switch table keeps on updating every few seconds for better processing.

Switch

A Switch is an intelligent device with filtering capabilities. It can discard the faulty data frames and will allow only the errorless data frames in the network. Also, it will forward the data frames to the specific node based on the MAC address(taken from the Switch table). A Switch has multiple collision domains, so it has least or no collisions in the transmission channel. In fact, every port of switch has a separate collision domain.

When a data frame arrives at the Switch, it first checks for any kind of error in the data frame. If the frame is error-free, it will search the MAC address of the destination in the Switch table. If the address is available in the switch table, it will forward the data frame to that specific node, else switch will register the MAC address in the switch table. If the destination address is not specified, it will broadcast the data frame to each node in the network.

A Switch can have 8/6/24/48 ports. The data transmission speed is slow in a switch(around 10-100 Mbps). Also, it has only one broadcasting domain.

There are mainly four types of Switches, they are:

Store and Forward Switch: It is the most widely and commonly used switch. It does not forward the data frames unless the frames are errorless and completely received in the switch buffer. It is reliable in nature.

Cut-through Switch: Cut-through switches have no error checking. Also, it starts sending the data frame to the destination node when it starts receiving it. It is unreliable in nature.

Fragment-Free Switch: It is a combination of store and forward, and cut-through switch. It checks only the starting 64 bytes(header information) of the data frame before transmitting the frame.

Adaptive Switch: It is the most advanced kind of switch which automatically chooses any of the above three switches as per the need.

Following are the advantages of using a Switch:

The implementation cost is medium.

It does not require any special system administration configuration. We can just plug and play it.

Improves security by limiting the scope of data frames.

It has the filtering capability.

It can be used in a large network.

It uses full-duplex mode of communication

It has multiple collision domains, so there are least or no collisions in the channel.

Following are the disadvantages of using a Switch:

It can connect devices of the same network only.

There is a delay in forwarding the frames due to error checking.

There is a need to maintain a Switch table.

Router

A Router is a layer-3 network connecting device, i.e., it works on the physical, data-link and network layer of the OSI model. It interprets data in the form of data packets. It is mainly an internetworking device, which can connect devices of different networks(implementing the same architecture and protocols). In other words, it can connect two physically and logically different network devices with each other. A Router is used to connect the networks or it routes traffic between the networks. In other words, a Router is the Gateway of a network.

Since, connecting two devices of different networks, the connecting device should implement an Internet Protocol(IP) address. So, the Router has a physical and logical(Internet Protocol) address for each of its interfaces. It routes or forwards the data packets from one network to another based on their IP addresses. It changes the physical address of the data packet(both source and destination) when it forwards the data packets.

Router

A router maintains a routing table using the routing algorithms. When a data packet is received at the router, it first checks the IP address. If the IP address is the same as the network's IP address, it receives the data packet, else it forwards the data packet to the destination IP address using the routing table. A router does not perform addressing. It can have 2/4/8 ports for connecting the devices. It can control both the collision domain(inside the network) and the broadcast domain(outside the network). It has a fast data transmission speed(up to 1 Gbps). A Router can be a Wireless Router, Core Routers, Edge Routers, Virtual Routers, etc.

Routers

There are mainly two types of routing performed by Routers, they are:

Static Routing: In Static Routing, the path for the data packets is manually set. It is generally used for small networks.

Dynamic Routing: In Dynamic Routing, various routing algorithms are used to find the best and shortest path for the data packets.

Following are the advantages of using a Router:

It can connect devices and provides routing facilities over different networks implementing the same protocol and structure.

Improves security by limiting the scope of data packets.

It has the filtering capability.

It can be used in a large network.

It uses full-duplex mode of communication

It has control over both the collision and broadcast domain.

Following are the disadvantages of using a Router:

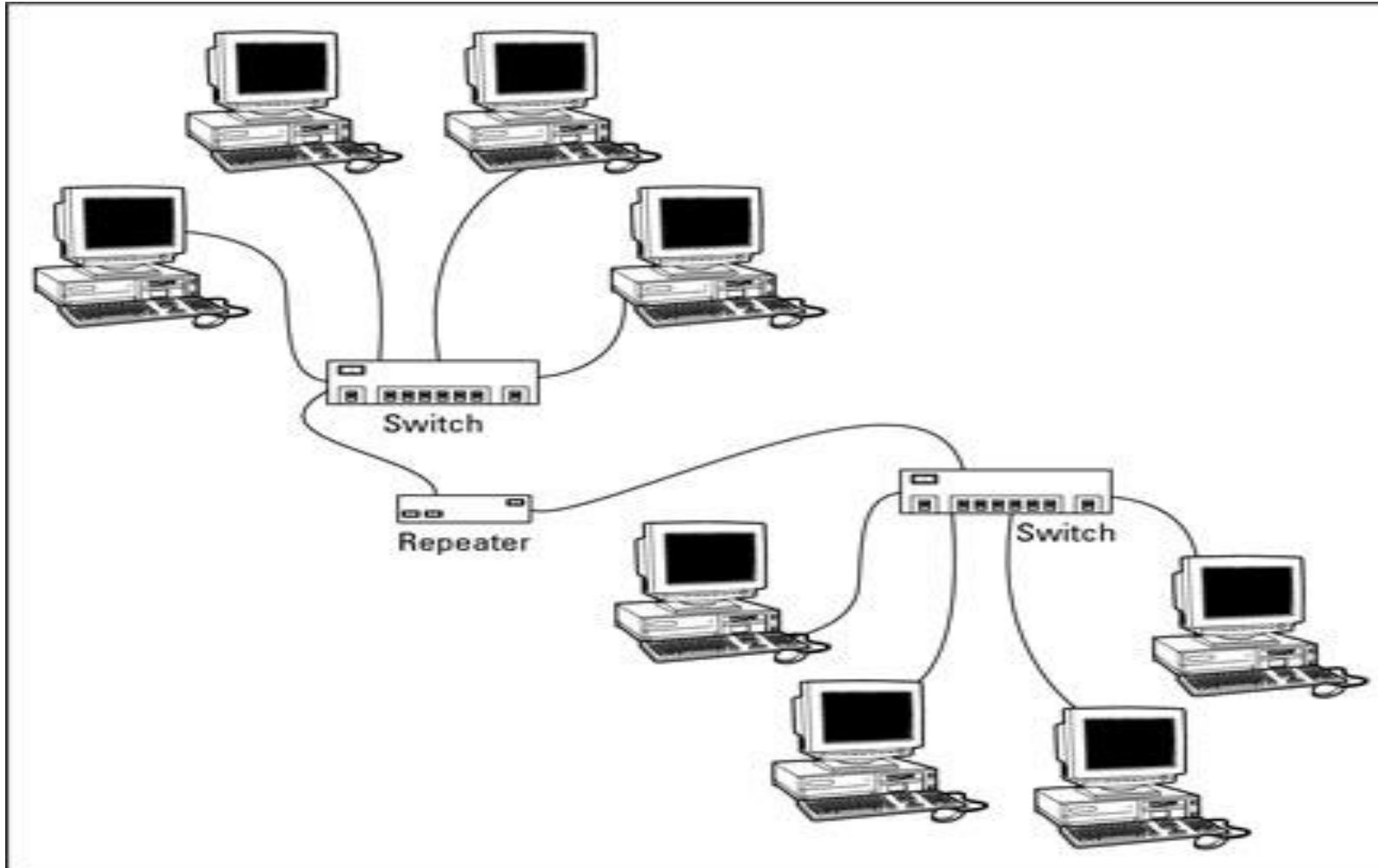
It is very complex to implement.

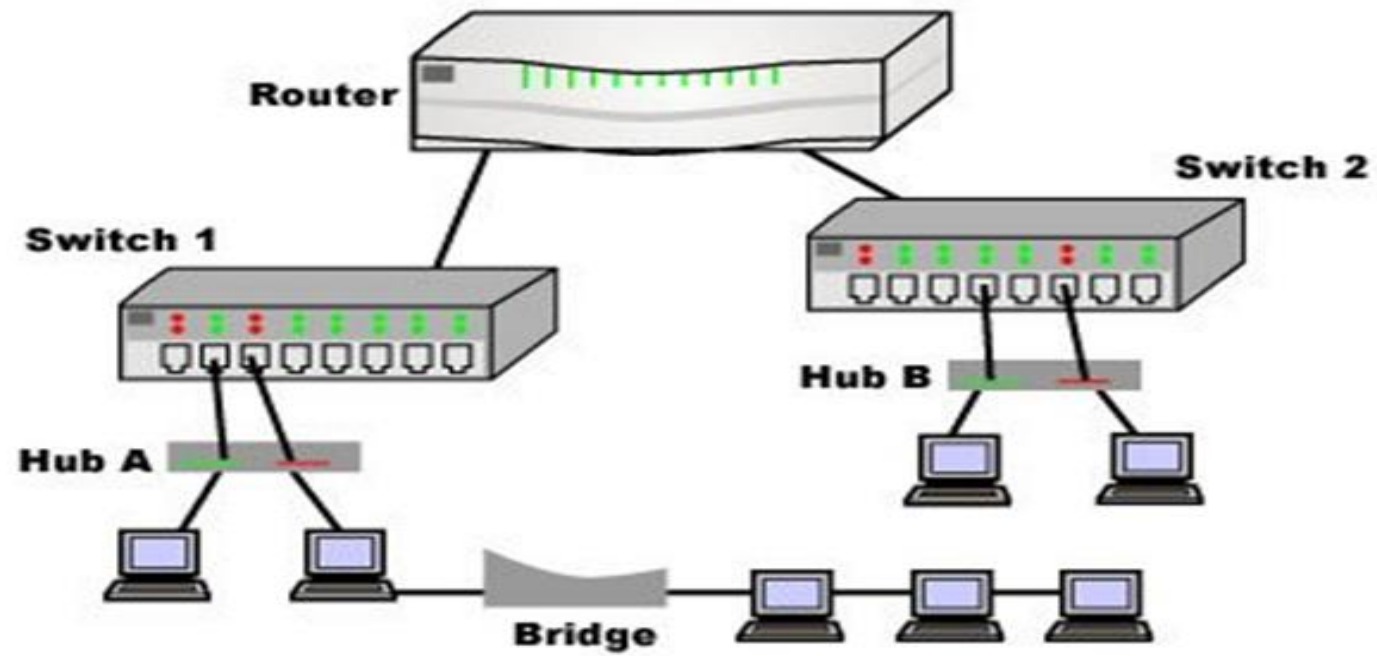
The implementation cost is quite high.

There is a need to maintain a Routing table.

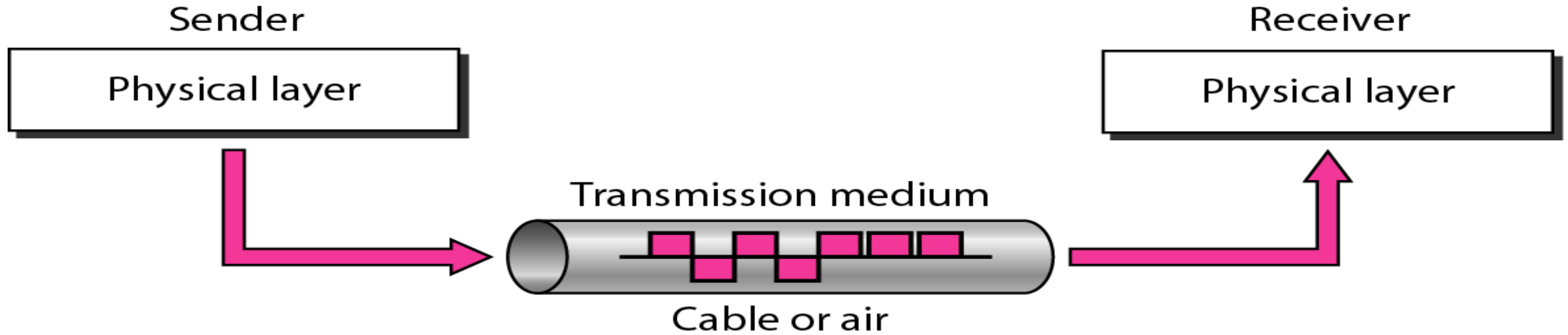
There is a delay in forwarding the packets due to error checking.

It requires a special system administration configuration.

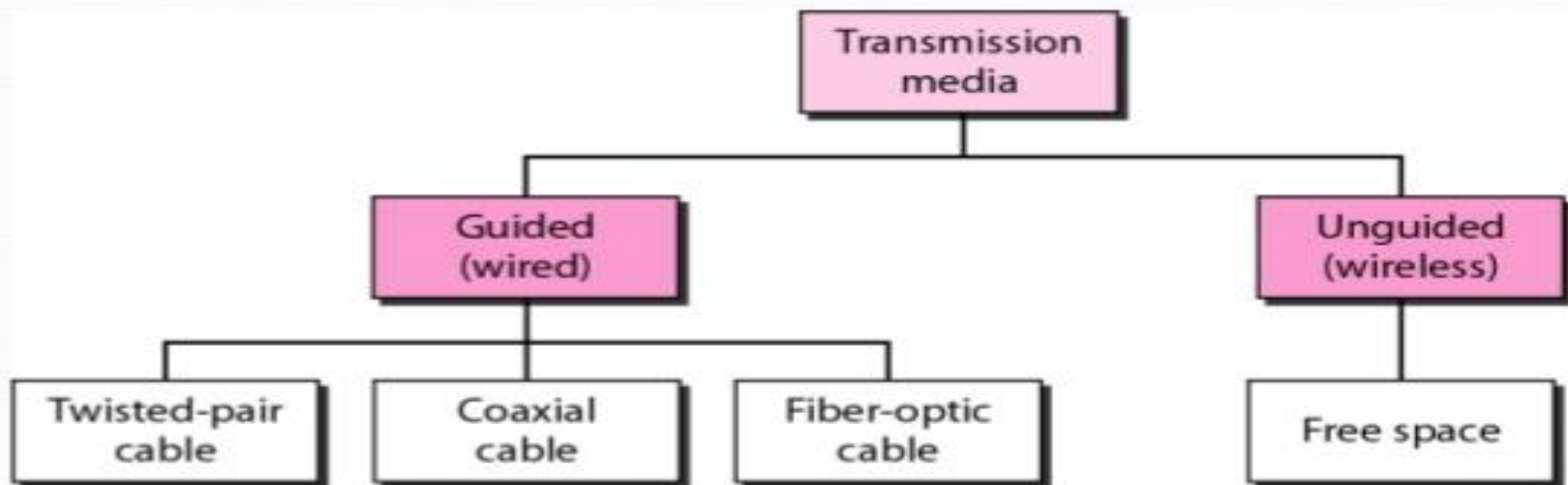




Transmission Media



Classes of transmission media



Transmission Media

- ❑ Guided Media (Wired)
 - Twisted-Pair Cable
 - Coaxial Cable
 - Fiber-Optic Cable
- ❑ Unguided Media (Wireless)
 - Radio Waves
 - Microwaves
 - Infrared

Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

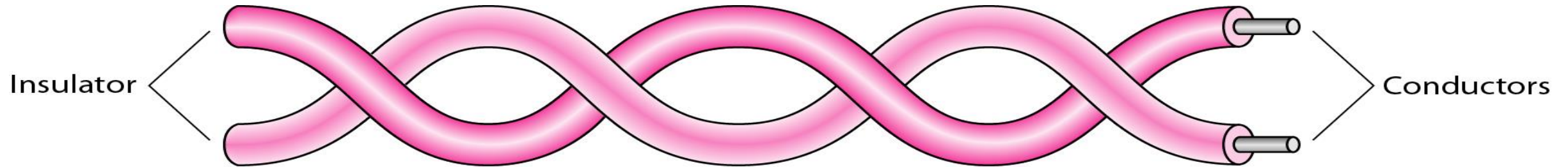
Topics discussed in this section:

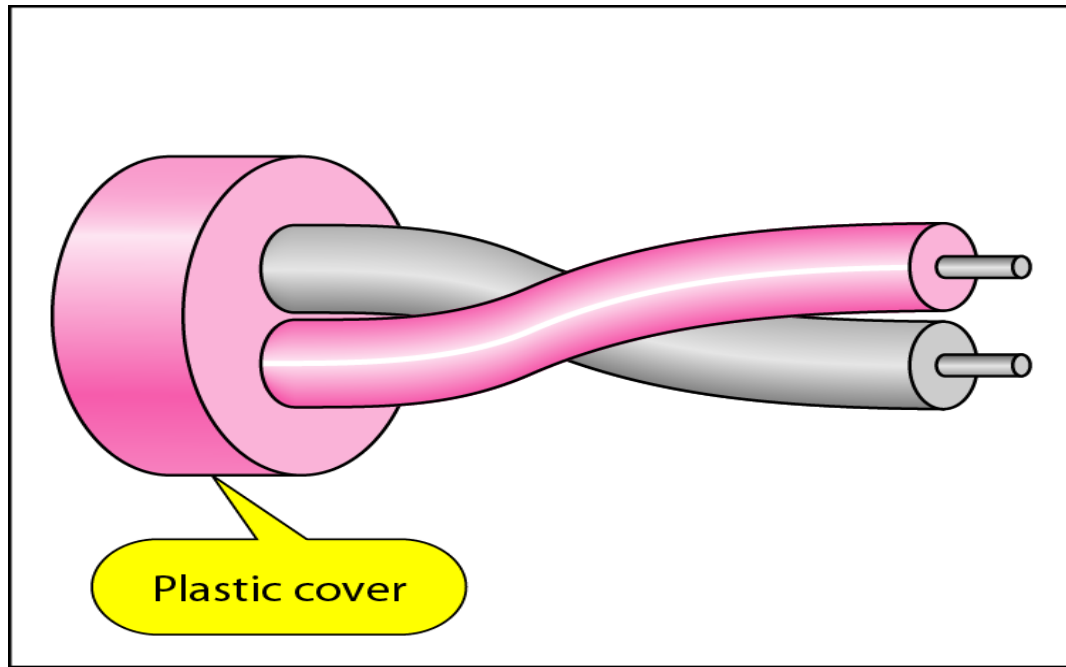
Twisted-Pair Cable

Coaxial Cable

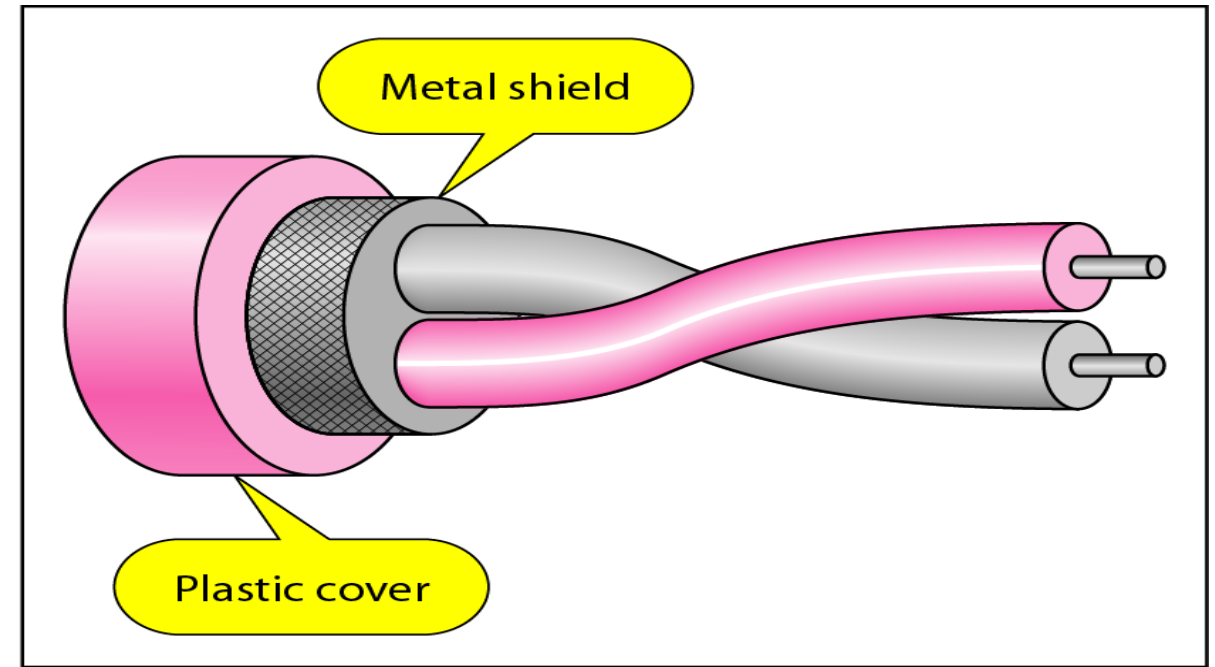
Fiber-Optic Cable

Figure 7.3 *Twisted-pair cable*





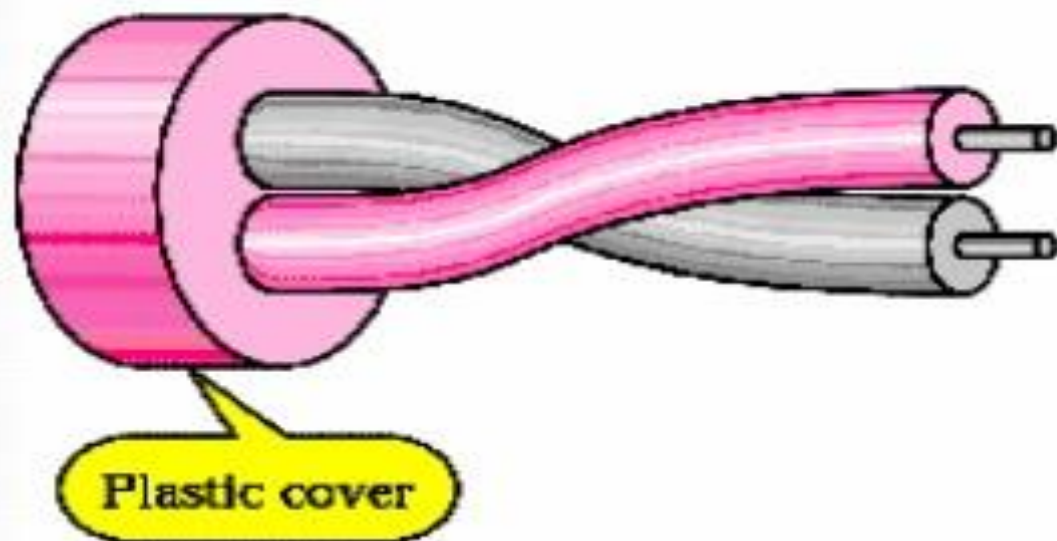
a. UTP



b. STP

Unshielded Twisted Pair (UTP)

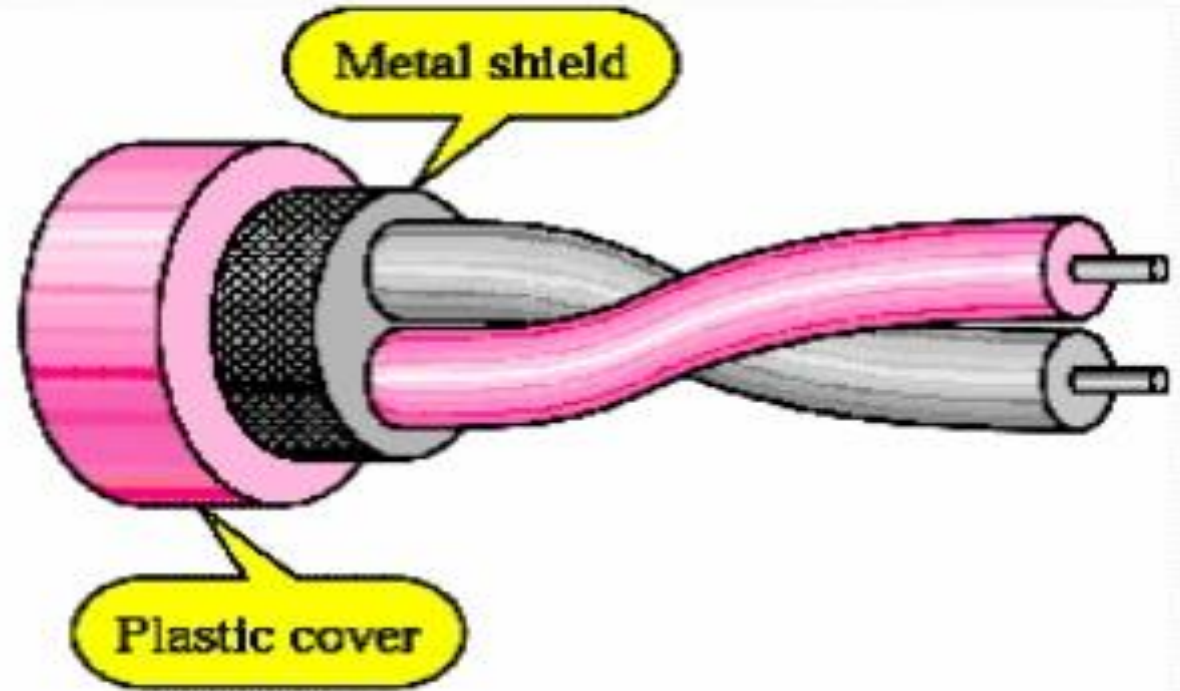
- Ordinary telephone wire
- Cheapest
- Easiest to install
- Suffers from external EM interference



a. UTP

Shielded Twisted Pair (STP)

- Metal braid or sheathing that reduces interference
- More expensive
- Harder to handle (thick, heavy)



b. STP

TWISTED PAIR

Connector = RJ45

Capacity = 10/100/1000/10000mbps

Coverage = 100mts.

STP (Shielded Twisted Pair)

UTP (Un-Shielded Twisted Pair)

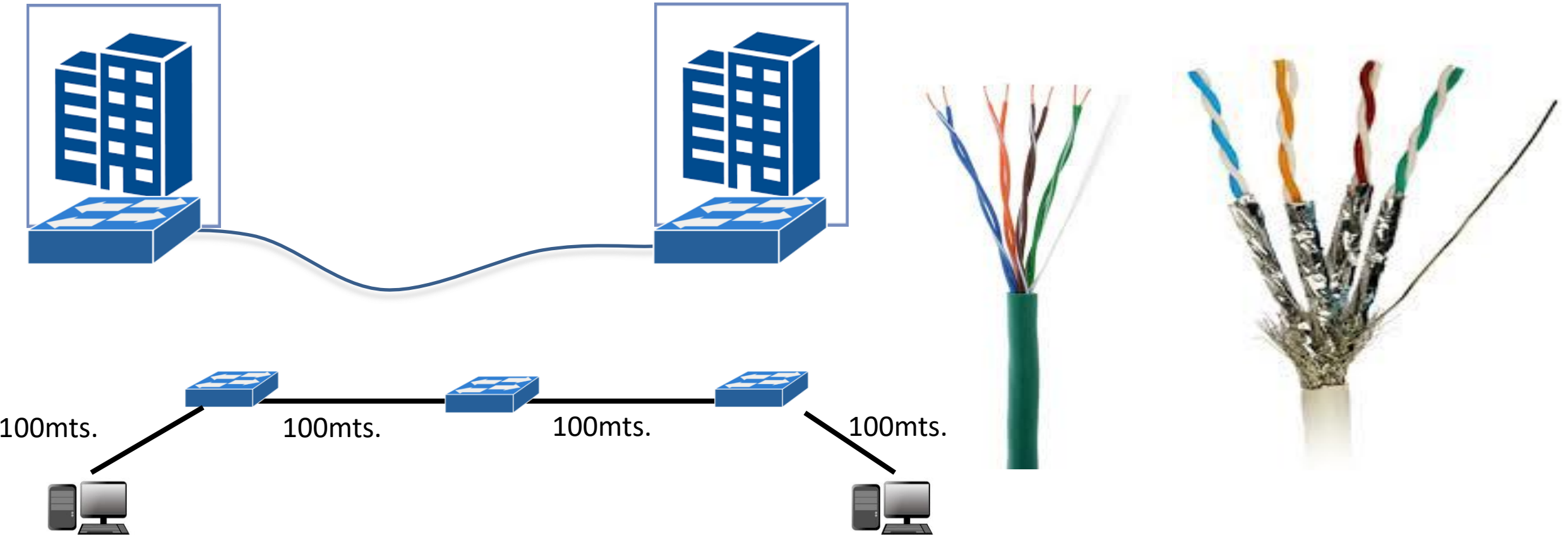
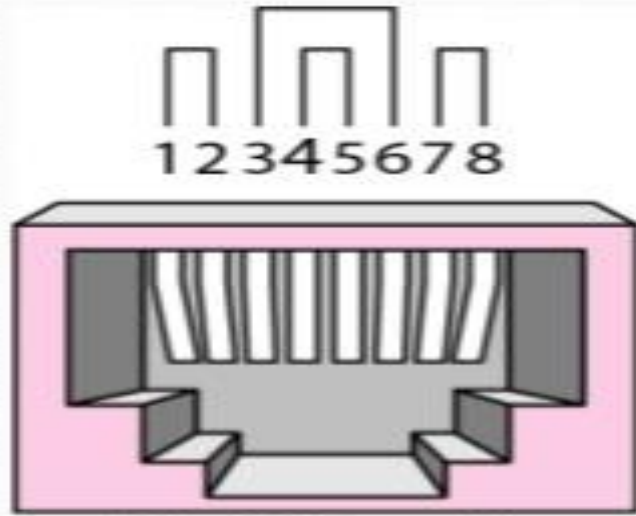


Table 7.1 Categories of unshielded twisted-pair cables

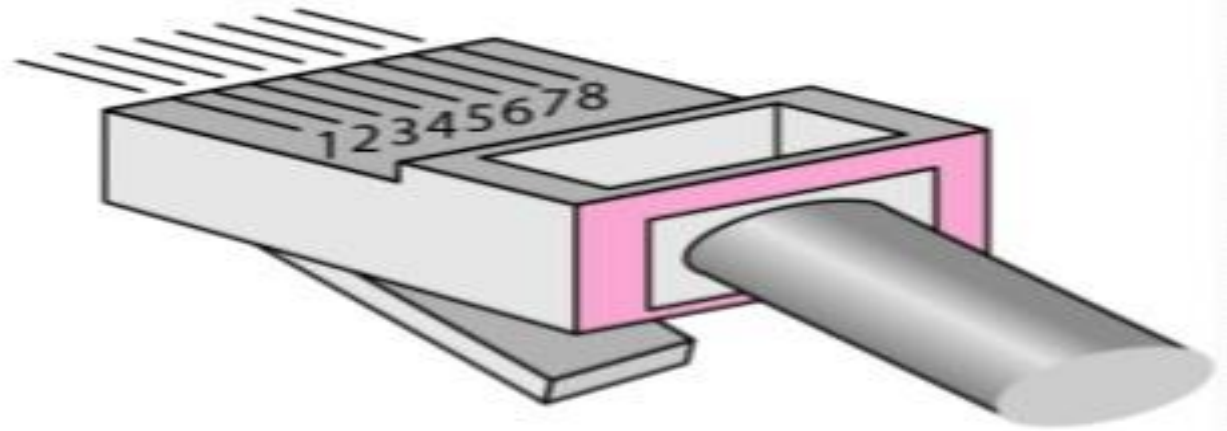
Category	Specification	Data Rate (Mbps)	Use
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T-lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called SSTP (shielded screen twisted-pair). Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

Guided Media – UTP

UTP Connector

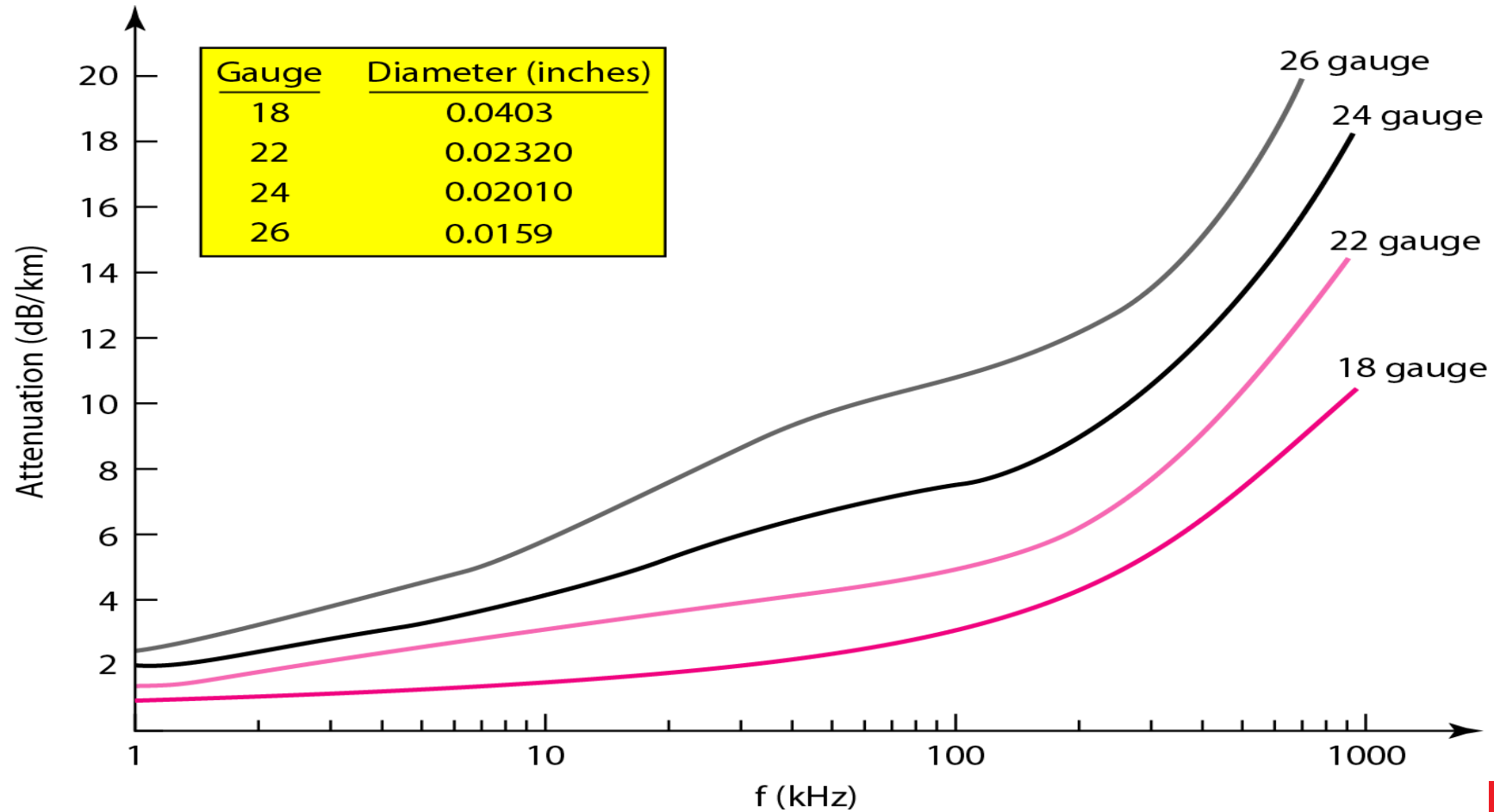


RJ-45 Female



RJ-45 Male

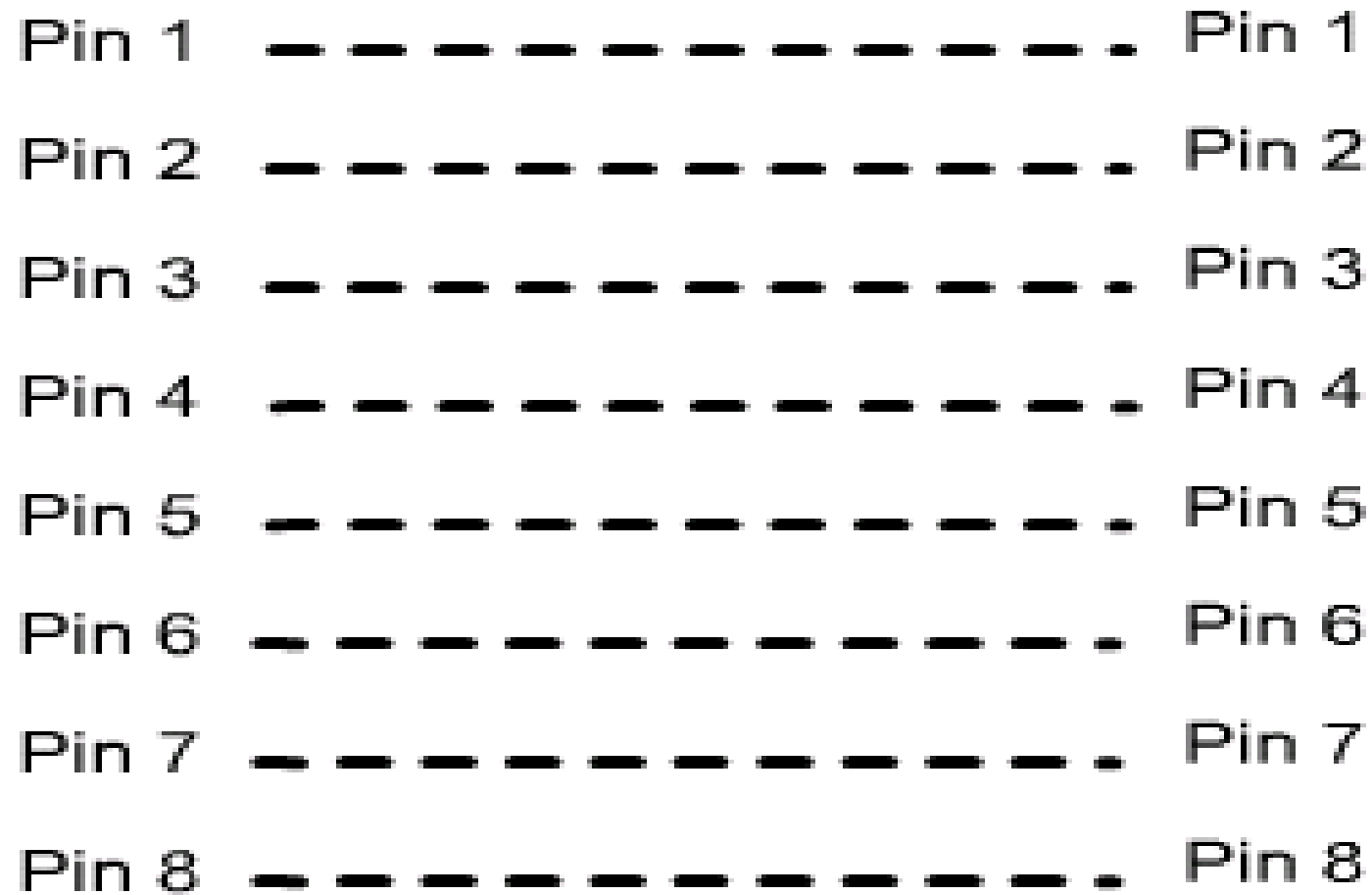
RJ stands for Registered Jack



Straight- through cable



Straight- through cable



Crossover cable

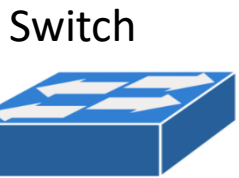


Crossover cable

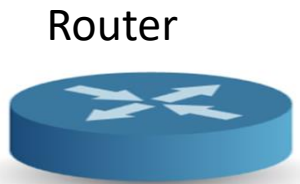
From	To
1	3
2	6
3	1
4	none
5	none
6	2
7	none
8	none



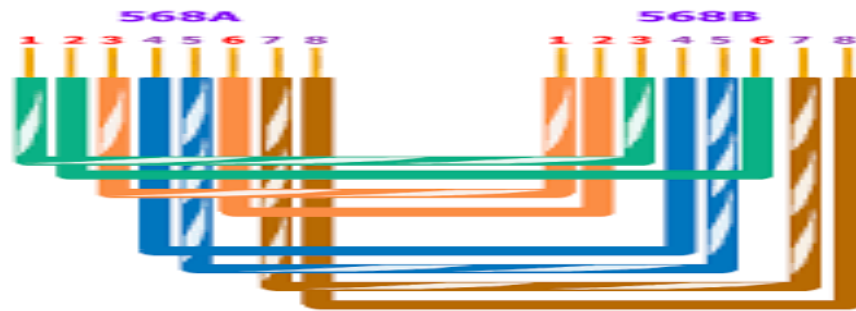
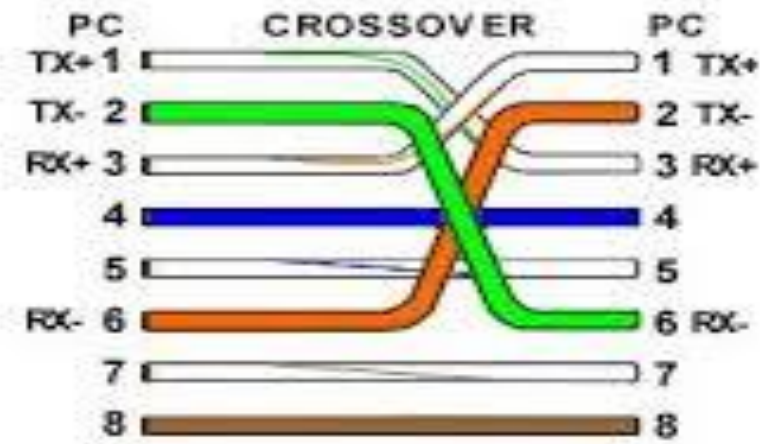
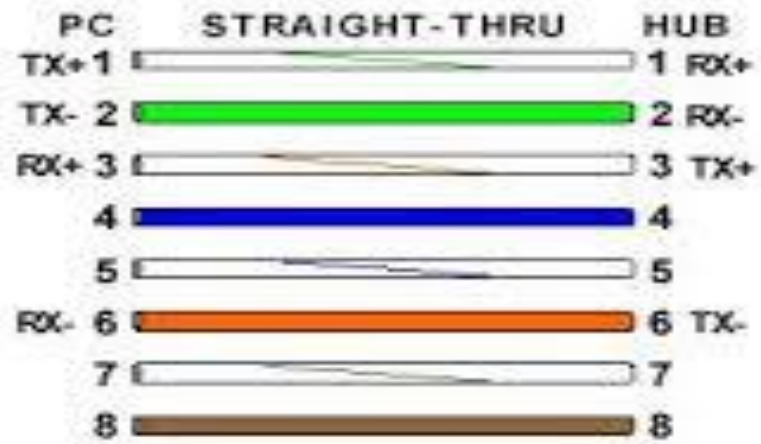
Edge Device



Access Device



Routing Device





Straight - Thru Cable



Straight - Thru Cable



Cross - over Cable



Cross - over Cable



Cross - over Cable

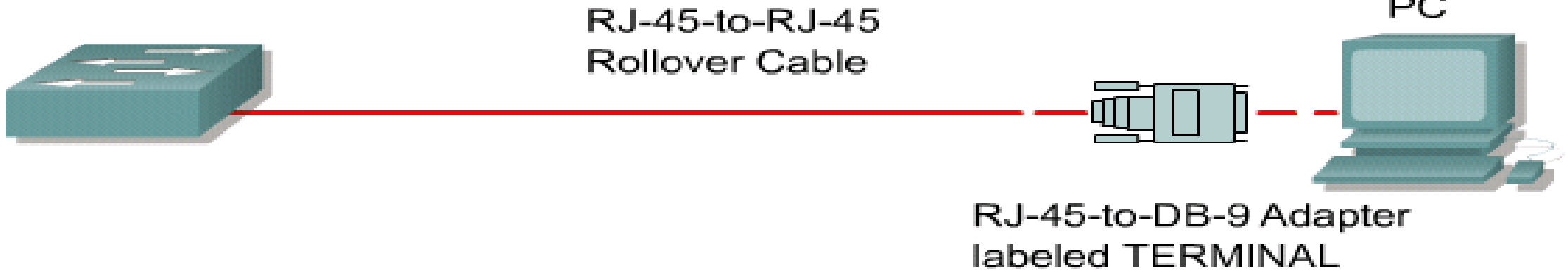


Cross - over Cable

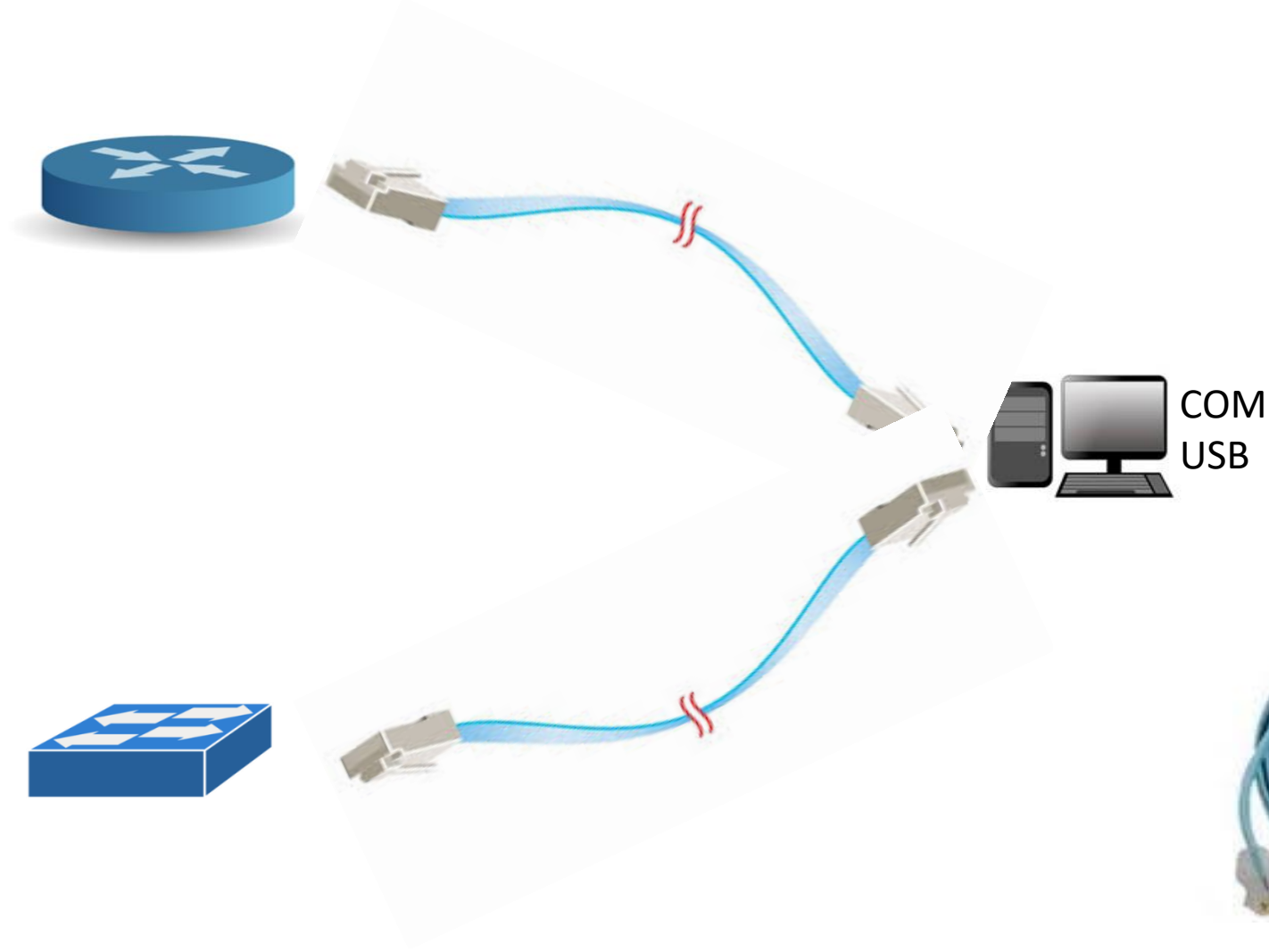


Rollover cable

Device with Console



- PCs require an RJ-45 to DB-9 or RJ-45 to DB-25 adapter.
- COM port settings are 9600 bps, 8 data bits, no parity, 1 stop bit, no flow control.
- This provides out-of-band console access.
- AUX switch port may be used for a modem-connected console.



Roll-Over Cable		
1	_____	8
2	_____	7
3	_____	6
4	_____	5
5	_____	4
6	_____	3
7	_____	2
8	_____	1

} RS232



CISCO
Console
Cable

Straight-Thru or Crossover

☐ Use straight-through cables for the following cabling:

- ☐ Switch to router
- ☐ Switch to PC or server
- ☐ Hub to PC or server

☐ Use crossover cables for the following cabling:

- ☐ Switch to switch
- ☐ Switch to hub
- ☐ Hub to hub
- ☐ Router to router
- ☐ PC to PC
- ☐ Router to PC

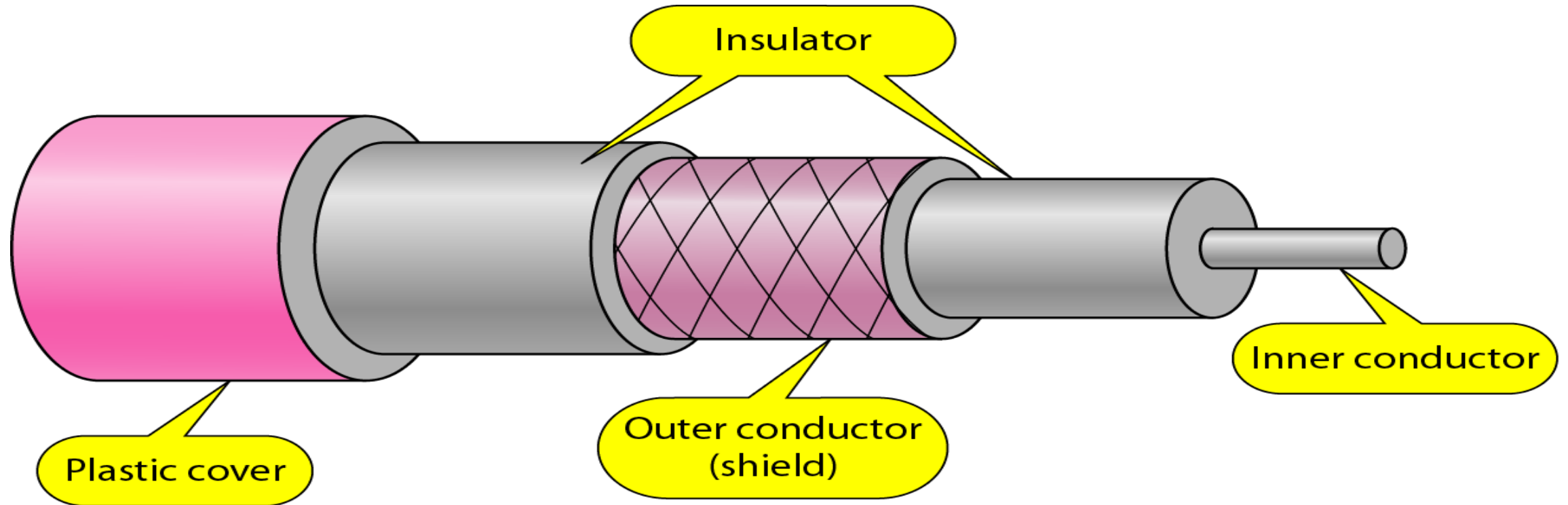
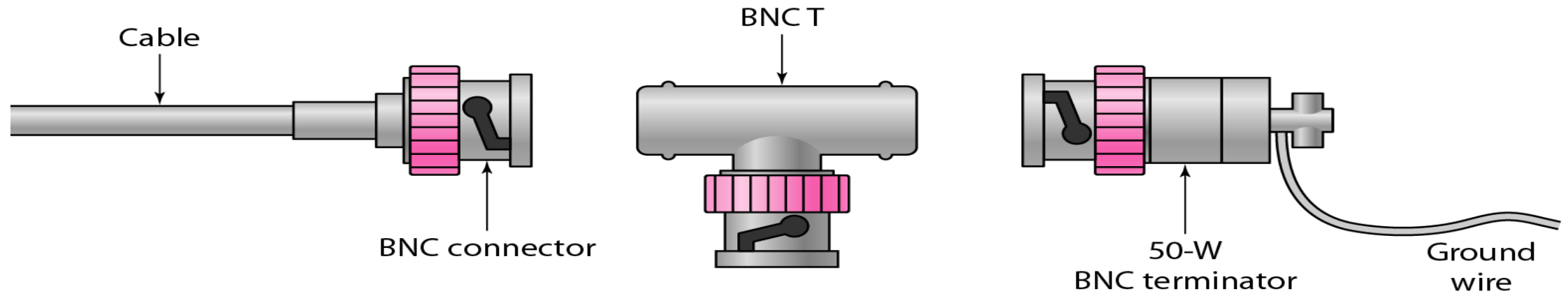


Table 7.2 *Categories of coaxial cables*

<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet



BNC Connector :(Bayonet
Neill-Concelman

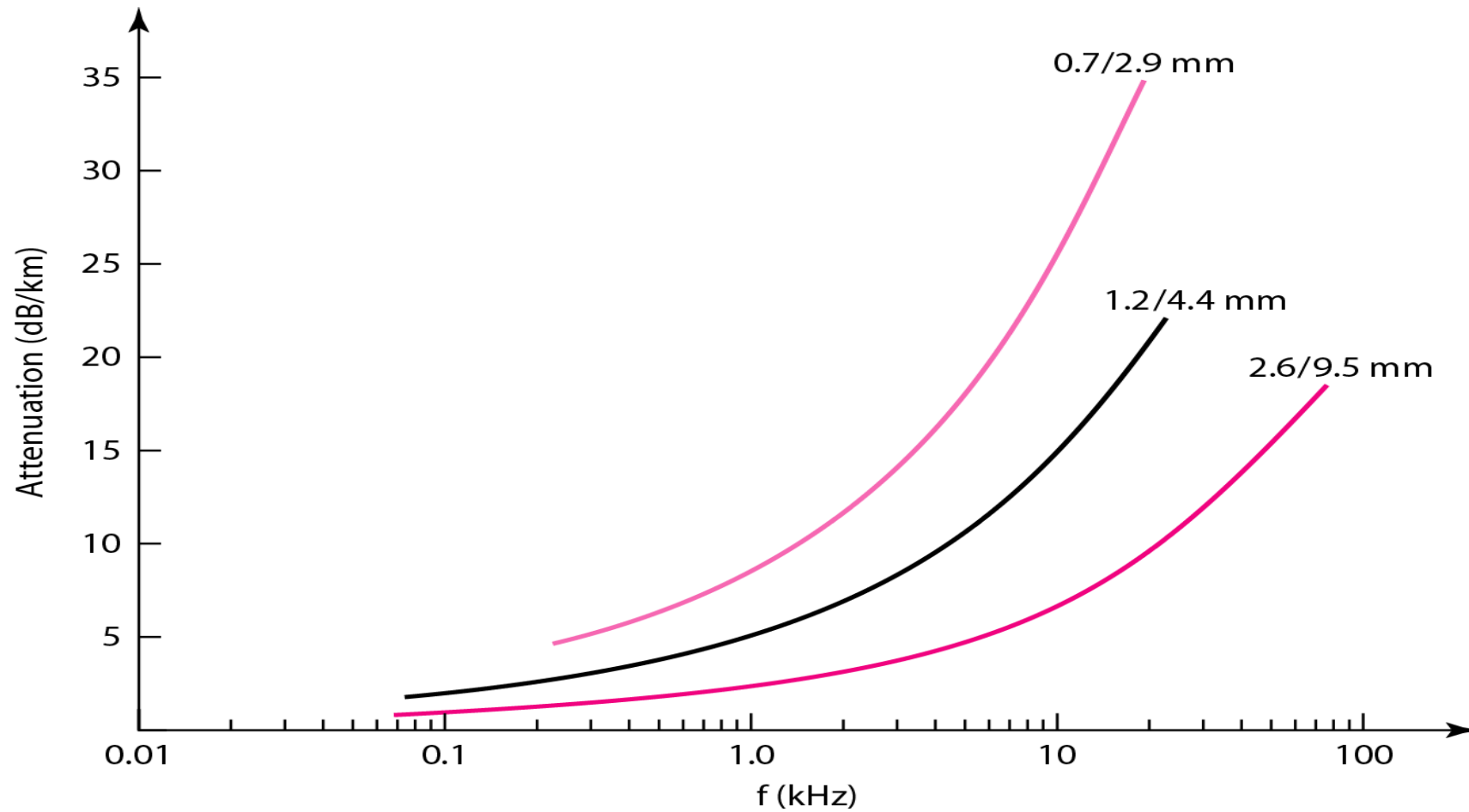
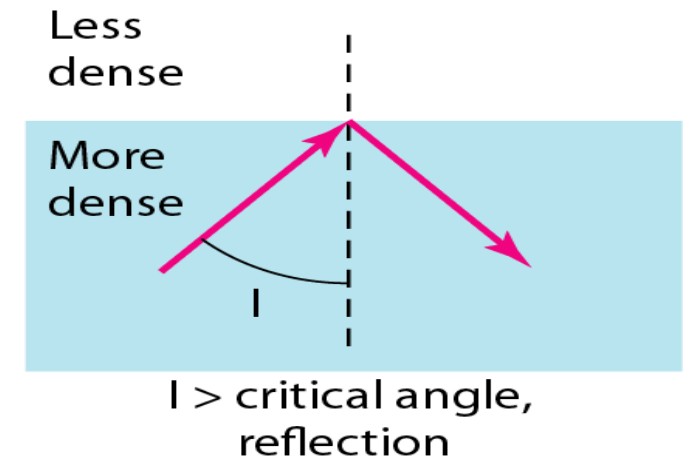
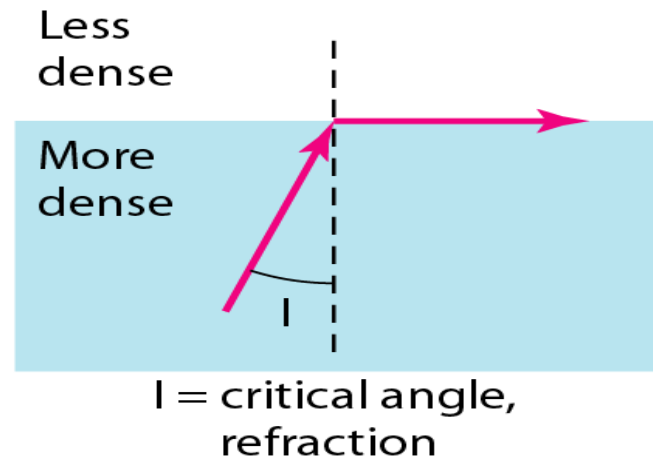
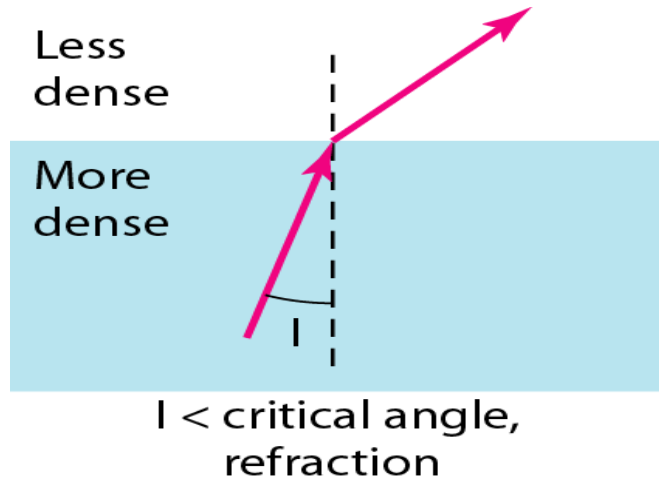


Figure 7.10 Fiber optics: *Bending of light ray*



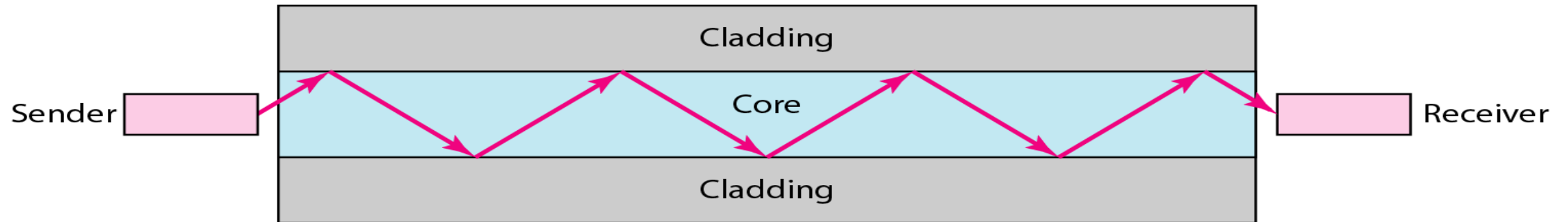
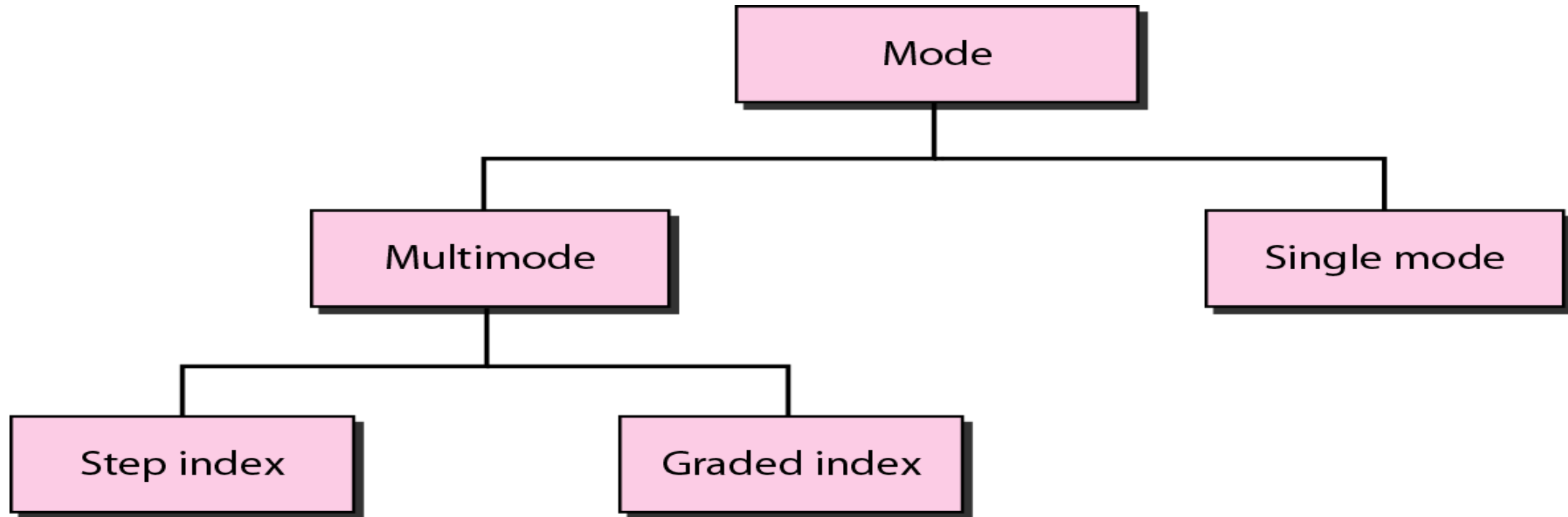
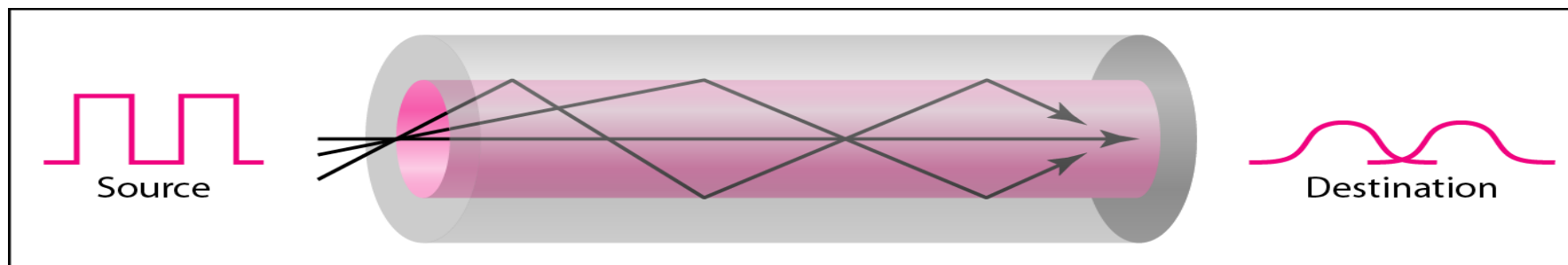
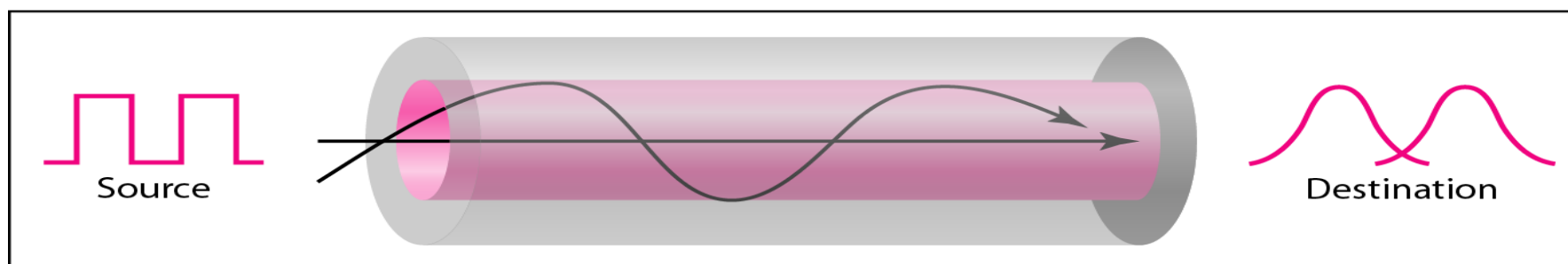


Figure 7.12 *Propagation modes*

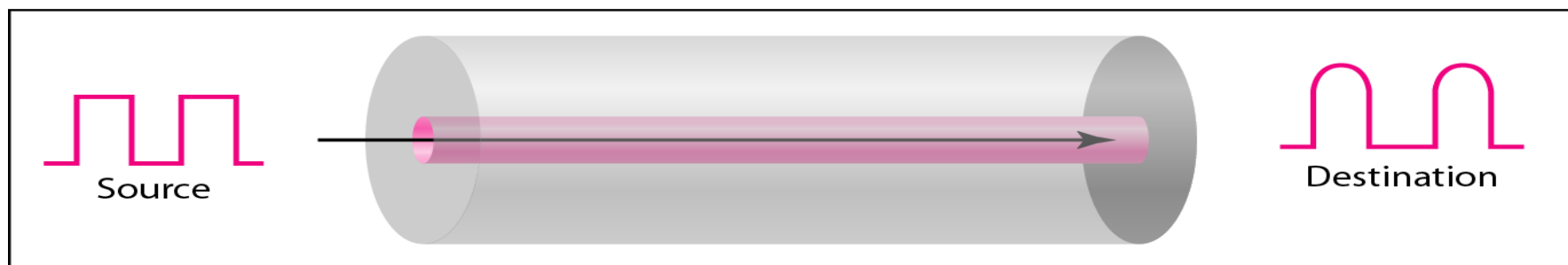




a. Multimode, step index



b. Multimode, graded index

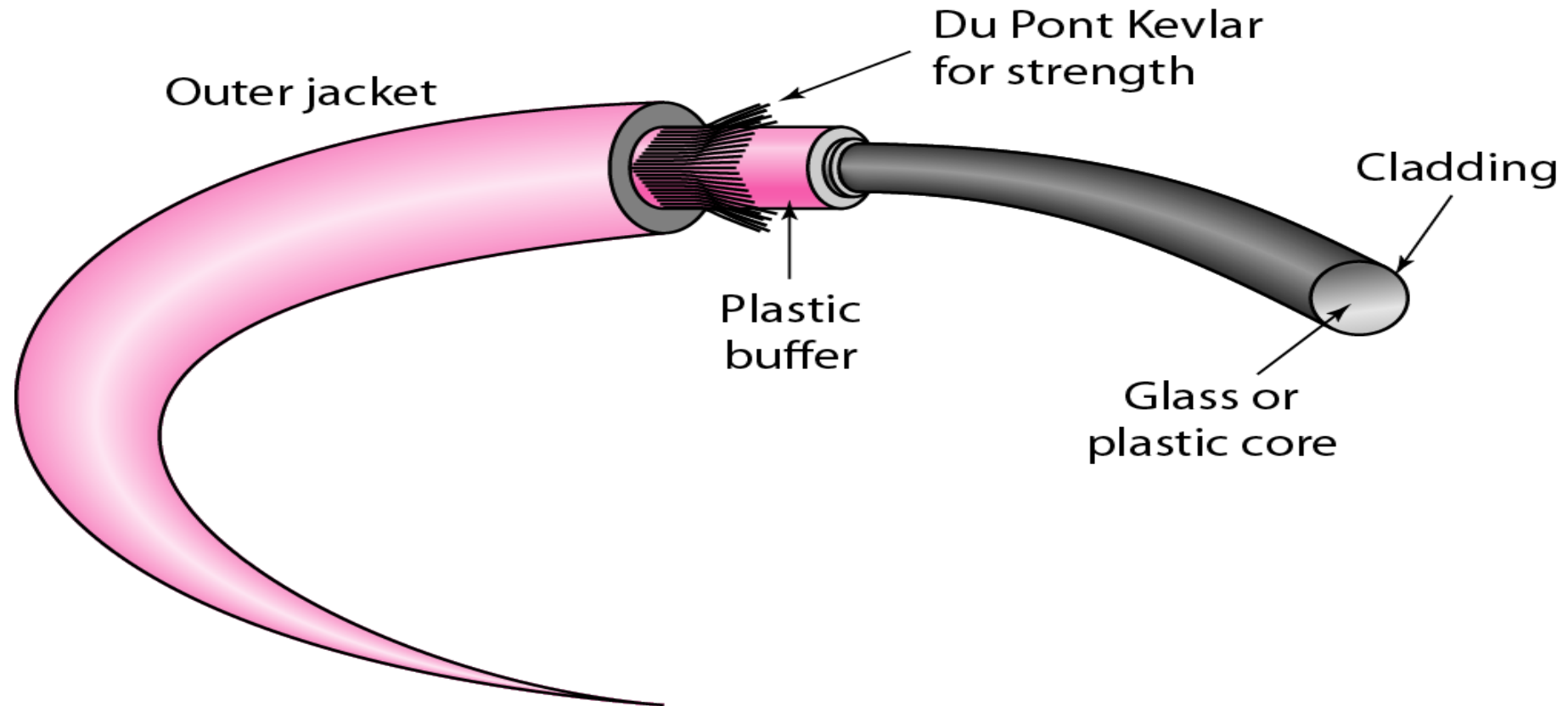


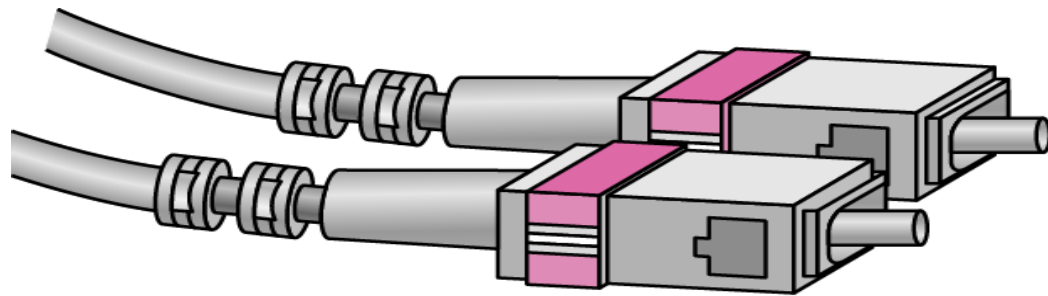
c. Single mode

Table 7.3 *Fiber types*

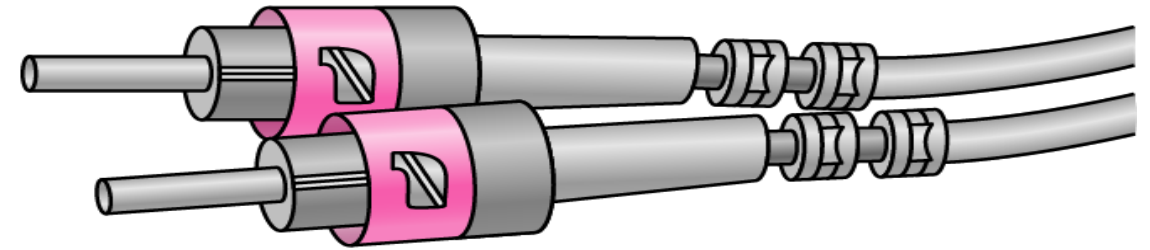
<i>Type</i>	<i>Core (μm)</i>	<i>Cladding (μm)</i>	<i>Mode</i>
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

Figure 7.14 *Fiber construction*

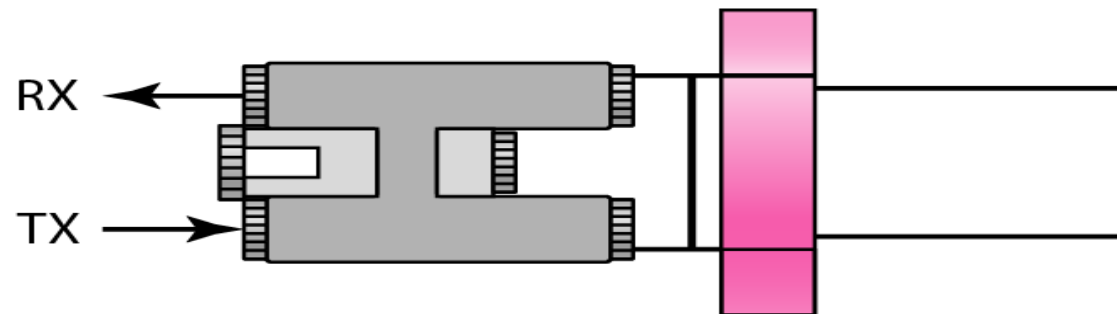




SC connector

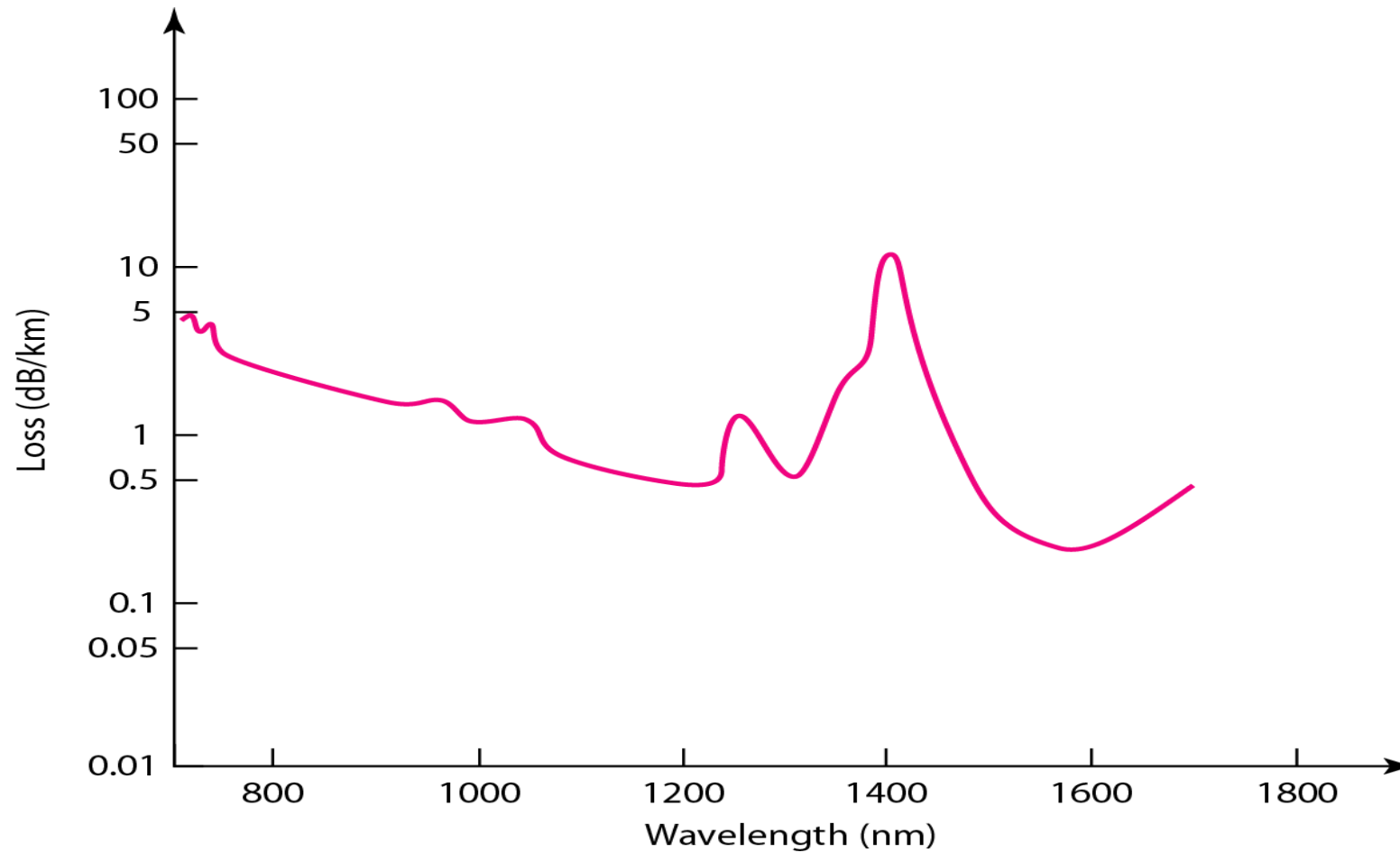


ST connector



MT-RJ connector

Figure 7.16 Optical fiber performance



Sr.no	Twisted Pair Cable	Co-axial Cable	Optical Fiber
1	Transmission of signals takes place in the electrical form over the metallic conducting wires.	Transmission of signals takes place in the electrical form over the inner conductor of the cable	Signal transmission takes place in an optical form over a glass fiber.
2	Noise immunity is low, Therefore more distortion.	Higher noise immunity than the twisted pair cable due to the presence of shielding conductor.	Highest noise immunity as the light rays are unaffected by the electrical noise.
3	Affected due to external magnetic field.	Less affected due to external magnetic field.	Not affected by the external magnetic field.
4	Short circuit between the two conductors is possible.	Short circuit between the two conductors is possible.	Short circuit is not possible.
5	Cheapest	Moderately expensive	Expensive
6	Can support low data rates	Moderately high bandwidth	Very high bandwidth
7	Power loss due to conduction and radiation.	Power loss due to conduction.	Power loss due to absorption, scattering, dispersion and bending.
8	Low Bandwidth	Moderately high bandwidth	Very high bandwidth
9	Node capacity per segment is 2	Node capacity per segment is 30 to 100.	Node capacity per segment is 2
10	Attenuation is very high	Attenuation is low	Attenuation is very low
11	Installation is easy	Installation is fairly easy	Installation is difficult
12	Electromagnetic Interference (EMI) takes place.	EMI is reduced due to shielding.	EMI is not present.

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.

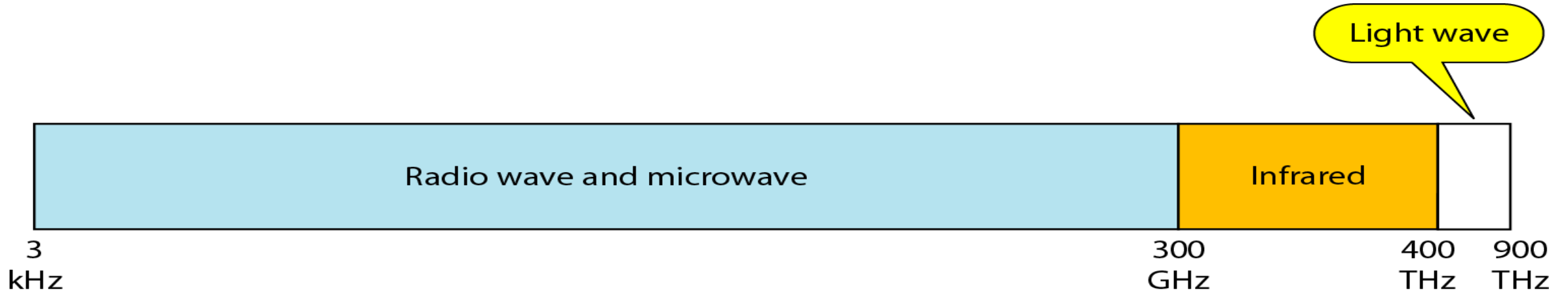
Topics discussed in this section:

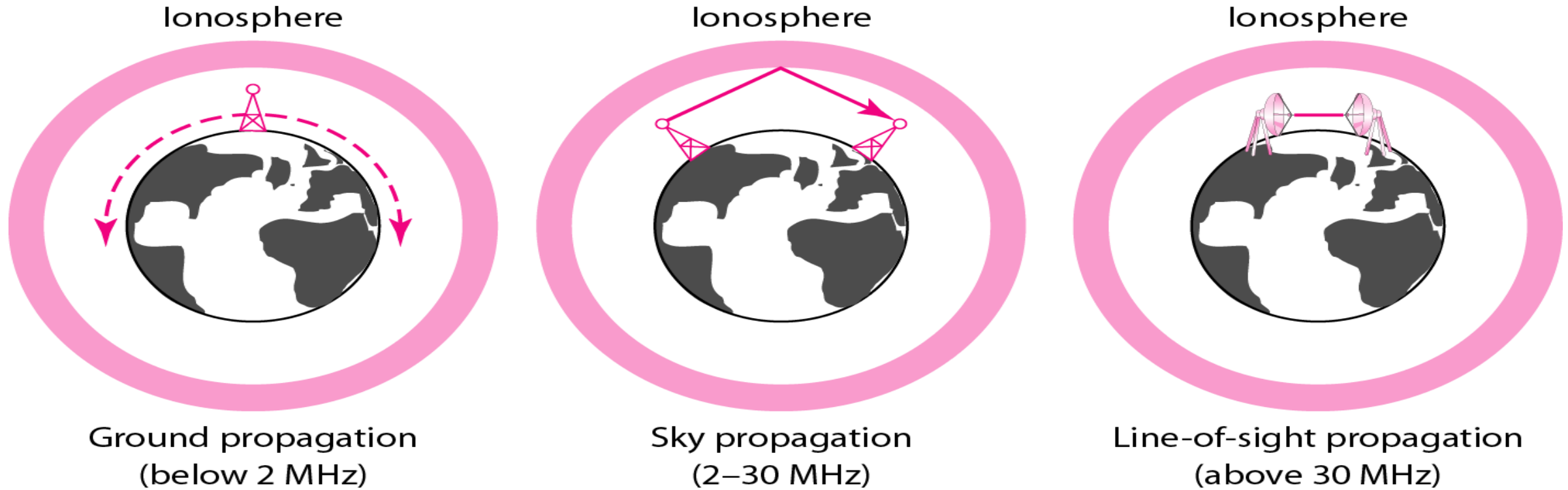
Radio Waves

Microwaves

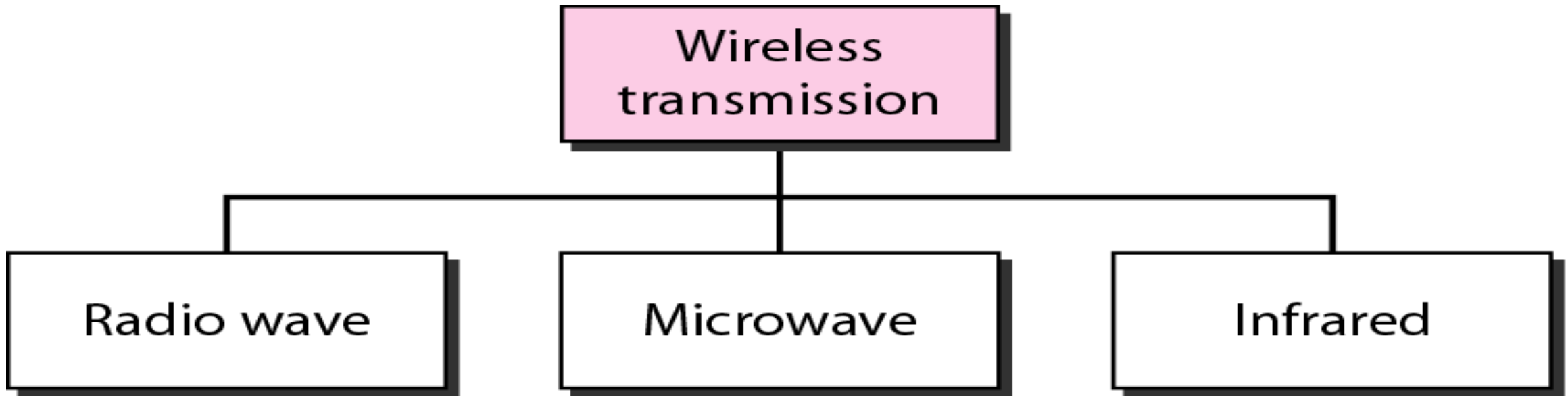
Infrared

Figure 7.17 *Electromagnetic spectrum for wireless communication*



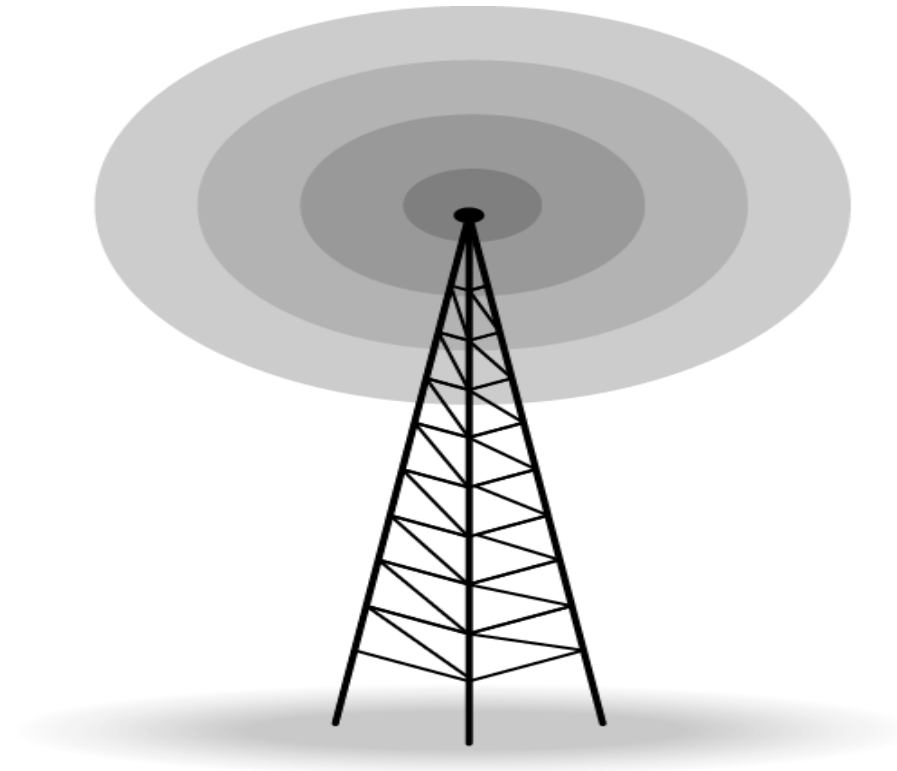


<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite



Note

Radio waves are used for multicast communications, such as radio and television, and paging systems. They can penetrate through walls. Highly regulated. Use omni directional antennas

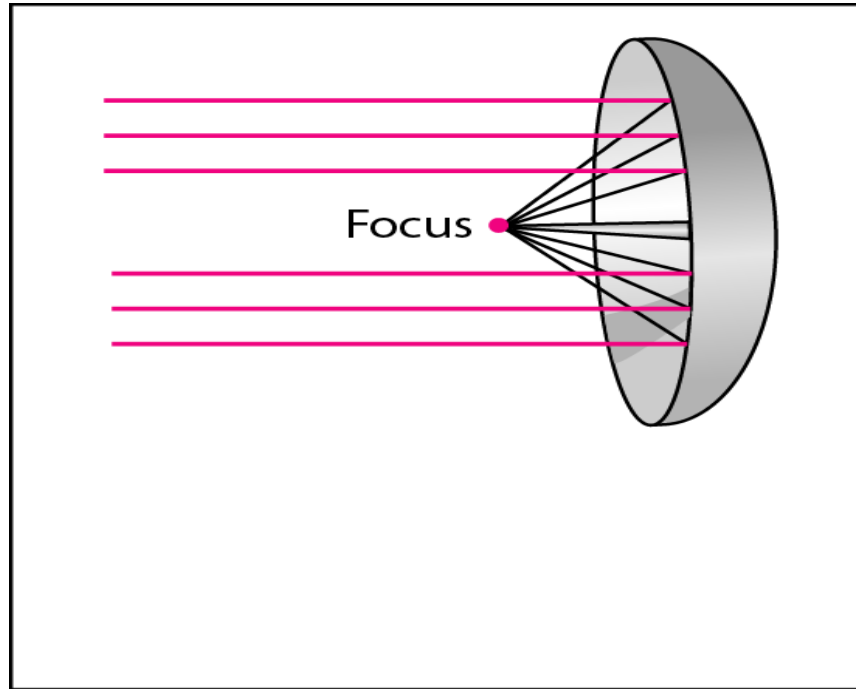


Note

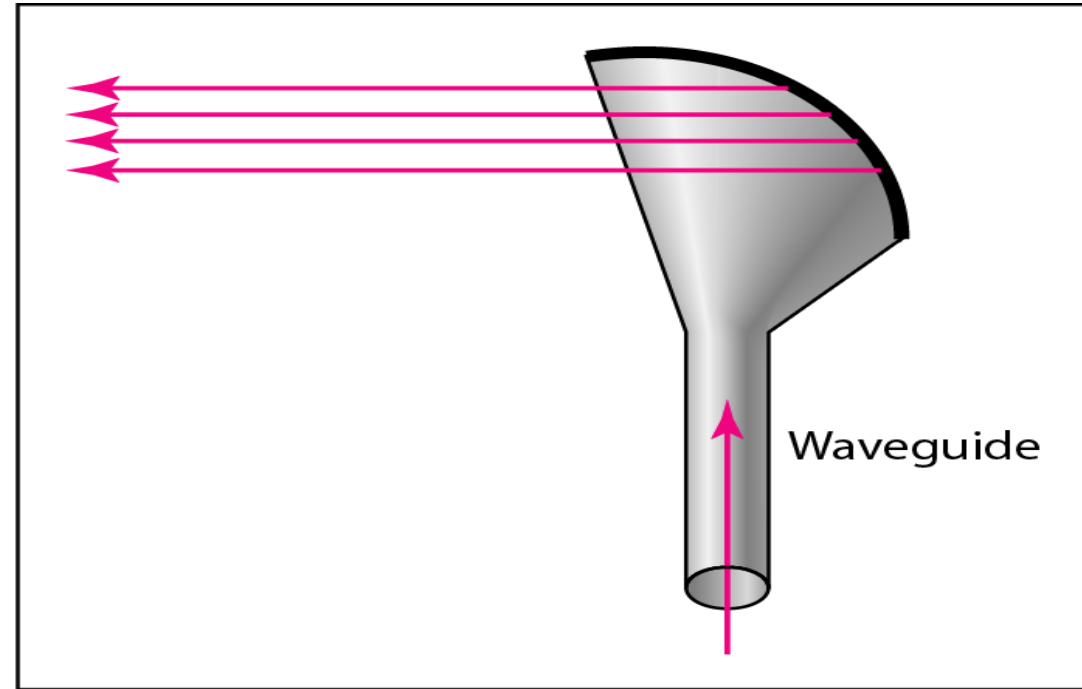
Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

**Higher frequency ranges cannot penetrate walls.
Use directional antennas - point to point line of sight communications.**

Figure 7.21 Unidirectional antennas



a. Dish antenna



b. Horn antenna

Note

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

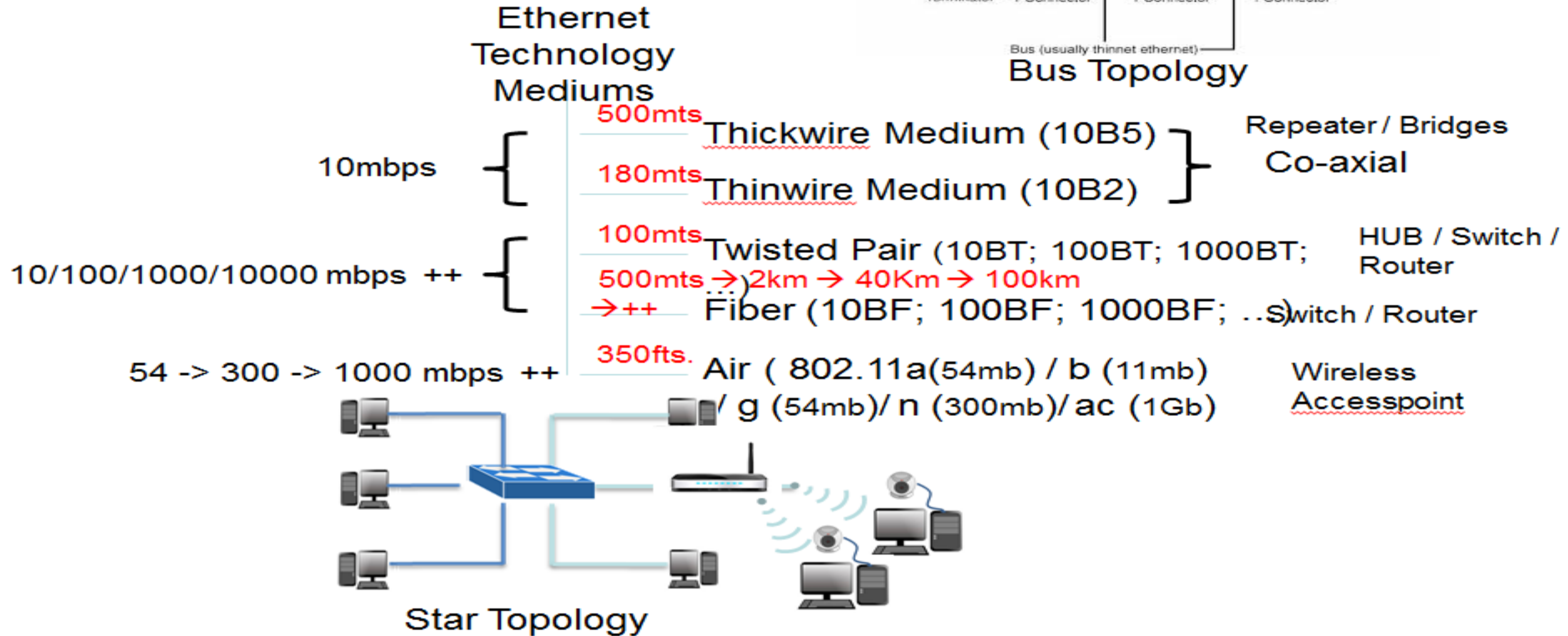
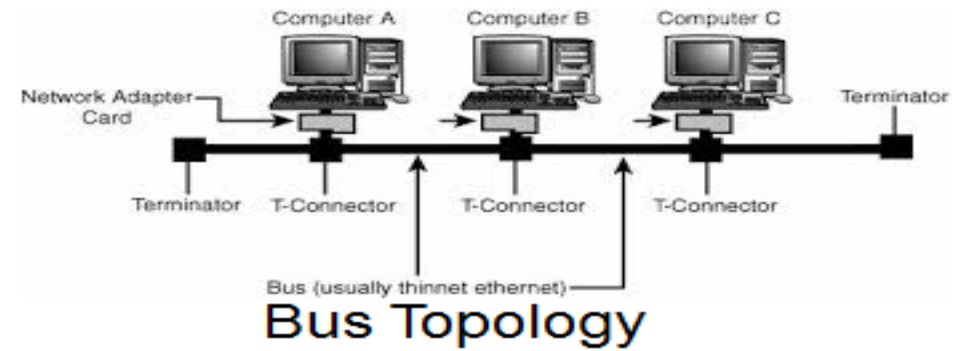
Wireless Channels

Are subject to a lot more errors than guided media channels.

Interference is one cause for errors, can be circumvented with high SNR.

The higher the SNR the less capacity is available for transmission due to the broadcast nature of the channel.

Channel also subject to fading and no coverage holes.



References

www.cisco.com

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References

- “Data Communication and Networking”, Behrouz Forouzan 5e
- “TCP/IP Protocol Suite “, Behrouz Forouzan 4e.
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